

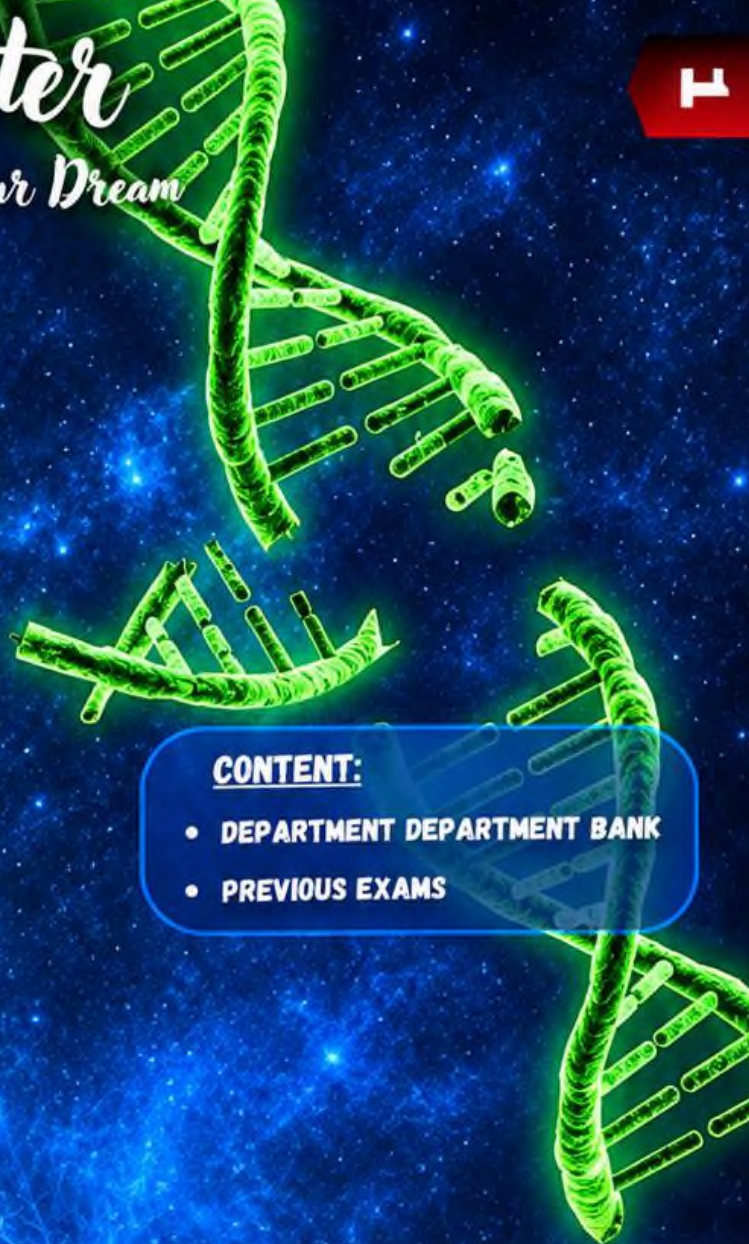
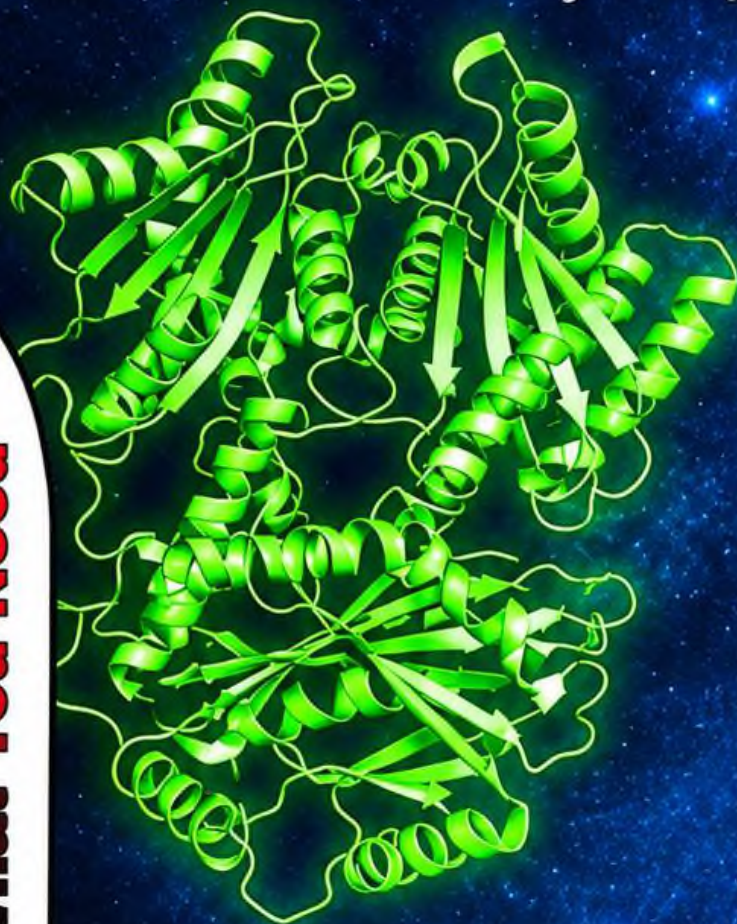
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All What You Need



CONTENT:

- DEPARTMENT DEPARTMENT BANK
- PREVIOUS EXAMS

BIOCHEMISTRY & GENETICS BOOKLET

2026



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1. Carbohydrates

1. In glucose the orientation of the -H and -OH groups around the carbon atom 5 adjacent to the terminal primary alcohol carbon determines [End Semester 2018 (دور اول)]
- D or L series
 - Dextrose or levorotatory
 - α and β anomers
 - Epimers
2. Which of the following is a reducing disaccharide? [End Semester 2018 (دور اول)]
- Sucrose
 - Trehalose
 - Isomaltose
 - Agar
3. All are polymers of glucose except [End Semester 2018 (دور اول)]
- Glycogen
 - insulin
 - Starch
 - Cellulose
4. Which of the following is an example of disaccharide? [End Semester 2018 (دور ثاني)]
- Fructose
 - Galactose
 - Glucose
 - Maltose
5. Which of the following is true about monosaccharides? [End Semester 2018 (دور ثاني)]
- They are insoluble in water
 - They can be hydrolyzed
 - They cannot be hydrolyzed
 - They have sour taste
6. In which form is Glucose stored in animals? [End Semester 2018 (دور ثاني)]
- Cellulose
 - Dextrins
 - Glycogen
 - Starch



7. The polysaccharide used in assessing glomerular filtration rate [End Semester 2018 (دور ثاني)]
- Agar
 - Glycogen
 - Hyaluronic acid
 - Insulin
8. Which of the following is the Aldose sugar? [End Module 2019]
- Sucrose
 - Ribose
 - Fructose
 - Ribulose
9. One of the following has reducing properties [End Module 2019]
- Glucuronic acid
 - Gluconic acid
 - Mucic acid
 - Glucaric acid
10. The complete name of the opposite structure is: [End Module 2019]
- α -D-gulopyranose
 - β -D-gulopyranose
 - α -D-gulofuranose
 - β -D-gulofuranose
11. ALL of these are correct combinations of monosaccharides to form disaccharides EXCEPT: [End Module 2019]
- Glucose + fructose = Sucrose
 - Glucose + galactose = Lactose
 - Glucose + Glucose = Maltose
 - Glucose + fructose = Lactose
12. The two sugar units of disaccharides are connected by which of the following bonds? [End Module 2019]
- O-glycosidic bond
 - N-glycosidic bond
 - Peptide bond
 - Phosphodiester bond



13. Which of the following is NOT glucosan? [End Module 2019]
- Inulin
 - Glycogen
 - Starch
 - Cellulose
14. What are the two structures making up starch? [End Module 2019]
- Amylose and cellulose
 - Amylose and amylopectin
 - Amylopectin and cellobiose
 - Cellobiose and cellulose
15. The polysaccharide used in assessing the glomerular filtration rate (GFR) is [End Module 2019]
- Glycogen
 - Agar
 - Inulin
 - Hyaluronic acid
16. A sugar alcohol is: [Formative Exam 2021]
- Mannitol
 - Trehalose
 - Xylulose
 - Arabinose
17. The sugar found in DNA is [Formative Exam 2021]
- Xylose
 - Ribose
 - Deoxyribose
 - Ribulose
18. Ribulose is a [Formative Exam 2021]
- Ketotetrose
 - Aldosterone
 - Ketopentose
 - Aldopentose



19. Spermatozoa in seminal fluid utilizes the following sugar for its metabolism

[Formative Exam 2021]

- a. Glucose
- b. Fructose
- c. Sucrose
- d. Mannose

20. α -D-glucose $\rightleftharpoons$ β -D-glucose (with a change in optical rotation over time) represents which of the following phenomena? [End Module 2022]

- a. Optical isomerism
- b. Mutarotation
- c. Epimerization
- d. D and L isomerism

21. Which of the following is not a polymer of glucose? [End Module 2022]

- a. Amylose
- b. Inulin
- c. Cellulose
- d. Dextrin

22. Which of the following is a non-reducing sugar? [End Module 2022]

- a. Isomaltose
- b. Maltose
- c. Lactose
- d. Trehalose

23. What is the glycosaminoglycan which does not contain uronic acid? [End Module 2022]

- a. Dermatan sulphate
- b. Chondroitin sulphate
- c. Keratan sulphate
- d. Heparan sulphate

24. Which of the following is an epimeric pair? [End Module 2022]

- a. Glucose and fructose
- b. Glucose and galactose
- c. Galactose and mannose
- d. Lactose and maltose



25. In glucose the orientation of the -H and -OH groups around the carbon atom 5 adjacent to the terminal primary alcohol carbon determines which of the following [End Module 2022]

- a. D or L series
- b. Dextro or levorotatory
- c. α and β anomers
- d. Epimers

26. What is the polysaccharide used in assessing the glomerular filtration rate (GFR)? [End Module 2022]

- a. Glycogen
- b. Agar
- c. Inulin
- d. Hyaluronic acid

27. Isomers differing as a result of variations in configuration of the OH and H on carbon atoms 2, 3 and 4 of glucose are known as [End Semester 2022]

- a. Epimers
- b. Anomers
- c. Optical isomers
- d. Stereoisomers

28. A disaccharide formed by a 1,1-glycosidic linkage between their monosaccharide units is [End Semester 2022]

- a. Lactose
- b. Maltose
- c. Trehalose
- d. Sucrose

29. Keratan sulphate I is found in abundance in which of the following tissues? [End Semester 2022]

- a. Heart muscle
- b. Liver
- c. Adrenal cortex
- d. Cornea



30. What is the carbon atom that becomes asymmetric when the straight chain form of a monosaccharide changes into ring form? [End Semester 2022]
- Anomeric carbon atom
 - Epimeric carbon atom
 - Isomeric carbon atom
31. The highest concentrations of fructose are found in which of the following? [End Semester 2022]
- Aqueous humor
 - Vitreous humor
 - Synovial fluid
 - Seminal fluid
32. Oxidation of the aldehyde group of glucose by dilute nitric acid produces which of the following compounds? [End Semester 2022]
- Glucuronic acid
 - Gluconic acid
 - Glucaric acid
 - Galactic acid
33. What is the branched part of the starch molecule? [End Semester 2022]
- Glucose
 - Amylopectin
 - Isomaltose
 - Amylose
34. Dietary fibers are rich in which of the following? [End Semester 2022]
- Cellulose
 - Glycogen
 - Starch
 - Proteoglycans
35. Which of the following is a non-reducing disaccharide? [Formative Exam 2023]
- Starch
 - Maltose
 - Sucrose
 - Glycogen



36. How many asymmetric carbon atoms are there in D-glucose? [Formative Exam 2023]
- Two
 - Three
 - Four
 - Five
37. How many carbon atoms are present in Dihydroxyacetone? [End Semester 2024]
- 0
 - 1
 - 2
 - 3
38. Which of the following describes sucrose as a non-reducing disaccharide? [End Semester 2024]
- It presents in both alpha form and beta form
 - It's formed of alpha fructose and beta glucose linked by $\alpha 1 \rightarrow \beta 2$ linkage
 - It can form characteristic osazone crystals
 - after hydrolysis the angle of rotation becomes inverted
39. Which disaccharide contains beta glucose? [End Semester 2024]
- Maltose
 - Isomaltose
 - Cellobiose
 - Trehalose
40. After hydrolysis of starch it gives Which of the following? [End Semester 2024]
- Alpha maltose
 - Beta maltose
 - Beta glucose
 - Alpha isomaltose
41. Which substance is used as a kidney function test? [End Semester 2024]
- Inulin
 - Chitin
 - agar
 - dextrose



42. What is the importance of the indigestible cellulose in our diet? [End Semester 2024]

- a. It's hydrolysed into glucose which is absorbed in the blood
- b. It's hydrolysed into cellobiose which is absorbed in the blood
- c. It's hydrolysed into cellobiose which protects against constipation
- d. As it is making stool soft and preventing constipation

43. How many carbon atoms are present in Dihydroxyacetone? [End Semester 2025]

- a. 0
- b. 1
- c. 2
- d. 3

44. Which of the following describes sucrose as a non-reducing disaccharide? [End Semester 2025]

- a. It presents in both alpha form and beta form
- b. It's formed of alpha fructose and beta glucose linked by $\alpha 1 \rightarrow \beta 2$ linkage
- c. It can form characteristic osazone crystals
- d. After hydrolysis the angle of rotation becomes inverted

45. Which disaccharide contains beta glucose? [End Semester 2025]

- a. Maltose
- b. Isomaltose
- c. Cellobiose
- d. Trehalose

46. After hydrolysis of starch it gives which of the following? [End Semester 2025]

- a. Alpha maltose
- b. Beta maltose
- c. Beta glucose
- d. Alpha isomaltose

47. Which substance is used as a kidney function test? [End Semester 2025]

- a. Inulin
- b. Chitin
- c. Agar
- d. Dextrose



48. What is the importance of the indigestible cellulose in our diet? [End Semester 2025]

- a. It's hydrolysed into glucose which is absorbed in the blood
- b. It's hydrolysed into cellobiose which is absorbed in the blood
- c. It's hydrolysed into cellobiose which protects against constipation
- d. As it makes stool soft and prevents constipation

49. Which of the following statements best describes glucose? [End Module 2026 (دور اول)]

- a. It is a C-4 epimer of galactose.
- b. It is a ketose and usually exists as a furanose ring in solution.
- c. It is produced from dietary starch by the action of α -amylase.
- d. It is utilized in biological systems only in the L-isomeric form.

50. Which of the following best describes starch? [End Module 2026 (دور اول)]

- a. It cannot be digested by humans.
- b. It is composed of unbranched amylose and branched amylopectin.
- c. It is composed of unbranched amylose and unbranched amylopectin.
- d. It is a D-glucose polymer.

51. Which of the following statements is characteristic for cellulose compared to starch? [End Module 2026 (دور اول)]

- a. It is a glucose polymer.
- b. It can be digested by humans.
- c. It is the main source of carbohydrates in diet.
- d. It can't be used as a source of energy.

52. In uncontrolled diabetes, which sugar alcohol tends to accumulate in tissues like the lens, contributing to complications (e.g., cataracts and neuropathy)? [End Module 2026 (دور اول)]

- a. Mannitol
- b. Xylitol
- c. Sorbitol
- d. Ribitol

53. Which polysaccharide consists of β -(1 $\rightarrow$ 4)-linked glucose units? [End Module 2026 (دور اول)]

- a. Amylose
- b. Glycogen
- c. Cellulose
- d. Amylopectin



54. Which of the following sugars lacks reducing properties? [End Module 2026 (دور أول)]

- a. Lactose
- b. Fructose
- c. Maltose
- d. Sucrose

55. Hydrolysis of sucrose by the enzyme sucrase (invertase) yields which monosaccharides? [End Module 2026 (دور أول)]

- a. Glucose and galactose
- b. Two molecules of glucose
- c. Fructose and galactose
- d. Glucose and fructose

56. Which of the following is an anomeric pair? [End Module 2026 (دور أول)]

- a. D-glucose and L-glucose
- b. D-glucose and D-fructose
- c. α -D-glucose and β -D-glucose
- d. α -D-glucose and α -L-glucose

57. What is the general formula of dextrose? [End Module 2026 (دور أول)]

- a. $C_5H_{10}O_5$
- b. $C_6H_{10}O_5$
- c. $C_6H_{10}O_6$
- d. $C_6H_{12}O_6$

58. Spermatozoa in seminal fluid utilizes which of the following sugars for its metabolism? [End Module 2026 (دور أول)]

- a. Glucose
- b. Fructose
- c. Sucrose
- d. Mannose

59. Which is a disaccharide linked by a β -1,4 glycosidic linkage? [End Module 2026 (دور أول)]

- a. Lactose
- b. Sucrose
- c. Cellulose
- d. Maltose



60. Which statement describes the use of glycogen during starvation? [End Module 2026 (دور أول)]
- Liver glycogen is conserved for muscle use
 - Muscle glycogen is converted to glucose for the brain
 - Liver glycogen is broken down to maintain blood glucose
 - Muscle glycogen increases due to lack of energy use
61. The conversion of α -D-Glucose (+112 $\rightarrow$ +52.5) and β -D-glucose (+19 $\rightarrow$ +52.5) is defined by which of the following? [End Module 2026 (دور أول)]
- Stereoisomerism
 - Optical isomerism
 - Mutarotation
 - Epimerization
62. What is the product(s) of reduction of glucose? [End Module 2026 (دور أول)]
- Dulcitol
 - Sorbitol
 - Mannitol
 - Mannitol and sorbitol
63. The main cause of lactose intolerance is directly attributed to which of the following? [End Module 2026 (دور أول)]
- An autoimmune reaction to casein in milk
 - A deficiency of the enzyme sucrase
 - A deficiency of the lactase enzyme
 - Impaired galactose transport into enterocytes
64. Which of the following statements is TRUE regarding invert sugar? [End Module 2026 (دور أول)]
- It is the hydrolytic product of lactose sugar.
 - It is a non-reducing mixture.
 - It exhibits mutarotation.
 - It deviates the angle of polarized light to the right.
65. Which of the following statements best describes glucose? [End Module 2026]
- It is a C-4 epimer of galactose.
 - It is a ketose and usually exists as a furanose ring in solution.
 - It is produced from dietary starch by the action of α -amylase.
 - It is utilized in biological systems only in the L-isomeric form.



66. Which of the following best describes for starch? [End Module 2026]
- It cannot be digested by humans.
 - It is composed of unbranched amylose and branched amylopectin.
 - It is composed of unbranched amylose and unbranched amylopectin.
 - It is a D-glucose polymer.
67. Which of the following statements is characteristic for cellulose compared to starch? [End Module 2026]
- It is a glucose polymer.
 - It can be digested by humans.
 - It is the main source of carbohydrates in diet.
 - It can't be used as a source of energy.
68. In uncontrolled diabetes, which sugar alcohol tends to accumulate in tissues like the lens, contributing to complications (e.g., cataracts and neuropathy)? [End Module 2026]
- Mannitol
 - Xylitol
 - Sorbitol
 - Ribitol
69. Which polysaccharide consists of β -(1 $\rightarrow$ 4)-linked glucose units? [End Module 2026]
- Amylose
 - Glycogen
 - Cellulose
 - Amylopectin
70. Which of the following sugars lacks the reducing properties? [End Module 2026]
- Lactose
 - Fructose
 - Maltose
 - Sucrose
71. Hydrolysis of sucrose by the enzyme sucrase (invertase) yields which monosaccharides? [End Module 2026]
- Glucose and galactose
 - Two molecules of glucose
 - Fructose and galactose
 - Glucose and fructose



72. Which of following is an anomeric pair? [End Module 2026]

- a. D-glucose and L-glucose
- b. D-glucose and D-fructose
- c. alpha-D-glucose and beta-D-glucose
- d. alpha-D-glucose and alpha-L-glucose

73. What is the general formula of dextrose? [End Module 2026]

- a. $C_3H_6O_3$
- b. $C_4H_8O_4$
- c. $C_5H_{10}O_5$
- d. $C_6H_{12}O_6$

74. Spermatozoa in seminal fluid utilizes which following sugar for its metabolism? [End Module 2026]

- a. Glucose
- b. Fructose
- c. Sucrose
- d. Mannose

75. Which is a disaccharide linked by beta-(1→4) glycosidic linkage? [End Module 2026]

- a. Lactose
- b. Sucrose
- c. Cellulose
- d. Maltose

76. Which statement describes the use of glycogen during starvation? [End Module 2026]

- a. Liver glycogen is conserved for muscle use.
- b. Muscle glycogen is converted to glucose for the brain.
- c. Liver glycogen is broken down to maintain blood glucose.
- d. Muscle glycogen increases due to lack of energy use.

77. The conversion of α -D-Glucose $+112 \rightarrow +52.5$, B-D-glucose $+19 \rightarrow +52.5$ is defined by which of the following? [End Module 2026]

- a. Stereoisomerism
- b. Optical isomerism
- c. Mutarotation
- d. Epimerization



78. What is the product (s) of reduction of glucose? [End Module 2026]

- a. Dulcitol
- b. Sorbitol
- c. Mannitol
- d. Mannitol and sorbitol

79. The main cause of lactose intolerance is directly attributed to which of the following? [End Module 2026]

- a. An autoimmune reaction to casein in milk.
- b. A deficiency of the enzyme sucrase.
- c. A deficiency of the lactase enzyme.
- d. Impaired galactose transport into enterocytes.

80. Which of the following statements is TRUE regarding invert sugar? [End Module 2026]

- a. It is the hydrolytic product of lactose sugar.
- b. It is a non-reducing mixture.
- c. It exhibits mutarotation.
- d. It deviates the angle of polarized light to the right.

81. A patient with constipation is advised to increase dietary fiber. Which polysaccharide increases stool bulk? [End Semester 2026 (حکومی)]

- a. Starch
- b. Glycogen
- c. Cellulose
- d. Inulin

82. Deoxyribose lacks oxygen at which carbon? [End Semester 2026 (حکومی)]

- a. C1
- b. C2
- c. C3
- d. C4



83. In the cyclic form of glucose (pyranose), the anomeric carbon is most directly involved in which of the following? [End Semester 2026 (حکومی)]

- a. Determining whether the sugar is a pentose or hexose
- b. Formation of alpha and beta anomers via the orientation of its hydroxyl group
- c. Defining the sugar as D or L based on its configuration
- d. Phosphorylation during glycolysis

84. Glucuronic acid is physiologically important primarily because of which of the following? [End Semester 2026 (حکومی)]

- a. It serves as a major energy source for brain cells.
- b. It is a key intermediate in glycolysis.
- c. It conjugates with toxins and drugs to increase their excretion
- d. It forms the backbone of cellulose in plant cell walls.

85. In terms of glycosidic bond composition, how do starch and glycogen differ? [End Semester 2026 (حکومی)]

- a. Starch contains only alpha-1,4 bonds, while glycogen contains both alpha-1,4 and alpha-1,6 bonds.
- b. Both contain alpha-1,4 and alpha-1,6 bonds, but glycogen has more frequent alpha-1,6 bonds.
- c. Glycogen contains beta-1,4 bonds, while starch contains alpha-1,4 bonds.
- d. Starch contains alpha-1,6 bonds at branching points, while glycogen contains only alpha-1,4 bonds.

86. A biology student is examining three polysaccharides: starch, cellulose, and glycogen. Which of the following statements accurately distinguishes their glycosidic linkages? [End Semester 2026 (حکومی)]

- a. Starch and cellulose contain beta-1,4 linkages, while glycogen contains alpha-1,4 linkages.
- b. Starch and glycogen contain alpha-1,4 linkages, while cellulose contains beta-1,4 linkages.
- c. All three contain alpha-1,4 linkages, but only glycogen has alpha-1,6 branches.
- d. Cellulose and glycogen contain both alpha-1,4 and alpha-1,6 linkages, while starch contains only beta-1,4 linkages.



87. What is the main glycosidic bond in starch? [End Semester 2026 (أهلية)]
- β -1,4 linkage
 - α -1,6 linkage
 - α -1,4 linkage
 - β -1,6 linkage
88. D-Glucose and D-Galactose are considered which of the following isomers? [End Semester 2026 (أهلية)]
- Anomers
 - Epimers
 - Enantiomers
 - Ketoses isomers
89. Which of the following is a ketose sugar? [End Semester 2026 (أهلية)]
- Glucose
 - Galactose
 - Fructose
 - Ribose
90. Cellulose is indigestible by humans because it contains: [End Semester 2026 (أهلية)]
- α -1,4 linkages
 - β -1,4 linkages
 - α -1,6 linkages
 - β -1,6 linkages
91. Why is the product of sucrose hydrolysis called "invert sugar"? [End Semester 2026 (أهلية)]
- Because it changes color
 - Because the optical rotation changes
 - Because it becomes a reducing sugar
 - Because it forms a crystalline structure
92. A patient with constipation is advised to increase dietary fiber. Which of the following carbohydrates is recommended in this case? [End Semester 2026 (دور ثاني)]
- Starch
 - Glycogen
 - Cellulose
 - Inulin



93. Deoxyribose lacks oxygen at which carbon? [End Semester 2026 (دور تاني)]

- a. C1
- b. C2
- c. C3
- d. C4

94. In the cyclic form of glucose (pyranose), the anomeric carbon is most directly involved in which of the following? [End Semester 2026 (دور تاني)]

- a. Determining whether the sugar is a pentose or hexose
- b. Formation of α and β anomers via the orientation of its hydroxyl group
- c. Defining the sugar as D or L based on its configuration
- d. Phosphorylation during glycolysis

95. A biology student is examining three polysaccharides: starch, cellulose, and glycogen. Which of the following statements accurately distinguishes their glycosidic linkages? [End Semester 2026 (دور تاني)]

- a. Starch and cellulose contain β -1,4 linkages, while glycogen contains α -1,4 linkages.
- b. Starch and glycogen contain α -1,4 linkages, while cellulose contains β -1,4 linkages.
- c. All three contain α -1,4 linkages, but only glycogen has α -1,6 branches.
- d. Cellulose and glycogen contain both α -1,4 and α -1,6 linkages, while starch contains only β -1,4 linkages.

96. In uncontrolled diabetes, which sugar alcohol tends to accumulate in tissues like the lens, contributing to complications (e.g., cataracts and neuropathy)? [End Semester 2026 (دور تاني)]

- a. Mannitol
- b. Xylitol
- c. Sorbitol
- d. Ribitol

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97. Which of the following sugars lacks the reducing properties? [End Semester 2026 (دور تاني)]
- Lactose
 - Fructose
 - Maltose
 - Sucrose
98. Which of the following statements is TRUE regarding invert sugar? [End Semester 2026 (دور تاني)]
- It is the hydrolytic product of lactose sugar.
 - It is a non-reducing mixture.
 - It exhibits mutarotation.
 - It deviates the angle of polarized light to the right.

Written Questions

1. Compare between the followings: [End Semester 2024]
Homopolysaccharides vs Heteropolysaccharides (Definition - Examples)
Liver glycogen vs Muscle glycogen (Function - Relation to blood glucose)
Enantiomers vs Epimers (Definition - Examples)
2. Compare between sucrose and invert sugar regarding structure and reducing properties. [End Semester 2026 (حكومي)]
3. Demonstrate an example of sugar alcohol of clinical importance. [End Semester 2026 (حكومي)]
4. List two differences between Amylose and amylopectin. [End Semester 2026 (حكومي)]
5. Compare between liver and muscle Glycogen as regard Fate and function in a table. [End Semester 2026 (أهلية)]
6. Demonstrate an example of sugar alcohol of clinical importance. [End Semester 2026 (أهلية)]
7. List Two differences between Amylose and amylopectin. [End Semester 2026 (أهلية)]
8. Compare between starch and cellulose regarding their structure, function and tabulate your answer. [End Semester 2026 (دور تاني)]
9. Describe the differences between liver and muscle glycogen regarding function, importance and fate of hydrolysis. [End Semester 2026 (دور تاني)]
10. Define polysaccharides with reference to its classification with examples. [End Semester 2026 (دور تاني)]



2. Lipids

1. A monounsaturated fatty acid is: [End Semester 2018 (دور أول)]
 - a. Oleic acid
 - b. Arachidonic acid
 - c. Palmitic acid
 - d. Linolenic acid
2. The degree of unsaturation of lipid can be measured by [End Semester 2018 (دور أول)]
 - a. Saponification number
 - b. Iodine number
 - c. Polenske number
 - d. Reichert Meissil Number
3. Free fatty acids are transported in the blood [End Semester 2018 (دور أول)]
 - a. Combined with albumin
 - b. Combined with fatty acid binding protein
 - c. Combined with β -lipoprotein
 - d. In unbound free salts
4. What is the condensed structural formula of Palmitic acid? [End Semester 2018 (دور ثاني)]
 - a. $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-COOH}$
 - b. $\text{CH}_3\text{-(CH}_2\text{)}_4\text{-COOH}$
 - c. $\text{CH}_3\text{-(CH}_2\text{)}_{14}\text{-COOH}$
 - d. $\text{CH}_3\text{-(CH}_2\text{)}_{16}\text{-COOH}$
5. $\Delta 9$ indicates a double bond between carbon atoms of the fatty acid [End Semester 2018 (دور ثاني)]
 - a. 8 and 9
 - b. 9 and 10
 - c. 9 and 11
 - d. 9 and 12
6. Unpleasant odors and taste in a fat (rancidity) can be delayed or prevented by the addition of which of the following? [End Semester 2018 (دور ثاني)]
 - a. Lead
 - b. Copper
 - c. Tocopherol
 - d. Ergosterol



7. Dietary fats after absorption appear in the circulation as [End Semester 2018 (دور ثانی)]
- HDL
 - VLDL
 - LDL
 - Chylomicrons
8. Condensed structural formula of Palmitic acid is [End Module 2019]
- $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-COOH}$
 - $\text{CH}_3\text{-(CH}_2\text{)}_4\text{-COOH}$
 - $\text{CH}_3\text{-(CH}_2\text{)}_{14}\text{-COOH}$
 - $\text{CH}_3\text{-(CH}_2\text{)}_{16}\text{-COOH}$
9. In mammals, the major fat in adipose tissues is [End Module 2019]
- Sphingolipids
 - Cholesterol
 - Triacylglycerol
 - Phospholipids
10. Which of the following structures is a 20:2 ($\Delta^{4,9}$) fatty acid? [End Module 2019]
- $\text{CH}_3(\text{CH}_2)_9\text{CH=CH}(\text{CH}_2)_3\text{CH=CH}(\text{CH}_2)_2\text{COOH}$
 - $\text{CH}_3(\text{CH}_2)_2\text{CH=CH}(\text{CH}_2)_3\text{CH=CH}(\text{CH}_2)_9\text{COOH}$
 - $\text{CH}_3(\text{CH}_2)_{10}\text{CH=CH}(\text{CH}_2)_3\text{CH=CHCH}_2\text{COOH}$
 - $\text{CH}_3\text{CH}_2\text{CH=CH}(\text{CH}_2)_3\text{CH=CH}(\text{CH}_2)_{10}\text{COOH}$
11. Which of the following is ω^9 fatty acid? [End Module 2019]
- Oleic acid
 - Linoleic acid
 - Linolenic acid
 - Stearic acid
12. Number of milligrams of KOH required to neutralize fatty acid present in 1 g of fat is called [End Module 2019] D o c t o r s
- Potassium number
 - Acid number
 - Saponification number
 - Iodine number



13. Hydrogenation of oleic acid produces.... [End Module 2019]
- Linoleic acid
 - Linolenic acid
 - Palmitic acid
 - Stearic acid
14. Gangliosides are complex glycosphingolipids found in [End Module 2019]
- Liver
 - Brain
 - Kidney
 - Muscle
15. Which of the following protects our heart and kidneys from injury? [Formative Exam 2021]
- Skin
 - Muscles
 - Lubricants
 - Fat
16. Fat storing cells of vertebrates are called [Formative Exam 2021]
- Hepatocytes
 - Astrocytes
 - Adipocytes
 - Melanocytes
17. Main function of preen glands in birds is [Formative Exam 2021]
- Hormone secretion
 - Fatty acid degradation
 - Wax secretion
 - Pigment synthesis
18. In mammals, what is the major fat in adipose tissue? [End Module 2022]
- Phospholipid
 - Cholesterol
 - Sphingolipids
 - Triacylglycerol



19. In humans, what is a dietary essential fatty acid? [End Module 2022]
- Palmitic acid
 - Stearic acid
 - Butyric acid
 - Linoleic acid
20. The importance of phospholipids as a constituent of cell membrane is because they possess: [End Module 2022]
- Fatty acids
 - Both polar and nonpolar groups
 - Glycerol
 - Phosphoric acid
21. Cholesterol is the precursor for the biosynthesis of which of the following? [End Module 2022]
- Fatty acid
 - Prostaglandins
 - Bile acids
 - Sphingomyelin
22. Which phospholipid acts as surfactant? [End Module 2022]
- Cephalin
 - Phosphatidylinositol
 - Lecithin
 - Phosphatidylserine
23. When choline of lecithin is replaced by ethanolamine, what is the product of such process? [End Module 2022]
- Sphingomyelin
 - Cephalin
 - Plasmalogens
 - Lysolecithin
24. What is the lipoprotein associated with a high incidence of coronary atherosclerosis? [End Semester 2022]
- LDL
 - VLDL
 - IDL
 - HDL



25. Depot fat is composed mainly of cholesterol. [End Semester 2022]
a. True
b. False
26. Glycerol is the type of alcohol present in wax. [End Semester 2022]
a. True
b. False
27. Hardening of oils nutritional value [Formative Exam 2023]
a. Increases
b. Decreases
28. Lung surfactant is formed mainly from the phospholipid dipalmitoyl lecithin. Which of following describes the structure of dipalmitoyl lecithin? [End Semester 2024]
a. Glycerol + 2 palmitic acids+ phosphoryl choline
b. Glycerol + 2 palmitic acids+phosphoryl ethanolamine
c. Sphingosine + 2 palmitic acids+ phosphoryl choline
d. Sphingosine + 2 palmitic acids + phosphoryl ethanolamine 2 palmitic acid
29. Which of the following is a derived lipid? [End Semester 2024]
a. Carotenes
b. Waxes
c. Vitamin D3
d. Ceramide
30. Lung surfactant is formed mainly from the phospholipid dipalmitoyl lecithin. Which of the following describes the structure of dipalmitoyl lecithin? [End Semester 2025]
a. Glycerol + 2 palmitic acids + phosphoryl choline
b. Glycerol + 2 palmitic acids + phosphoryl ethanolamine
c. Sphingosine + 2 palmitic acids + phosphoryl choline
d. Sphingosine + 2 palmitic acids + phosphoryl ethanolamine
31. Which of the following is a derived lipid? [End Semester 2025]
a. Carotenes
b. Waxes
c. Vitamin D3
d. Ceramide

32. The iodine number of fat or oil is a direct measure of its [End Semester 2026 (حکومی)]

- a. Average chain length of fatty acids
- b. Degree of unsaturation
- c. Saponification value
- d. free fatty acid content

33. Which lipid is a component of the inner mitochondrial membrane? [End Semester 2026 (حکومی)]

- a. Phosphatidylcholine
- b. Phosphatidylethanolamine
- c. Cardiolipin
- d. Sphingomyelin

34. Phospholipids are an essential structural component of which of the following? [End Semester 2026 (حکومی)]

- a. Liver glycogen stores
- b. Lung surfactant
- c. Hemoglobin in red blood cells
- d. Bone matrix

35. Which of the following is a storage form of lipid in human body? [End Semester 2026 (حکومی)]

- a. Phospholipid
- b. Glycolipid
- c. Triacylglycerol
- d. Cholesterol

36. Which of the following is a monounsaturated fatty acid? [End Semester 2026 (حکومی)]

- a. Oleic acid
- b. Linoleic acid
- c. Linolenic acid
- d. Stearic acid

37. Which of the following is essential fatty acid? [End Semester 2026 (حکومی)]

- a. Oleic acid
- b. Linolenic
- c. Palmitic acid
- d. Stearic acid



38. Unpleasant odour and taste in a fat (rancidity) can be prevented by addition of which of the following? [End Semester 2026 (حکومی)]
- Lead
 - Copper
 - Vitamin E
 - Ergosterol
39. Which of the following is a derived lipid? [End Semester 2026 (حکومی)]
- Fatty acid
 - Triacylglycerol
 - Phospholipid
 - Glycolipid
40. Complete hydrogenation of linolenic produces which of the following? [End Semester 2026 (حکومی)]
- Arachidonic acid
 - Linoleic acid
 - Oleic acid
 - Stearic acid
41. Linolenic acid belongs to which of the following classes? [End Semester 2026 (حکومی)]
- omega-12
 - omega-9
 - omega-6
 - omega-3
42. Phosphatidic acid on hydrolysis yields which of the following contents? [End Semester 2026 (حکومی)]
- Glycerol, fatty acids, phosphoric acid, choline
 - Glycerol, fatty acids, phosphoric acid
 - Glycerol, fatty acids, phosphoric acid, Glucose
 - Sphingosine, fatty acids, phosphoric acid
43. In humans, which of the following is a dietary essential fatty acid? [End Semester 2026 (أهلية)]
- Palmitic acid
 - Stearic acid
 - Oleic acid
 - Linoleic acid



44. What is the nitrogenous base in lecithin? [End Semester 2026 (أهلية)]

- a. Ethanolamine
- b. Choline
- c. Serine
- d. Betaine

45. In cephalin, choline is replaced by which of the following? [End Semester 2026 (أهلية)]

- a. Serine
- b. Ethanolamine
- c. Betaine
- d. Sphingosine

46. Which of the following is a health risk associated with hydrogenated fats? [End Semester 2026 (أهلية)]

- a. It produces toxic gases
- b. It increases the risk of atherosclerosis
- c. It makes the oil rancid
- d. It destroys vitamins

47. Which type of fatty acid is most susceptible to rancidity? [End Semester 2026 (أهلية)]

- a. Oleic acid
- b. Stearic acid
- c. Palmitic acid
- d. Linolenic acid

48. Which of the following is a primary component of structural fat? [End Semester 2026 (أهلية)]

- a. Triglycerides
- b. Cholesterol
- c. Free fatty acids
- d. Glycerol

49. Which lipid is a component of the inner mitochondrial membrane? [End Semester 2026 (دور ثاني)]

- a. Phosphatidylcholine
- b. Phosphatidylethanolamine
- c. Cardiolipin
- d. Sphingomyelin



50. Which of the following is a storage form of lipid in human body? [End Semester 2026 (دور تاني)]
- Phospholipid
 - Glycolipid
 - Triacylglycerol
 - Sulfolipid
51. Which of the following is an essential fatty acid? [End Semester 2026 (دور تاني)]
- Oleic acid
 - Linolenic acid
 - Palmitic acid
 - Stearic acid
52. Which of the following is a derived lipid? [End Semester 2026 (دور تاني)]
- Fatty acid
 - Triacylglycerol
 - Phospholipid
 - Glycolipid
53. Complete hydrogenation of linolenic produces which of the following? [End Semester 2026 (دور تاني)]
- Arachidonic acid
 - Linoleic acid
 - Oleic acid
 - Stearic acid
54. Linolenic acid belongs to which of the following classes? [End Semester 2026 (دور تاني)]
- ω -12
 - ω -9
 - ω -6
 - ω -3
55. Hydrolysis of phosphatidic acid by phospholipases yields which of the following contents? [End Semester 2026 (دور تاني)]
- Glycerol, fatty acids, phosphoric acid, choline
 - Glycerol, fatty acids, phosphoric acid
 - Glycerol, fatty acids, phosphoric acid, Glucose
 - Sphingosine, fatty acids, phosphoric acid

Written Questions

1. Look at the opposite structure of two triglyceride molecules and answer the following questions: (2 marks) [End Semester 2024]

[Triglyceride توجد صورة لجزئين من الـ]

Is triglyceride 1 mixed or simple?

Is the fatty acid at position 3 of triglyceride 1 omega 3 or omega 6?

What is the name of the fatty acid in triglyceride 2?

Explain why triglyceride 1 has a higher iodine value than triglyceride 2?

2. Mention the importance of cholesterol for the human body? (2 marks) [End Semester 2024]

3. Compare between Omega 3 and omega 6 fatty acids. [End Semester 2026 (حكومي)]

4. List two main differences between triacylglycerol and wax. [End Semester 2026 (حكومي)]

5. Rancidity: Define and enumerate factors protecting against it. [End Semester 2026 (أهلية)]

6. Enumerate two important functions of phospholipid. [End Semester 2026 (أهلية)]

7. Define rancidity and describe causes and effects. [End Semester 2026 (دور ثاني)]

8. List two main differences between triacylglycerol and wax in a table. [End Semester 2026 (دور ثاني)]

Fighter
Doctors



3. Proteins

1. **Cysteine, cystine and methionine are:** [End Semester 2018 (دور اول)]
 - a. Essential amino acids
 - b. Non-protein amino acids
 - c. Acidic amino acids
 - d. Sulfur containing amino acids
2. **Rotation of plain polarized light is caused by solution of all the following except:** [End Semester 2018 (دور اول)]
 - a. Alanine
 - b. Glycine
 - c. Leucine
 - d. Serine
3. **Isoelectric pH is the pH at which:** [End Semester 2018 (دور اول)]
 - a. Protein molecules is denatured
 - b. Protein molecules carries no net charges
 - c. Protein molecules is most soluble
 - d. The disulfide bonds in protein molecules are disrupted
4. **In proteins the α -helix and β -pleated sheet are examples of** [End Semester 2018 (دور اول)]
 - a. Primary structure
 - b. Secondary structure
 - c. Tertiary structure
 - d. Quaternary structure
5. **Denaturation of proteins results in** [End Semester 2018 (دور اول)]
 - a. Disruption of primary structure
 - b. Breakdown of peptide bonds
 - c. Destruction of hydrogen bonds
 - d. Reversible changes in the molecule
6. **Facilitated diffusion:** [End Semester 2018 (دور اول)]
 - a. Requires a transmembrane protein
 - b. Requires ATP
 - c. Can move molecules against a concentration gradient
 - d. Is typically used to transport small nonpolar molecules



7. **Isoelectric pH is the pH at which** [End Semester 2018 (دور ثانی)]
- Protein molecules carry no net charges
 - Protein molecules are denatured
 - Protein molecules are most soluble
 - The disulfide bonds in protein molecules are disrupted
8. **Lysine is.....** [End Semester 2018 (دور ثانی)]
- Neutral amino acid
 - Acidic amino acid
 - Basic amino acid
 - Aromatic amino acid
9. **Which of the following is a tripeptide** [End Semester 2018 (دور ثانی)]
- Aspartame
 - Aspartate
 - Glutamate
 - Glutathione
10. **During denaturation of proteins, all of the following are disrupted except..** [End Semester 2018 (دور ثانی)]
- Primary structure
 - Secondary structure
 - Tertiary structure
 - Quaternary structure
11. **The optically inactive amino acid is** [End Module 2019]
- Glycine
 - Serine
 - Threonine
 - Valine
12. **All the following are sulphur containing amino acids found in proteins except** [End Module 2019]
- Cysteine
 - Cystine
 - Methionine
 - Threonine



13. During denaturation of proteins, all of the following are disrupted except [End Module 2019]

- a. Primary structure
- b. Secondary structure
- c. Tertiary structure
- d. Quaternary structure

14. Which of the following is a tripeptide? [End Module 2019]

- a. Anserine
- b. Oxytocin
- c. Glutathione
- d. Kallidin

15. Tertiary structure of a protein describes.. [End Module 2019]

- a. The order of amino acids
- b. Location of disulphide bonds
- c. Loop regions of proteins
- d. The ways of protein folding

16. Cysteine, cystine and methionine are: [Formative Exam 2021]

- a. Essential amino acids
- b. Non-protein amino acids
- c. Acidic amino acids
- d. Sulfur containing amino acids

17. The basic amino acid is: [Formative Exam 2021]

- a. Lysine
- b. Bile acids
- c. Glycine
- d. Alanine

18. An -OH group is present in the side chain of [Formative Exam 2021]

- a. Serine
- b. Arginine
- c. Lysine
- d. Proline



19. Which of the following is a Sulphur containing amino acid? [End Module 2022]
- Methionine
 - Leucine
 - Valine
 - Asparagine
20. Which of the following is an optically inactive amino acid? [End Semester 2022]
- Glycine
 - Serine
 - Threonine
 - Valine
21. Which of the following is an amino acid with a nonpolar side chain? [End Semester 2022]
- Serine
 - Valine
 - Asparagine
 - Threonine
22. At isoelectric pH, an amino acid exists in which of the following states? [End Semester 2022]
- Anion
 - Cation
 - Zwitterion
 - Undissociated molecule
23. Primary structure of a protein is formed by which of the following bonds? [End Semester 2022]
- Hydrogen bonds
 - Peptide bonds
 - Disulphide bonds
 - Hydrophobic bonds
24. Basic amino acids are positively charged at isoelectric point. [Formative Exam 2023]
- True
 - False



25. Which of the following amino acids would more possibly be on the surface of a membrane protein? [End Semester 2024]

- a. Aspartate
- b. Lysine
- c. Valine
- d. Serine

26. At the isoelectric point, the amino acid carry? [End Semester 2024]

- a. Cation
- b. Anion
- c. Un charged
- d. Dipolar

27. Which of the following describes the forces that stabilize the tertiary level of protein structure? [End Semester 2024]

- a. Hydrogen bonds between components of the peptide bond is one of these forces
- b. Interaction between two cysteines forms disulfide bridges
- c. Interaction between valine, leucine, and isoleucine forms ionic interactions
- d. Interaction between aspartate and lysine forming hydrophobic interactions

28. Which level of protein structure describes how different protein subunits bind together? [End Semester 2024]

- a. 1ry structure
- b. 2ry structure
- c. 3ry structure
- d. 4ry structure

29. Which of the following is considered a derived protein? [End Semester 2024]

- a. Peptone
- b. Albumin
- c. Histone
- d. casein



30. Which of the following bonds stabilize the primary level of protein structure? [End Semester 2024]

- a. Hydrogen bonds
- b. Peptide bonds
- c. Ionic bonds
- d. Disulfide bonds

31. Which of the following amino acids would more possibly be on the surface of a membrane protein? [End Semester 2025]

- a. Aspartate
- b. Lysine
- c. Valine
- d. Serine

32. At the isoelectric point, the amino acid carries? [End Semester 2025]

- a. Cation
- b. Anion
- c. Uncharged (zwitterion)
- d. Dipolar

33. Which of the following describes the forces that stabilize the tertiary level of protein structure? [End Semester 2025]

- a. Hydrogen bonds between components of the peptide bond is one of these forces
- b. Interaction between two cysteines forms disulfide bridges
- c. Interaction between valine, leucine, and isoleucine forms ionic interactions
- d. Interaction between aspartate and lysine forming hydrophobic interactions

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- c. 3ry structure
- d. 4ry structure



35. Which of the following is considered a derived protein? [End Semester 2025]

- a. Peptone
- b. Albumin
- c. Histone
- d. Casein

36. Which of the following bonds stabilize the primary level of protein structure? [End Semester 2025]

- a. Hydrogen bonds
- b. Peptide bonds
- c. Ionic bonds
- d. Disulfide bonds

37. Ionic (Electrostatic) bond can be formed between side chains of which of the following? [End Module 2026 (دور أول)]

- a. Alanine & leucine
- b. Lysine & aspartate
- c. Leucine & valine
- d. Aspartate & glutamate

38. What is the protein bond that is stable even in denaturation? [End Module 2026 (دور أول)]

- a. Peptide bond
- b. H-bond
- c. Disulphide bond
- d. Hydrophobic interaction

39. Methionine & threonine are classified as which of the following? [End Module 2026 (دور أول)]

- a. Phenol containing amino acids
- b. Acidic amino acids
- c. Basic amino acids
- d. Essential amino acids

40. Which is considered an optically inactive amino acid? [End Module 2026 (دور أول)]

- a. Serine
- b. Valine
- c. Glycine
- d. Alanine



41. Which amino acid contains an imino group? [End Module 2026 (دور أول)]
- Serine
 - Valine
 - Proline
 - Alanine
42. In prion diseases, what is characteristic for the infectious form of prion protein (PrP^{Sc})? [End Module 2026 (دور أول)]
- Differs from PrP^C in primary amino acid sequence
 - Is more susceptible to proteolytic degradation
 - Contains a higher proportion of β -pleated sheet structure
 - Is encoded by a different gene than PrP^C
43. Which of the following protein abnormalities is most characteristic of Alzheimer disease? [End Module 2026 (دور أول)]
- Conversion of α -helices into β -sheets in prion protein
 - Accumulation of neurofibrillary tangles composed of hyperphosphorylated tau protein
 - Formation of Lewy bodies composed of α -synuclein
 - Accumulation of misfolded prion protein (PrP^{Sc}) in neurons
44. Which level of protein structure is defined by the linear sequence of amino acids? [End Module 2026 (دور أول)]
- Primary
 - Secondary
 - Tertiary
 - Quaternary
45. In a protein engineering experiment, a scientist replaces all the internal leucine residues (hydrophobic) with glutamate (negatively charged). What is the most likely effect on the protein's tertiary structure? [End Module 2026 (دور أول)]
- Protein misfolding and loss of function.
 - Increased stability due to new salt bridges.
 - Conversion of the protein from globular to fibrous.
 - No significant change.



46. Considering the isoelectric point (I.P), what would be the net charge of a protein with an I.P of 8.5 when placed in a solution at pH 7.4? [End Module 2026 (دور اول)]
- Positive
 - Negative
 - Neutral (Zwitterion)
 - Cannot be determined
47. Which of the following is a set of hydrophobic amino acids? [End Module 2026 (دور اول)]
- Aspartic - valine - isoleucine
 - Arginine - lysine - methionine
 - Hydroxyproline - histidine - Glutamine
 - Valine - methionine - isoleucine
48. What is a key difference between a native protein and its denatured form, regarding its ability to: [End Module 2026 (دور اول)]
- Increase the viscosity of a solution.
 - Act as an antigen and induce an immune response.
 - Expose its hydrophobic groups to the solvent.
 - Be composed of a sequence of amino acids.
49. Which of the following levels of protein structure describes the relationship among several subunits? [End Module 2026 (دور اول)]
- Primary
 - Secondary
 - Tertiary
 - Quaternary
50. Amino acids containing one amino group and one carboxylic group are known as ... [End Module 2026 (دور اول)]
- Polar
 - Acidic
 - Neutral
 - Basic



51. Ionic (Electrostatic) bond can be formed between side chains of which of the following? [End Module 2026]
- Alanine & leucine
 - Lysine & aspartate
 - Leucine & valine
 - Aspartate & glutamate
52. What is the protein bond that is stable even in denaturation? [End Module 2026]
- Peptide bond
 - H-bond
 - Disulphide bond
 - Hydrophobic interaction
53. Methionine & threonine are classified as which of the following? [End Module 2026]
- Phenol containing amino acids
 - Acidic amino acids
 - Basic amino acids
 - Essential amino acids
54. Which is considered optically inactive amino acid? [End Module 2026]
- Serine
 - Valine
 - Glycine
 - Alanine
55. Which amino acids contains imino group? [End Module 2026]
- Serine
 - Valine
 - Proline
 - Alanine
56. In prion diseases, What is characteristic for the infectious form of prion protein (PrP^{Sc})? [End Module 2026]
- Differs from PrP^C in primary amino acid sequence.
 - Is more susceptible to proteolytic degradation.
 - Contains a higher proportion of beta-pleated sheet structure.
 - Is encoded by a different gene than PrP^C.



- 57. Which of the following protein abnormalities is most characteristic of Alzheimer disease?** [End Module 2026]
- a. Conversion of alpha-helices into beta-sheets in prion protein.
 - b. Accumulation of neurofibrillary tangles composed of hyperphosphorylated tau protein.
 - c. Formation of Lewy bodies composed of alpha-synuclein.
 - d. Accumulation of misfolded prion protein PrPSc in neurons.
- 58. Which level of protein structure is defined by the linear sequence of amino acids?** [End Module 2026]
- a. Primary
 - b. Secondary
 - c. Tertiary
 - d. Quaternary
- 59. In a protein engineering experiment, a scientist replaces all the internal leucine residues (hydrophobic) with glutamate (negatively charged). What is the most likely effect on the protein's tertiary structure?** [End Module 2026]
- a. Protein misfolding and loss of function.
 - b. Increased stability due to new salt bridges.
 - c. Conversion of the protein from globular to fibrous.
 - d. No significant change.
- 60. Considering the isoelectric point (I.P), what would be the net charge of a protein with a I.P of 8.5 when placed in a solution at pH 7.4?** [End Module 2026]
- a. Positive
 - b. Negative
 - c. Neutral (Zwitterion)
 - d. Cannot be determined
- 61. Which of the following is a set of hydrophobic amino acids?** [End Module 2026]
- a. Aspartic - valine - isoleucine
 - b. Arginine - lysine - methionine
 - c. Hydroxyproline - histidine - Glutamine
 - d. Valine - methionine - isoleucine



62. What is key difference between a native protein and its denatured form, regarding its ability to: [End Module 2026]

- a. Increase the viscosity of a solution.
- b. Act as an antigen and induce an immune response.
- c. Expose its hydrophobic groups to the solvent.
- d. Be composed of a sequence of amino acids.

63. Which of the following levels of protein structure describes the relationship among several subunits? [End Module 2026]

- a. Primary
- b. Secondary
- c. Tertiary
- d. Quaternary

64. Amino acids containing one amino group and one carboxylic group are known as [End Module 2026]

- a. Polar
- b. Acidic
- c. Neutral
- d. Basic

65. Phenylketonuria usually is due to a defect in the enzyme phenylalanine hydroxylase and this results in the reduced synthesis of an aromatic hydroxyamino acid. The levels of which of the following amino acids is decreased in the body in this condition? [End Semester 2026 (حکومی)]

- a. Tyrosine
- b. Tryptophan
- c. Histidine
- d. Serine

66. Which of the following amino acids differs from the other in the nature of their amino group? [End Semester 2026 (حکومی)]

- a. Proline
- b. Glycine
- c. Leucine
- d. Alanine



67. In Sickle Cell Anemia, the nucleotide alteration leads to substitution of a glutamate residue with valine at the 7th position in the β -globin chain. Which of the following best describes this substitution? [End Semester 2026 (حکومی)]

- a. A hydroxy-amino acid is replaced by a simple amino acid
- b. A polar amino acid is replaced by a non-polar amino acid
- c. An aliphatic amino acid is replaced by an aromatic amino acid
- d. A dietary essential amino acid is replaced by a non-essential amino acid

68. A particular point mutation results in disruption of the α -helical structure in a segment of the mutant protein. What is the most likely change in the primary structure of the mutant protein? [End Semester 2026 (حکومی)]

- a. glutamate to aspartate.
- b. lysine to arginine.
- c. methionine to proline.
- d. valine to alanine.

69. Which of the following interactions is NOT involved in the stabilization of tertiary structure of proteins? [End Semester 2026 (حکومی)]

- a. Peptide bonds
- b. Hydrophobic interactions
- c. Hydrogen bonds
- d. Ionic interactions

70. A scientist observes that a normally soluble globular protein becomes insoluble and loses its biological function after being heated. What is the explanation for this case? [End Semester 2026 (حکومی)]

- a. The primary structure was hydrolyzed
- b. The protein denatured, exposing hydrophobic groups
- c. New peptide bonds were formed
- d. The protein gained a quaternary structure

71. Which amino acid is a site for N-linked glycosylation of proteins? [End Semester 2026 (حکومی)]

- a. Serine
- b. Threonine
- c. Asparagine
- d. Tyrosine



72. Denaturation of proteins involves loss of which of the following? [End Semester 2026 (حکومی)]

- a. Primary structure
- b. Ionic interactions
- c. Amino acid sequence
- d. Peptide bonds

73. A medical researcher is performing electrophoresis on a mixture of amino acids at pH 6.0. Glycine (pI ~6.0) remains stationary, while aspartic acid (pI ~3.0) migrates toward the anode. This is because: [End Semester 2026 (أهلیة)]

- a. Both amino acids are zwitterions at pH 6.0
- b. Glycine has a net negative charge and aspartic acid has a net positive charge at pH 6.0
- c. Glycine has no net charge, while aspartic acid has a net negative charge at pH 6.0
- d. Glycine has no net charge, while aspartic acid has a net positive charge at pH 6.0

74. The imino group is present in which of the following amino acids? [End Semester 2026 (أهلیة)]

- a. Serine
- b. Valine
- c. Proline
- d. Alanine

75. All of the following are basic essential amino acids except: [End Semester 2026 (أهلیة)]

- a. Histidine
- b. Leucine
- c. Lysine
- d. Arginine

76. Which of the following amino acids is known as aliphatic? [End Semester 2026 (دور ٢)]

- a. Tyrosine
- b. Tryptophan
- c. Histidine
- d. Serine



77. In Sickle Cell Anemia, the nucleotide alteration leads to substitution of a glutamate residue with valine at the 6th position in the β -globin chain. Which of the following best describes this substitution? [End Semester 2026 (دور تاني)]

- a. A hydroxy-amino acid is replaced by a simple amino acid
- b. A polar amino acid is replaced by a non-polar amino acid
- c. An aliphatic amino acid is replaced by a non-essential amino acid
- d. A dietary essential amino acid is replaced by an aromatic amino acid

78. A particular point mutation results in disruption of the α -helical structure in a segment of the mutant protein. What is the most likely change in the primary structure of the mutant protein? [End Semester 2026 (دور تاني)]

- a. glutamate to aspartate
- b. lysine to arginine
- c. methionine to proline
- d. valine to alanine

79. Which of the following amino acids differs relatively to its structure? [End Semester 2026 (دور تاني)]

- a. Proline
- b. Glycine
- c. Leucine
- d. Alanine

80. What is the form of an amino acid at its isoelectric point? [End Semester 2026 (دور تاني)]

- a. Cation
- b. Anion
- c. Zwitterion
- d. Not charged form

81. The helical structure of any polypeptide chain is mainly derived from which of the following? [End Semester 2026 (دور تاني)]

- a. Hydrogen bonds
- b. Disulfide linkage
- c. Hydrophobic interaction
- d. Peptide linkage



82. Denaturation of proteins involves loss of which of the following? [End Semester 2026 (دور تاني)]

- a. Primary structure
- b. Ionic interactions
- c. Amino acid sequence
- d. Peptide bonds

83. A patient with a high fever is brought to the emergency department. The elevated body temperature is likely denaturing many plasma and cellular proteins. Which level of protein structure would remain intact even after the hydrogen bonds, disulfide bridges, and hydrophobic interactions stabilizing the protein's native conformation are disrupted by the heat? [Formative 2026]

- a. Secondary structure
- b. Quaternary structure
- c. Tertiary structure
- d. Primary structure

84. A researcher is studying a mutant hemoglobin protein where a single point mutation replaces a neutral amino acid with a charged one. This creates a novel interaction between two residues on the protein's surface that disrupts its quaternary structure. This new, pathological bond would most likely form between the side chains of which pair of amino acids? [Formative 2026]

- a. Alanine & leucine
- b. Leucine & valine
- c. Aspartate & glutamate
- d. Leucine and aspartate

85. A structural biologist is examining the sharp turn in a protein's backbone, which is facilitated by the unusual cyclic structure of one of its constituent amino acids. This amino acid's side chain is bonded to its own amino group, making it a secondary amine. Which amino acid contains this imino group? [Formative 2026]

- a. Proline
- b. Valine
- c. Alanine
- d. Serine



86. A 55-year-old man presents with rapidly progressive dementia and myoclonus. Creutzfeldt-Jakob disease (CJD) is suspected. A brain biopsy is analyzed, and the pathogenic form of the prion protein (PrP^{Sc}) is found to be highly resistant to degradation by proteases. This resistance is best explained by which of the following structural properties of PrP^{Sc}? [Formative 2026]

- a. It has a different amino acid sequence than the normal cellular prion protein (PrP^C).
- b. It contains a higher proportion of β -pleated sheet structure.
- c. It is more susceptible to proteolytic degradation.
- d. It is encoded by a different gene than PrP^C. Final Page.

Written Questions

1. Explain with examples hydrophilic uncharged amino acids? (2 marks) [End Semester 2024]
2. Describe the process of protein denaturation (Definition & effects on proteins) (3 marks) [End Semester 2024]
3. Describe what is meant by quaternary level of protein structure and enumerate its stabilizing force [End Semester 2024]
4. Enumerate two biochemical defects in Alzheimer's disease. [End Semester 2026 (حكومي)]
5. List two differences between alpha helix and beta sheet. [End Semester 2026 (حكومي)]
6. Enumerate two biochemical defects in Alzheimer's disease. [End Semester 2026 (أهلية)]
7. List two differences between alpha helix and beta sheet. [End Semester 2026 (أهلية)]
8. Enumerate two biochemical defects or changes that occur in protein denaturation. [End Semester 2026 (دور ثاني)]
9. List two differences between primary and secondary structure of polypeptide chain. [End Semester 2026 (دور ثاني)]



4. Enzymes

1. Enzymes, catalyzing the synthetic reactions with ATP utilization, are included into class of: [End Semester 2018 (دور أول)]

- a. Oxidoreductases
- b. Transferases
- c. Lyases
- d. Ligases

2. When $[S]$ is equal to K_m , which of the following conditions exist? [End Semester 2018 (دور أول)]

- a. Half the enzyme molecules are bound to substrate
- b. The velocity of the reaction is equal to V_{max}
- c. The velocity of the reaction is independent of substrate concentration
- d. Enzyme is completely saturated with substrate

3. A competitive inhibitor of an enzyme has which of the following properties? [End Semester 2018 (دور أول)]

- a. Is frequently a feedback inhibitor
- b. Becomes covalently attached to an enzyme
- c. Decreases the V_{max}
- d. Interferes with substrate binding to the enzyme

4. Enzymes, catalyzing the synthetic reactions with ATP utilization, are included into the class of [End Semester 2018 (دور ثاني)]

- a. Ligases
- b. Lyases
- c. Oxidoreductases
- d. Transferases

5. An allosteric enzyme influences the enzyme activity by which of the following mechanisms? [End Semester 2018 (دور ثاني)]

- a. Changing the conformation of the enzyme by binding to a site other than catalytic site
- b. Changing the nature of the products formed
- c. Changing the specificity of the enzyme for the substrate
- d. Competing for the catalytic site with the substrate



6. A competitive inhibitor of an enzyme has which of the following properties? [End Semester 2018 (دور ثاني)]
- It becomes covalently attached to an enzyme
 - It decreases the V_{max}
 - It interferes with substrate binding to the enzyme
 - It is frequently a feedback inhibitor
7. Apoenzyme is... [End Module 2019]
- The inorganic part of the holoenzyme
 - The metal part of holoenzyme
 - The organic (vitamin derivative) part of holoenzyme
 - The protein part of holoenzyme
8. Which of the following statements is TRUE about noncompetitive inhibitors? [End Module 2019]
- They can bind to the enzyme alone or the enzyme-substrate complex
 - They mainly bind to the active site of enzyme
 - They have structural similarity between the inhibitor and the substrate
 - Noncompetitive inhibitors are always reversible
9. At V_{max} , of the enzyme molecules are occupied by the substrate. [End Module 2019]
- 25%
 - 50%
 - 75%
 - 100%
10. Which of the following statements is true about allosteric enzyme? [End Module 2019]
- They are presented by a sigmoidal curve when V_0 is plotted against substrate concentration
 - They are inhibited by substrate
 - They are enzymes that contain active site only
 - They are enzymes with a higher K_m than other enzymes
11. Lactase enzyme belongs to which of the following enzyme classes? [End Module 2022]
- Hydrolase
 - Lyase
 - Transferase
 - Epimerase



12. What is meant by reversible non-competitive enzyme inhibition activity? [End Module 2022]

- a. V_{max} is increased
- b. K_m is increased
- c. K_m is decreased
- d. Concentration of active enzyme is reduced

13. LDH1 and LDH2 are elevated in which of the following conditions? [End Module 2022]

- a. Myocardial infarction
- b. Liver disease
- c. Kidney disease
- d. Brain disease

14. What is the CK isoform present in cardiac muscle? [End Module 2022]

- a. BB and MB
- b. MM and MB
- c. BB only
- d. MB only

15. DNA polymerase belongs to which of the following classes of enzymes? [End Semester 2022]

- a. Transferases
- b. Isomerases
- c. Lyases
- d. Ligases

16. Enzymes change the equilibrium of the reaction. [End Semester 2022]

- a. True
- b. False

17. Non-competitive inhibitor binds to the active site of the enzyme. [End Semester 2022]

- a. True
- b. False



18. Phosphorylation and dephosphorylation of the enzymes are known as which of the in terms? [End Semester 2024]

- a. Feedback inhibition
- b. long term regulation
- c. Allosteric regulation
- d. Covalent modification

19. A 60-year-old man is diagnosed with hypertension and is started on an angiotensin-converting enzyme (ACE) inhibitor. ACE inhibitors bind to the active site of the enzyme. What be inhibition does this describe? [End Semester 2024]

- a. Competitive inhibition
- b. Uncompetitive inhibition
- c. Non-competitive inhibition
- d. Feedback inhibition

20. The initial rate of an enzyme-catalyzed reaction increases with which of the following? [End Semester 2024]

- a. Decreasing enzyme concentration
- b. Increasing the product concentration
- c. Increasing the substrate concentration
- d. Decreasing the pH or the temperature

21. What is the expected change of the apparent K_m that occurs in the presence of a noncompetitive inhibitor? [End Semester 2024]

- a. Increases
- b. Decreases
- c. Remains unchanged
- d. Becomes zero

22. What is the functional form of the conjugated enzymes? [End Semester 2024]

- a. Apoenzyme
- b. Coenzyme
- c. Cofactor
- d. Holoenzyme



23. Pepsinogen belongs to which of the following enzymatic categories? [End Semester 2024]

- a. Zymogens
- b. Allosteric enzymes
- c. Isoenzymes
- d. Holoenzymes

24. Phosphorylation and dephosphorylation of enzymes are known as which of the following terms? [End Semester 2025]

- a. Feedback inhibition
- b. Long term regulation
- c. Allosteric regulation
- d. Covalent modification

25. A 60-year-old man is diagnosed with hypertension and is started on an angiotensin-converting enzyme (ACE) inhibitor. ACE inhibitors bind to the active site of the enzyme. What type of inhibition does this describe? [End Semester 2025]

- a. Competitive inhibition
- b. Uncompetitive inhibition
- c. Non-competitive inhibition
- d. Feedback inhibition

26. The initial rate of an enzyme-catalyzed reaction increases with which of the following? [End Semester 2025]

- a. Decreasing enzyme concentration
- b. Increasing the product concentration
- c. Increasing the substrate concentration
- d. Decreasing the pH or the temperature

27. What is the expected change of the apparent K_m that occurs in the presence of a noncompetitive inhibitor? [End Semester 2025]

- a. Increases
- b. Decreases
- c. Remains unchanged
- d. Becomes zero



28. What is the functional form of conjugated enzymes? [End Semester 2025]

- a. Apoenzyme
- b. Coenzyme
- c. Cofactor
- d. Holoenzyme

29. Pepsinogen belongs to which of the following enzymatic categories? [End Semester 2025]

- a. Zymogens
- b. Allosteric enzymes
- c. Isoenzymes
- d. Holoenzymes

30. Which of the following best describes a functional plasma enzyme? [End Semester 2026 (حکومی)]

- a. Its concentration in plasma is normally very low compared to tissues.
- b. It increases in plasma during tissue damage or disease.
- c. Its substrate is always present in the blood.
- d. It has no known physiological function in the blood.

31. A patient with myocardial infarction has elevated creatine kinase MB (CKMB). Which enzyme class does creatine kinase belong to? [End Semester 2026 (حکومی)]

- a. Oxidoreductase
- b. Transferase
- c. Hydrolase
- d. Ligase

32. The "lock and key" model of enzyme action proposes that: [End Semester 2026 (حکومی)]

- a. The active site changes shape to fit the substrate
- b. The substrate and active site are complementary in shape from the start
- c. Enzymes are only specific for one type of reaction, not substrate shape
- d. Cofactors are necessary for substrate binding



33. Which of the following correctly describes a coenzyme? [End Semester 2026 (حكومي)]
- An inorganic metal ion that activates enzymes
 - An organic helper molecule, often a vitamin derivative
 - The holoenzyme with its non-protein part
 - The protein portion of an enzyme
34. Enzyme A has a K_m of 0.1 mM for its substrate, while Enzyme B has a K_m of 10 mM for the same substrate. At a substrate concentration of 0.5 mM: [End Semester 2026 (حكومي)]
- Both enzymes will operate at their maximum velocities
 - Enzyme A will have a higher activation level than Enzyme B
 - Enzyme B will have a higher activation level than Enzyme A
 - Both enzymes will have equal activity
35. Which of the following best describes a functional plasma enzyme? [End Semester 2026 (أهلية)]
- Its concentration in plasma is normally very low compared to tissues.
 - It increases in plasma during tissue damage or disease.
 - Its substrate is always present in the blood.
 - It has no known physiological function in the blood.
36. Which of the following best describes the kinetics of an enzyme-catalyzed reaction in the presence of a non-competitive inhibitor? [End Semester 2026 (أهلية)]
- The inhibitor can bind to both free enzyme and the ES complex; K_m unchanged; V_{max} decreased
 - The inhibitor can bind to both free enzyme and the ES complex; K_m decreased; V_{max} unchanged
 - The inhibitor can bind only to free enzyme; K_m unchanged; V_{max} decreased
 - The inhibitor can bind only to free enzyme; K_m decreased; V_{max} unchanged
37. The difference between holoenzyme and apoenzyme is that holoenzyme: [End Semester 2026 (أهلية)]
- Contains its cofactor
 - Is fully active
 - Is phosphorylated
 - Is glycosylated



38. Which of the following best describes the "induced fit" model of enzyme-substrate interaction? [End Semester 2026 (أهلية)]

- a. The substrate exactly fits into the rigid active site
- b. The enzyme changes conformation upon substrate binding
- c. The substrate changes shape to match the enzyme
- d. Only the correct substrate can enter the active site

39. Enzyme A has a K_m of 0.1 mM for its substrate, while Enzyme B has a K_m of 10 mM for the same substrate. At a substrate concentration of 0.5 mM: [End Semester 2026 (أهلية)]

- a. Both enzymes will operate at their maximum velocities
- b. Enzyme A will have a higher activation level than Enzyme B
- c. Enzyme B will have a higher activation level than Enzyme A
- d. Both enzymes will have equal activity

40. What is the organic non-protein part of conjugated enzymes? [End Semester 2026 (أهلية)]

- a. Coenzyme
- b. Zymogen
- c. Co-factor
- d. Holoenzyme

41. What is the shape of the curve for allosteric enzyme kinetics? [End Semester 2026 (أهلية)]

- a. Hyperbolic
- b. Straightline
- c. Sigmoidal (S-shaped)
- d. U-shaped

42. Which of the following best describes about the kinetics of an enzyme catalyzed reaction in the presence of a non-competitive inhibitor? [End Semester 2026 (دور ثاني)]

- a. The inhibitor can bind to both free Enzyme and the ES complex; K_m unchanged; V_{max} decreased
- b. The inhibitor can bind to both free Enzyme and the ES complex; K_m decreased; V_{max} unchanged
- c. The inhibitor can bind only to free Enzyme; K_m unchanged; V_{max} decreased
- d. The inhibitor can bind only to free Enzyme; K_m decreased; V_{max} unchanged



43. What is the difference between holoenzyme and apoenzyme is that holoenzyme:

[End Semester 2026 (دور تاني)]

- a. Contains its cofactor
- b. Is fully active
- c. Is phosphorylated
- d. Is glycosylated

44. Which of the following best describes a functional plasma enzyme? [End Semester 2026 (دور تاني)]

- a. Its concentration in plasma is normally very low compared to tissues.
- b. It increases in plasma during tissue damage or disease.
- c. Its substrate is always present in the blood.
- d. It has no known physiological function in the blood.

45. To which of the following classes lactase enzyme belongs to? [End Semester 2026 (دور تاني)]

- a. Oxidoreductases
- b. Transferases
- c. Hydrolases
- d. Ligases

46. Enzyme A has a K_m of 0.1 mM for its substrate, while Enzyme B has a K_m of 10 mM for the same substrate. At a substrate concentration of 0.5 mM: [End Semester 2026 (دور تاني)]

- a. Both enzymes will operate at their maximum velocities
- b. Enzyme A will have a higher activation level than Enzyme B
- c. Enzyme B will have a higher activation level than Enzyme A
- d. Both enzymes will have equal activity

47. What is the organic non-protein part of the conjugated enzymes? [End Semester 2026 (دور تاني)]

- a. Coenzyme
- b. Zymogen
- c. Co-factor
- d. Holoenzyme

48. Enzyme inhibition caused by a substance resembling substrate molecule is known as which of the following? [End Semester 2026 (دور تاني)]

- a. Competitive inhibition
- b. Non-competitive inhibition
- c. Feedback inhibition
- d. Allosteric inhibition

Written Questions

1. The following picture describes the relationship between the enzyme action and the substrate. Describe this relationship and demonstrate how to increase the rate of the reaction in figure (panel). (2 marks) [End Semester 2024]

[توجد صورتان توضحان الإنزيمات والركائز] (Substrate and Enzyme)]

2. Compare between competitive and non-competitive inhibitors. (2 marks) [End Semester 2024]

3. Enumerate four mechanisms of enzyme catalysis. [End Semester 2026 (حكومي)]

4. You are studying the enzyme dihydrofolate reductase, a target for cancer chemotherapy. You test two potential drugs, Drug X and Drug Y, for their inhibitory effects. The following kinetic data are obtained: [End Semester 2026 (حكومي)]

Without inhibitor: $K_m = 1.5 \mu M$, $V_{max} = 60 \text{ pmol/min}$

With Drug X: $K_m = 6.0 \mu M$, $V_{max} = 60 \text{ pmol/min}$

With Drug Y: $K_m = 1.5 \mu M$, $V_{max} = 20 \text{ pmol/min}$

Classify the type of inhibition exhibited by Drug X and Drug Y. Justify your answer based on the changes in K_m and V_{max} .



5. A 55-year-old male presents to the emergency department with chest pain. Serum enzyme analysis shows elevated total lactate dehydrogenase (LDH) and creatine kinase (CK). Further isoenzyme analysis reveals: [End Semester 2026 (أهلية)]

LDH1: 45% (normal: 15-30%)

LDH2: 30% (normal: 30-40%)

LDH3: 15% (normal: 15-25%)

LDH4: 5% (normal: 5-10%)

LDH5: 5% (normal: 5-10%)

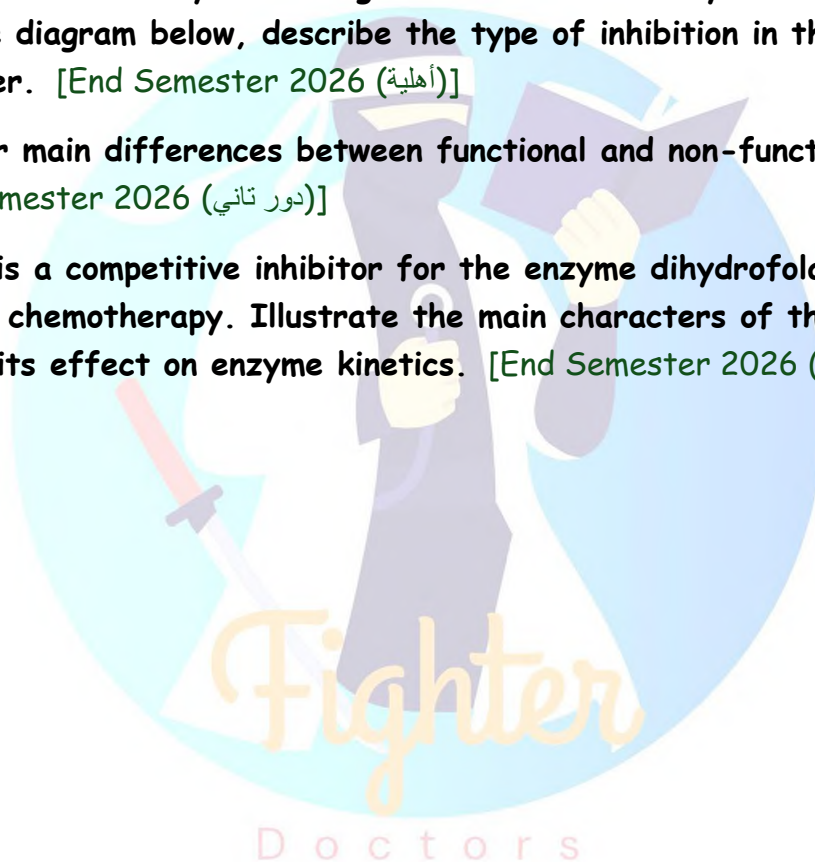
CK-MB: 12% of total CK (normal: <5%)

Which isoenzyme pattern is indicative of myocardial infarction? Explain and define the structure of the dominant isoenzyme.

6. Methotrexate is a commonly used drug that inhibits the dihydrofolate reductase enzyme. From the diagram below, describe the type of inhibition in this case and justify your answer. [End Semester 2026 (أهلية)]

7. Enumerate four main differences between functional and non-functional plasma enzymes. [End Semester 2026 (دور ثاني)]

8. Methotrexate is a competitive inhibitor for the enzyme dihydrofolate reductase, a target for cancer chemotherapy. Illustrate the main characters of this inhibitor with the reference to its effect on enzyme kinetics. [End Semester 2026 (دور ثاني)]





5. Nucleotides & DNA Structure

1. What is the chemical name of Adenine? [End Semester 2018 (دور أول)]
- 6-Amino purine
 - 2-Amino-6-oxypurine
 - 2-Oxy-4-aminopyrimidine
 - 2,4-Dioxypyrimidine
2. The analysis of the base composition of unlabeled nucleic acid samples showed 10% adenine, 40% cytosine, 30% thymine and 20% guanine. According to such results, the source of this nucleic acid is most likely to be: [End Semester 2018 (دور أول)]
- Bacterial chromosome
 - Bacterial plasmid
 - Nuclear chromosome
 - Viral genome
3. In DNA, three hydrogen bonds are formed between..... [End Semester 2018 (دور ثاني)]
- Adenine and guanine
 - Adenine and thymine
 - Guanine and cytosine
 - Thymine and cytosine
4. In a DNA molecule the thymine concentration is 30%, the guanosine concentration will be [End Module 2019]
- 10%
 - 20%
 - 30%
 - 40%
5. In Uridine monophosphate (UMP), the glycosidic linkage is formed between C1 of the pentose sugar and [End Module 2019]
- N1
 - N5
 - N7
 - N9



6. Which of the following statements is TRUE about melting temperature (T_m)? [End Module 2019]
- At T_m , all of the double helical structure is lost
 - DNA rich in G and C has low melting temperature
 - Unfolded DNA has a higher absorption at 260 nm UV light
 - DNA rich in A and T has high melting temperature
7. Uracil is [End Module 2019]
- 6-Amino purine
 - 2-Amino-6-oxypurine
 - 2-Oxy-4-aminopyrimidine
 - 2,4-Dioxypyrimidine
8. Histones are [End Module 2019]
- Basic proteins formed of 3 classes
 - Strongly negatively charged proteins
 - Strong positively charged protein
 - Needed to stabilize single stranded DNA structure
9. RNA does not contain [Formative Exam 2021]
- Uracil
 - Adenine
 - Thymine
 - Ribose
10. The sugar moiety present in RNA is [Formative Exam 2021]
- Ribulose
 - Arabinose
 - Ribose
 - Deoxyribose
11. In DNA, the complementary base of adenine is [Formative Exam 2021]
- Guanine
 - Cytosine
 - Uracil
 - Thymine
12. DNA rich in A-T pairs have [Formative Exam 2021]
- 1 Hydrogen bond
 - 2 Hydrogen bonds
 - 3 Hydrogen bonds
 - 4 Hydrogen bonds



13. A nucleoside consists of [Formative Exam 2021]

- a. Nitrogenous base
- b. Purine or pyrimidine base + sugar
- c. Purine or pyrimidine base + phosphorous
- d. Purine + pyrimidine base + sugar + phosphorous

14. A nucleotide consists of [Formative Exam 2021]

- a. A nitrogenous base like choline
- b. Purine or pyrimidine base + sugar + phosphorous
- c. Purine or pyrimidine base + sugar
- d. Purine or pyrimidine base + phosphorous

15. A purine nucleotide is [Formative Exam 2021]

- a. AMP
- b. UMP
- c. CMP
- d. TMP

16. What is the chemical name of guanine? [End Module 2022]

- a. 2,4-Dioxy-5-methylpyrimidine
- b. 2-Amino-6-oxypurine
- c. 2-Oxy-4-aminopyrimidine
- d. 2,4-Dioxypyrimidine

17. In mRNA, if the guanine percentage is 20%, what is the percentage of cytosine?

[End Module 2022]

- a. 20%
- b. 30%
- c. 40%
- d. Cannot be estimated

18. Which of the following statements is true about histones? [End Module 2022]

- a. They are basic proteins so they can interact with positively charged DNA
- b. They are basic proteins that carry negative charge
- c. They are basic proteins so they can interact with negatively charged DNA
- d. They are basic proteins that have no role in regulation of gene expression



19. Which of the following does not require the presence of a palindromic sequence?

[End Semester 2022]

- a. Methylation sites during replication
- b. Denatured site in gene promoter
- c. Termination of gene transcription
- d. Sites for restriction endonucleases

20. How many hydrogen bonds are there between guanine and cytosine? [End Semester 2022]

- a. One
- b. Two
- c. Three
- d. Four

21. DNA unfolding is monitored by measuring the absorption of UV light at which of the following wavelengths? [End Semester 2022]

- a. 250 nm
- b. 260 nm
- c. 270 nm
- d. 280 nm

22. Which of the following DNA forms is the broadest type? [End Semester 2022]

- a. A form
- b. B form
- c. D form
- d. Z form

23. At T_m , of the total DNA is unwound. [End Semester 2022]

- a. 100%
- b. 75%
- c. 50%
- d. 25%

24. A researcher studying cancer cells finds that certain mutations cause an increase in G-C content of the DNA. What is the consequence of increased G-C content on the DNA structure? [End Semester 2024]

- a. Increased number of hydrogen bonds, making the DNA more stable
- b. Increased flexibility of the DNA helix
- c. Decreased number of hydrogen bonds, making the DNA less stable
- d. No significant effect on DNA stability



25. During denaturation of DNA, which bonds are disrupted? [End Semester 2024]
- Phosphodiester bonds
 - Glycosidic bonds
 - Hydrogen bonds
 - Peptide bond
26. A researcher studying cancer cells finds that certain mutations cause an increase in G-C content of the DNA. What is the consequence of increased G-C content on the DNA structure? [End Semester 2025]
- Increased number of hydrogen bonds, making the DNA more stable
 - Increased flexibility of the DNA helix
 - Decreased number of hydrogen bonds, making the DNA less stable
 - No significant effect on DNA stability
27. During denaturation of DNA, which bonds are disrupted? [End Semester 2025]
- Phosphodiester bonds
 - Glycosidic bonds
 - Hydrogen bonds
 - Peptide bond
28. If a DNA sample from a species contains 20% Thymine, what is the percentage of Guanine? [End Module 2026 (دور اول)]
- 20%
 - 30%
 - 40%
 - 60%
29. What is the main function of histone H1? [End Module 2026 (دور اول)]
- Forming the core of the nucleosome.
 - Binding to linker DNA.
 - Acetylating other histone proteins.
 - Serving as the scaffold for the 30nm fiber.



30. Which of the following DNA forms is a left-handed helix? [End Module 2026 (دور اول)]
- A form
 - B form
 - Z form
 - A and B form
31. In a DNA sample, which result would be consistent with Chargaff's findings? [End Module 2026 (دور اول)]
- $A = G$; $G = T$
 - $A + G = C + T$
 - $A + T = G + T$
 - $A + C = G + C$
32. Which of the following represents the chemical structure of adenine? [End Module 2026 (دور اول)]
- 6-aminopurine
 - 2-amino-6-oxypurine
 - 6-oxypurine
 - 2,6-dioxypurine
33. A DNA sequence reads 5'-ATCG-3' on one strand. What is the sequence of the complementary strand? [End Module 2026 (دور اول)]
- 5'-TAGC-3'
 - 5'-GCTA-3'
 - 5'-CGAT-3'
 - 5'-ATCG-3'
34. Which of the following events does NOT occur when DNA temperature is raised to 90°C? [End Module 2026 (دور اول)]
- Increased UV absorbance at 260 nm
 - Breaking of phosphodiester linkage
 - Disruption of hydrogen bonds
 - DNA double strand unfolding
35. Which of the following statements is true about the nucleotide structure? [End Module 2026 (دور اول)]
- The pentose sugar is present in the a configuration
 - The pyrimidine nitrogenous base is a bicyclic structure.
 - The glycosidic linkage is formed between N1 of purine and C1 of sugar.
 - The phosphate group is attached to carbon 5 on the pentose sugar.



36. Which of the following statements is true about the phosphodiester bond? [End Module 2026 (دور أول)]

- a. It is only present in the structure of DNA not RNA.
- b. It is responsible for formation of the DNA double helical structure.
- c. It is a condensation reaction between the sugars of two nucleotides.
- d. It is an attraction force between the nitrogenous bases of successive nucleotides.

37. Which of the following is NOT true about the DNA double helical structure? [End Module 2026 (دور أول)]

- a. The two strands are held together through base pairing between purine and pyrimidine bases.
- b. The pyrimidine content represents 50% of the total content of nitrogenous bases.
- c. The hydrogen bonds between cytosine and guanine are more than those between adenine and thymine.
- d. Annealing of DNA strands means separation and loss of double helical structure.

38. What is the functional significance of A- and T-rich DNA compared to G- and C-rich DNA? [End Module 2026 (دور أول)]

- a. It has a relatively higher melting temperature
- b. DNA structure is much stabilized in these regions
- c. They represent more important regulatory regions
- d. They are areas for DNA denaturation starting sites

39. Which of the following enzymes requires a palindromic sequence? [End Module 2026 (دور أول)]

- a. Rho protein
- b. Dam methylase
- c. Reverse transcriptase
- d. Primase

40. What is the amino acid commonly entering into the structure of histones? [End Module 2026 (دور أول)]

- a. Arginine
- b. Aspartate
- c. Glutamate
- d. Alanine



41. If a DNA sample from a species contains 20% Thymine, what is the percentage of Guanine? [End Module 2026]
- 20%
 - 30%
 - 40%
 - 60%
42. What is the main function of histone H1? [End Module 2026]
- Forming the core of the nucleosome.
 - Binding to linker DNA.
 - Acetylating other histone proteins.
 - Serving as the scaffold for the 30nm fiber.
43. Which of the following DNA forms is left handed helix? [End Module 2026]
- A form
 - B form
 - Z form
 - A and B form
44. In a DNA sample, which result would be consistent with Chargaff's findings? [End Module 2026]
- $A = G; G = T$
 - $A + G = C + T$
 - $A + T = G + T$
 - $A + C = G + C$
45. Which of the following represents the chemical structure of adenine? [End Module 2026]
- 6-aminopurine
 - 2-amino 6-oxy purine
 - 6-oxypurine
 - 2, 6 dioxypurine
46. A DNA sequence reads 5'-ATCG-3' on one strand. What is the sequence of the complementary strand? [End Module 2026]
- 5'-TAGC-3'
 - 5'-GCTA-3'
 - 5'-CGAT-3'
 - 5'-ATCG-3'



47. Which of the following events does NOT occur When DNA temperature raised into 90°C? [End Module 2026]

- a. Increased UV absorbance at 260 nm.
- b. Breaking phosphodiester linkage.
- c. Disruption of hydrogen bonds.
- d. DNA double strand unfolding.

48. Which of the following statements is true about the nucleotide structure? [End Module 2026]

- a. The pentose sugar is present in the alpha configuration.
- b. The pyrimidine nitrogenous base is a bicyclic structure.
- c. The glycosidic linkage is formed between N1 of purine and C1 of sugar.
- d. The phosphate group is attached to carbon 5 on pentose sugar.

49. Which of the following statements is true about phosphodiester bond? [End Module 2026]

- a. It is only present in the structure of DNA not RNA.
- b. It is responsible for formation of DNA double helical structure.
- c. It is a condensation reaction between the sugars of two nucleotides.
- d. It is an attraction force between the nitrogenous bases of successive nucleotides.

50. Which of the following is NOT true about the DNA double helical structure? [End Module 2026]

- a. The two strands are held together through base pairing between purine and pyrimidine bases.
- b. The pyrimidine content represents 50% of the total content of nitrogenous bases.
- c. The hydrogen bonds between cytosine and guanine is more than that between adenine and thymine.
- d. Annealing of DNA strands means separation and loss of double helical structure.

51. What is the functional significance of A- and T-rich DNA compared to G- and C-rich DNA? [End Module 2026]

- a. It has a relatively higher melting temperature.
- b. DNA structure is much stabilized in these regions.
- c. They represent more important regulatory regions.
- d. They are areas for DNA denaturation starting sites.



52. Which of the following enzymes requires a palindromic sequence? [End Module 2026]
- Rho protein
 - Dam methylase
 - Reverse transcriptase
 - Primase
53. What is the amino acid commonly entering in the structure of histones? [End Module 2026]
- Arginine
 - Aspartate
 - Glutamate
 - Alanine
54. A palindromic DNA sequence is characterized by which of the following? [End Semester 2026 (حكومي)]
- Reading the same forward and backward on the same strand
 - Self-complementary within each strand with twofold symmetry
 - Containing only one type of nucleotide
 - Being present only in viral DNA
55. A researcher heats a solution of double-stranded DNA and observes a sudden increase in UV absorbance at 260 nm. Which of the following best explains this phenomenon? [End Semester 2026 (حكومي)]
- The phosphodiester bonds are breaking, releasing free nucleotides
 - The hyperchromic effect due to unstacking of bases
 - The DNA is being hydrolyzed into nucleosides
 - The DNA is forming Z-DNA conformation
56. What type of bond is cleaved when DNA is hydrolyzed by nucleases? [End Semester 2026 (حكومي)]
- Glycosidic bonds
 - Hydrogen bonds
 - Phosphodiester bonds
 - Disulfide bonds



- 57. How is the phosphodiester bond formed between two nucleotides in a DNA strand?** [End Semester 2026 (حكومي)]
- Between the 5' phosphate of one nucleotide and the 3' OH of the next
 - Between the 3' phosphate of one nucleotide and the 5' OH of the next
 - Between two nitrogenous bases via hydrogen bonds
 - Between the sugar of one nucleotide and the base of the next
- 58. Which of the following best describes the composition of the nucleosome core particle?** [End Semester 2026 (حكومي)]
- DNA wrapped around histone H1 only
 - DNA wrapped around a tetramer of histones H3 and H4
 - DNA wrapped around an octamer of histones H2A, H2B, H3, and H4
 - DNA wrapped around non-histone scaffolding proteins
- 59. Histones are rich in which amino acids?** [End Semester 2026 (أهلية)]
- Acidic amino acids (Asp, Glu)
 - Basic amino acids (Lys, Arg)
 - Sulfur-containing amino acids (Cys, Met)
 - Aromatic amino acids (Phe, Tyr)
- 60. A palindromic DNA sequence is characterized by:** [End Semester 2026 (أهلية)]
- Reading the same forward and backward on the same strand
 - Self-complementarity within each strand with twofold symmetry
 - Containing only one type of nucleotide
 - Being present only in viral DNA
- 61. A researcher heats a solution of double-stranded DNA and observes a sudden increase in UV absorbance at 260 nm. Which of the following best explains this phenomenon?** [End Semester 2026 (دور ثاني)]
- The phosphodiester bonds are breaking, releasing free nucleotides
 - The hyperchromic effect due to unstacking of bases
 - The DNA is being hydrolyzed into nucleosides
 - The DNA is forming Z-DNA conformation



62. What type of bond is cleaved by restriction endonucleases? [End Semester 2026

(دور تاني)]

- a. Glycosidic bonds
- b. Hydrogen bonds
- c. Phosphodiester bonds
- d. Disulfide bonds

63. How is the phosphodiester bond formed between two nucleotides in a DNA strand?

[End Semester 2026 (دور تاني)]

- a. Between the 5' phosphate of one nucleotide and the 3' OH of the next
- b. Between the 3' phosphate of one nucleotide and the 5' OH of the next
- c. Between two nitrogenous bases via hydrogen bonds
- d. Between the sugar of one nucleotide and the base of the next

64. What is the main function of histone H1? [End Semester 2026 (دور تاني)]

- a. Forming the core of the nucleosome.
- b. Binding to linker DNA.
- c. Acetylating other histone proteins.
- d. Serving as the scaffold for the 30nm fiber.

65. Which of the following represents the chemical structure of adenine? [End

Semester 2026 (دور تاني)]

- a. 6-aminopurine
- b. 2-amino 6-oxypurine
- c. 6-oxypurine
- d. 2, 6 dioxypurine

66. In dTMP, what is the atom in the nitrogenous base forming glycosidic linkage with ribose sugar? [End Semester 2026 (دور تاني)]

- a. Carbon no 1
- b. Carbon no 5
- c. Nitrogen no 1
- d. Nitrogen no 9

Written Questions

1. Describe the basic structural unit of chromatin structure. (2 marks) [End Semester 2024]

2. What is the functional significance of A- and T-rich DNA compared to G- and C-rich DNA? [End Semester 2026 (حکومی)]
3. What is the functional significance of palindromic sequence? [End Semester 2026 (دور (تانی)]





6. DNA Replication

1. DNA fragments are sealed by [End Semester 2018 (دور اول)]
 - a. DNA polymerase II
 - b. DNA ligase
 - c. DNA gyrase
 - d. DNA topoisomerase II
2. The unwound strands of DNA are held apart by which of the following proteins? [End Semester 2018 (دور ثانی)]
 - a. Single strand binding protein
 - b. Double strand binding protein
 - c. Rep protein
 - d. DNA-a protein
3. Deoxyribonucleotides are added to RNA primer by which of the following? [End Semester 2018 (دور ثانی)]
 - a. DNA polymerase I
 - b. DNA polymerase II
 - c. DNA polymerase III holoenzyme
4. In mammalian cell cycle, synthesis of DNA occurs during..... [End Semester 2018 (دور ثانی)]
 - a. S phase
 - b. G1 phase
 - c. M Phase
 - d. G2 phase
5. Which of the following statements is NOT true about synthesis of lagging strand? [End Module 2019]
 - a. It needs single primer
 - b. It is synthesized continuously
 - c. Its template runs from 3'-5'
 - d. DNA polymerase synthesizes it in 5'-3' direction
6. Removal of RNA primer in replication requires: [End Module 2019]
 - a. 5'-3' exonuclease activity of DNA polymerase I
 - b. 5'-3' polymerase activity of DNA polymerase I
 - c. 3'-5' polymerase activity of DNA polymerase I
 - d. 3'-5' exonuclease activity of DNA polymerase I



7. Which of the following statements is true about Helicase? [End Module 2019]
- It causes unwinding of the DUE region of OriC
 - It runs in the direction of 5'-3'
 - It stabilizes single stranded DNA during formation of replication fork
 - It usually binds to single stranded DNA
8. In DNA replication the enzyme required in the first step is [Formative Exam 2021]
- DNA directed polymerase
 - Unwinding proteins
 - DNA polymerase
 - DNA ligase
9. DNA fragments are sealed by [Formative Exam 2021]
- DNA polymerase II
 - DNA ligase
 - DNA gyrase
 - DNA topoisomerase II
10. For unwinding of double helical DNA, [Formative Exam 2021]
- Energy is provided by ATP
 - Energy is provided by GTP
 - Energy can be provided by either ATP or GTP
 - No energy is required
11. Telomerase is normally active in which of the following cells? [End Module 2022]
- Germ cells
 - Liver cells
 - Pancreatic cells
 - Cancer cells
12. What is the function of the 5'-3' exonuclease activity of DNA polymerase I? [End Module 2022]
- Proof-reading activity and replacement of mismatched nucleotides during replication
 - Removal and replacement of mismatched nucleotides after replication
 - Removal of RNA primers during replication
 - Replacement of RNA primers during replication



13. The supercoils generated during DNA unwinding are relieved by which of the following enzymes? [End Module 2022]

- a. Polymerase
- b. Topoisomerase
- c. Telomerase
- d. Helicase

14. Which of the following is NOT presented as an unwinding element? [End Semester 2022]

- a. DUE region
- b. Pribnow box
- c. Hogness box
- d. CAAT box

15. Which of the following statements is correct about synthesis of the lagging strand? [End Semester 2022]

- a. It is synthesized continuously in 5'-3' direction
- b. It is synthesized specifically by DNA polymerase I only
- c. It requires multiple primers in their synthesis
- d. Its DNA template runs in the direction of 3'-5'

16. Which of the following statements is TRUE about synthesis of the lagging strand? [Formative Exam 2023]

- a. It is synthesized continuously
- b. It requires one primer
- c. It grows towards the replication fork
- d. It is synthesized in 5'-3' direction

17. A patient undergoing chemotherapy with a topoisomerase inhibitor, This drug likely works by which of the following mechanisms? [End Semester 2024]

- a. Preventing the unwinding of DNA strands
- b. Inhibiting the addition of nucleotides to the leading strand
- c. Blocking the formation of RNA primers
- d. Preventing the joining of Okazaki fragments

18. Which enzyme is responsible for unwinding the DNA helix during replication? [End Semester 2024]

- a. DNA polymerase
- b. Helicase
- c. Ligase
- d. Primase



19. What statement describes the synthesis of the lagging strand during DNA replication? [End Semester 2024]

- a. Continuously
- b. Discontinuously in Okazaki fragments
- c. Without the need for a primer
- d. In the 3' to 5' direction

20. The fidelity of DNA replication is maintained by the proofreading activity of DNA polymerase which of the following activities? [End Semester 2024]

- a. 3' to 5' exonuclease activity
- b. 5' to 3' exonuclease activity
- c. 5' to 3' polymerase activity
- d. 3' to 5' polymerase activity

21. A patient undergoing chemotherapy with a topoisomerase inhibitor: this drug likely works by which of the following mechanisms? [End Semester 2025]

- a. Preventing the unwinding of DNA strands
- b. Inhibiting the addition of nucleotides to the leading strand
- c. Blocking the formation of RNA primers
- d. Preventing the joining of Okazaki fragments

22. Which enzyme is responsible for unwinding the DNA helix during replication? [End Semester 2025]

- a. DNA polymerase
- b. Helicase
- c. Ligase
- d. Primase

23. What statement describes the synthesis of the lagging strand during DNA replication? [End Semester 2025]

- a. Continuously
- b. Discontinuously in Okazaki fragments
- c. Without the need for a primer
- d. In the 3' to 5' direction



24. The fidelity of DNA replication is maintained by the proofreading activity of DNA polymerase, which of the following activities? [End Semester 2025]

- a. 3' to 5' exonuclease activity
- b. 5' to 3' exonuclease activity
- c. 5' to 3' polymerase activity
- d. 3' to 5' polymerase activity

25. A mutation in the gene for the DnaA protein in E. coli that renders it unable to bind ATP would most directly affect which stage of replication? [End Module 2026 (دور اول)]]

- a. Elongation
- b. Termination
- c. Initiation
- d. Proofreading

26. What is the enzyme that joins the small fragments of the lagging strand together into a continuous strand? [End Module 2026 (دور اول)]]

- a. DNA gyrase
- b. DNA ligase
- c. DNA polymerase
- d. Primase

27. Telomerase is most active in which of the following cells? [End Module 2026 (دور اول)]]

- a. Somatic cells
- b. Neurons
- c. Stem cells
- d. Senescent cells

28. What is the role of single-stranded binding proteins (SSBs) in DNA replication? [End Module 2026 (دور اول)]]

- a. Synthesize the new DNA strand.
- b. Prevent the annealing of DNA strands.
- c. Unwind the DNA double helix.
- d. Join Okazaki fragments.



29. A mutation that inactivates the 3' to 5' exonuclease activity of DNA polymerase would result in? [End Module 2026 (دور أول)]
- A slower rate of DNA synthesis.
 - An increased rate of mutation.
 - Inability to remove RNA primers.
 - Failure to join Okazaki fragments.
30. Which of the following correctly lists the order of enzyme action at the replication fork? [End Module 2026 (دور أول)]
- Helicase → Primase → DNA Polymerase → Ligase
 - Primase → Helicase → DNA Polymerase → Ligase
 - DNA Polymerase → Primase → Helicase → Ligase
 - Ligase → Primase → DNA Polymerase → Helicase
31. In eukaryotes, DNA replication occurs during which phase of the cell cycle? [End Module 2026 (دور أول)]
- G1 phase
 - S phase
 - G2 phase
 - M phase
32. A mutation in the gene for the DnaA protein in E. coli that renders it unable to bind ATP would most directly affect which stage of replication? [End Module 2026]
- Elongation
 - Termination
 - Initiation
 - Proofreading
33. What is the enzyme that joins the small fragments of the lagging strand together into a continuous strand? [End Module 2026]
- DNA gyrase
 - DNA ligase
 - DNA polymerase
 - primase
34. Telomerase is most active in which of the following cells? [End Module 2026]
- Somatic cells
 - Neurons
 - Stem cells
 - Senescent cells



35. What is the role of single-stranded binding proteins (SSBs) in DNA replication?

[End Module 2026]

- a. Synthesize the new DNA strand.
- b. Prevent the annealing of DNA strands.
- c. Unwind the DNA double helix.
- d. Join Okazaki fragments.

36. A mutation that inactivates the 3' to 5' exonuclease activity of DNA polymerase would result in? [End Module 2026]

- a. A slower rate of DNA synthesis.
- b. An increased rate of mutation.
- c. Inability to remove RNA primers.
- d. Failure to join Okazaki fragments.

37. Which of the following correctly lists the order of enzyme action at the replication fork? [End Module 2026]

- a. Helicase → Primase → DNA Polymerase → Ligase
- b. Primase → Helicase → DNA Polymerase → Ligase
- c. DNA Polymerase → Primase → Helicase → Ligase
- d. Ligase → Primase → DNA Polymerase → Helicase

38. In eukaryotes, DNA replication occurs during which phase of the cell cycle? [End Module 2026]

- a. G1 phase
- b. S phase
- c. G2 phase
- d. M phase

39. During DNA replication, which of the following best explains why DNA polymerases require an RNA primer, whereas RNA polymerases do not? [End Semester 2026 (حکومی)]

- a. Unlike RNA polymerase, DNA polymerases only add nucleotides to an existing 3'-OH group.
- b. RNA primers provide higher fidelity for DNA synthesis.
- c. DNA polymerases lack the ability to unwind DNA, unlike RNA polymerases.
- d. RNA polymerases use ribonucleotides that do not require priming.



40. A new antibiotic is designed to selectively inhibit bacterial DNA replication by targeting the enzyme that synthesizes RNA primers. Which enzyme would be the target? [End Semester 2026 (حكومي)]

- a. DNA polymerase I
- b. DNA polymerase III
- c. Primase
- d. DNA ligase

41. How does the function of DNA polymerase I differ from DNA polymerase III during lagging strand synthesis? [End Semester 2026 (حكومي)]

- a. Pol III synthesizes the Okazaki fragments, while Pol I synthesizes the RNA primers.
- b. Pol III removes the RNA primers, while Pol I fills the resulting gaps with DNA.
- c. Pol III synthesizes the Okazaki fragments, while Pol I removes and replaces RNA primers.
- d. Pol III joins Okazaki fragments, while Pol I proofreads the newly synthesized DNA.

42. Which of the following enzymes has 5'-3' exonuclease activity that removes RNA primers? [End Semester 2026 (أهلية)]

- a. DNA polymerase I
- b. DNA polymerase III
- c. Primase
- d. Ligase

43. In prokaryotes, the main enzyme responsible for DNA elongation is: [End Semester 2026 (أهلية)]

- a. DNA polymerase I
- b. DNA polymerase II
- c. DNA polymerase III
- d. DNA polymerase A

44. A mutation in DNA polymerase III that eliminates its 3'-5' exonuclease activity would result in: [End Semester 2026 (أهلية)]

- a. Increased replication speed
- b. Decreased mutation rate
- c. Increased mutation rate
- d. Inability to synthesize DNA

45. Which checkpoint ensures DNA replication is complete before mitosis? [End

Semester 2026 (أهلية)]

- a. G1 checkpoint
- b. G2 checkpoint
- c. M checkpoint
- d. S checkpoint

46. The difference between leading and lagging strand synthesis is primarily due to:

[End Semester 2026 (أهلية)]

- a. Different DNA polymerases on each strand
- b. The antiparallel nature of DNA and the 5' → 3' synthesis direction
- c. Different primer requirements
- d. Different proofreading mechanisms

47. During DNA replication, proofreading occurs by: [End Semester 2026 (أهلية)]

- a. 3' → 5' exonuclease activity of DNA polymerase
- b. Nucleotide excision repair
- c. 5' → 3' exonuclease activity of DNA polymerase
- d. Mismatch repair enzyme

48. Single-stranded binding proteins (SSBs) function to: [End Semester 2026 (أهلية)]

- a. Synthesize RNA primers
- b. Prevent reannealing of separated strands
- c. Join Okazaki fragments
- d. Proofread newly synthesized DNA

49. A mutation in the gene for the DnaA protein in E. coli that renders it unable to bind ATP would most directly affect which stage of replication? [End Semester 2026

(دور ثاني)]

- a. Elongation
- b. Termination
- c. Initiation
- d. Proofreading

50. A mutation that inactivates the 3' to 5' exonuclease activity of DNA polymerase would result in? [End Semester 2026 (دور ثاني)]

- a. A slower rate of DNA synthesis.
- b. An increased rate of mutation.
- c. Inability to remove RNA primers.
- d. Failure to join Okazaki fragments.



51. A new antibiotic is designed to selectively inhibit bacterial DNA replication by targeting the enzyme that synthesizes RNA primers. Which enzyme would be the target? [End Semester 2026 (دور تاني)]

- a. DNA polymerase I
- b. DNA polymerase III
- c. Primase
- d. DNA ligase

52. How does the function of DNA polymerase I differ from DNA polymerase III during lagging strand synthesis? [End Semester 2026 (دور تاني)]

- a. Pol III synthesizes the Okazaki fragments, while Pol I synthesizes the RNA primers.
- b. Pol III removes the RNA primers, while Pol I fills the resulting gaps with DNA.
- c. Pol III synthesizes the Okazaki fragments, while Pol I removes and replaces RNA primers.
- d. Pol III joins Okazaki fragments, while Pol I proofreads the newly synthesized DNA.

53. During DNA replication, which of the following best explains why DNA polymerases require an RNA primer, whereas RNA polymerases do not? [End Semester 2026 (دور تاني)]

- a. Unlike RNA polymerase, DNA polymerases only add nucleotides to an existing 3'-OH group.
- b. RNA primers provide higher fidelity for DNA synthesis.
- c. DNA polymerases lack the ability to unwind DNA, unlike RNA polymerases.
- d. RNA polymerases use ribonucleotides that do not require priming.

54. A 7-year-old boy presents with short stature, a bird-like facial appearance, and premature graying. Genetic testing confirms a mutation in a protein involved in the removal of RNA primers during DNA replication and in DNA repair. Which enzyme is most likely deficient? [Formative 2026]

- a. DNA ligase
- b. Primase
- c. DNA polymerase III
- d. DNA polymerase I



55. A new antiviral drug is designed to be a nucleoside analog that lacks a 3' hydroxyl group. When incorporated into a growing DNA strand by a viral polymerase, it terminates further elongation. Which property of DNA polymerase explains why a similar drug would be highly toxic to human cells if it were incorporated? [Formative 2026]

- a. DNA polymerase has 5' to 3' polymerase activity.
- b. DNA polymerase can only synthesize DNA in the 5' to 3' direction on both strands.
- c. DNA polymerase requires a primer with a free 3' hydroxyl group to add new nucleotides.
- d. DNA polymerase has 3' to 5' proofreading exonuclease activity.

Written Questions

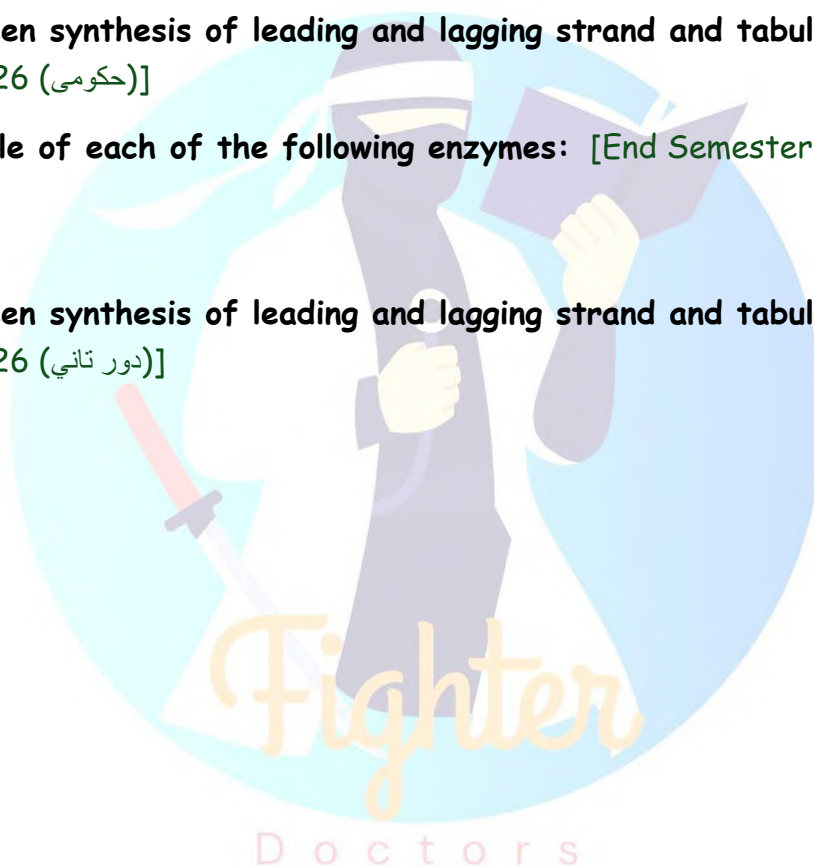
1. Compare between synthesis of leading and lagging strand and tabulate your answer. [End Semester 2026 (حكومي)]

2. Mention the role of each of the following enzymes: [End Semester 2026 (أهلية)]

Helicase

Primase

3. Compare between synthesis of leading and lagging strand and tabulate your answer. [End Semester 2026 (دور ثاني)]





7. DNA Repair

1. Exposure of DNA to ultraviolet radiation can lead to the formation of [End Semester 2018 (دور اول)]
 - a. Adenine dimers
 - b. Guanine dimers
 - c. Thymine dimers
 - d. Uracil dimers
2. Damaged DNA caused by ultraviolet radiation can be repaired by..... [End Semester 2018 (دور ثاني)]
 - a. AP endonuclease
 - b. DNA glycosylase
 - c. Mut proteins
 - d. uvr ABC excinuclease
3. Which of the following enzymes is NOT required for repair of depurinated bases? [End Module 2019]
 - a. DNA glycosylase
 - b. AP endonuclease
 - c. DNA polymerase I
 - d. DNA ligase
4. What is the repair system defected in hereditary nonpolyposis colon cancer? [End Module 2019]
 - a. Mismatch repair system
 - b. Double strand break system
 - c. Nucleotide excision repair
 - d. Base excision repair
5. In hereditary non-polyposis colorectal cancer (HNPCC), what is the defect in the patient? [End Module 2022]
 - a. Inability to remove thymine dimers from DNA
 - b. Inability to remove the base U from DNA
 - c. Low DNA ligase activity
 - d. Inability to correct mismatched bases in newly synthesized DNA
6. Defects caused by UV radiation are corrected by which of the following repair systems? [End Semester 2022]
 - a. Base excision repair
 - b. Nucleotide excision repair
 - c. Mismatch repair
 - d. Homologous recombination



7. Depurinated sites on DNA are corrected by which of the following factors? [End Semester 2022]

- a. ABC excinuclease
- b. DNA glycosylase
- c. AP endonuclease
- d. Mut proteins

8. Which of the following events occurs in response to defective mut proteins? [End Semester 2024]

- a. Inability to correct pyrimidine dimers.
- b. Insensitivity to ultraviolet rays.
- c. Development of cancer in tissues with high rate of regeneration.
- d. Inability to correct deaminated cytosine.

9. Which of the following events occurs in response to defective mut proteins? [End Semester 2025]

- a. Inability to correct pyrimidine dimers
- b. Insensitivity to ultraviolet rays
- c. Development of cancer in tissues with high rate of regeneration
- d. Inability to correct deaminated cytosine

10. Which of the following is a key difference between Base Excision Repair (BER) and Nucleotide Excision Repair (NER)? [End Module 2026 (دور اول)]

- a. BER requires DNA ligase, while NER does not.
- b. BER fixes bulky, helix-distorting lesions, while NER fixes small, non-helix-distorting lesions.
- c. BER is error-prone, while NER is error-free.
- d. BER fixes a single damaged base, while NER fixes an oligonucleotide fragment.

11. A patient with a genetic disorder shows severe sensitivity to ionizing radiation and has defects in repairing double-strand breaks. What is the most likely defective pathway? [End Module 2026 (دور اول)]

- a. Nucleotide Excision Repair
- b. Base Excision Repair
- c. Homologous Recombination
- d. Direct Repair



12. A defect in which repair system is associated with Hereditary Nonpolyposis Colorectal Cancer? [End Module 2026 (دور اول)]

- a. Nucleotide Excision Repair
- b. Base Excision Repair
- c. Mismatch Repair
- d. Non-Homologous End Joining

13. Which deamination indicated below would lead to a mutation in a resulting protein if not repaired? [End Module 2026 (دور اول)]

- a. T to U
- b. C to U
- c. G to A
- d. U to C

14. Which of the following is a key difference between Base Excision Repair (BER) and Nucleotide Excision Repair (NER)? [End Module 2026]

- a. BER requires DNA ligase, while NER does not.
- b. BER fixes bulky, helix-distorting lesions, while NER fixes small, non-helix-distorting lesions.
- c. BER is error-prone, while NER is error-free.
- d. BER fixes single damaged base, while NER fixes an oligonucleotide fragment.

15. A patient with a genetic disorder shows severe sensitivity to ionizing radiation and has defects in repairing double-strand breaks. What is the most likely defective pathway? [End Module 2026]

- a. Nucleotide Excision Repair
- b. Base Excision Repair
- c. Homologous Recombination
- d. Direct Repair

16. A defect in which repair system is associated with Hereditary Nonpolyposis Colorectal Cancer? [End Module 2026]

- a. Nucleotide Excision Repair
- b. Base Excision Repair
- c. Mismatch Repair
- d. Non-Homologous End Joining



17. Which deamination indicated below would lead to a mutation in a resulting protein if not repaired? [End Module 2026]

- a. T to U
- b. C to U
- c. G to A
- d. U to C

18. In a patient with hereditary nonpolyposis colorectal cancer (HNPCC), DNA sequencing reveals a mutation in the hMSH2 gene. Which of the following replication errors would accumulate most prominently in this patient's cells? [End Semester 2026 (حكومي)]

- a. Thymine dimers from UV exposure
- b. 8-oxoguanine lesions from oxidative damage
- c. Mismatched bases
- d. Double-strand breaks from ionizing radiation

19. Which of the following DNA lesions is primarily repaired by nucleotide excision repair? [End Semester 2026 (حكومي)]

- a. Deaminated cytosine
- b. Thymine dimers
- c. Single-strand breaks
- d. Mismatched base pairs

20. In a patient with hereditary nonpolyposis colorectal cancer (HNPCC), DNA sequencing reveals a mutation in the MSH2 gene. Which replication error would accumulate most prominently in this patient's cells? [End Semester 2026 (أهلية)]

- a. Thymine dimers from UV exposure
- b. 8-oxoguanine lesions from oxidative damage
- c. Mismatched bases
- d. Double-strand breaks from ionizing radiation

21. Spontaneous deamination of cytosine results in which of the following? [End Semester 2026 (أهلية)]

- a. Thymine
- b. Uracil
- c. Guanine
- d. Adenine



22. In mismatch repair (MMR), how does the cell distinguish the newly synthesized strand from the template strand in *E. coli*? [End Semester 2026 (أهلية)]

- a. By the presence of RNA primers
- b. By methylation of the parental strand
- c. By the presence of single-strand binding proteins
- d. By the direction of DNA synthesis

23. In a patient with hereditary nonpolyposis colorectal cancer (HNPCC), DNA sequencing reveals a mutation in the hMSH2 gene. Which of the following replication errors would accumulate most prominently in this patient's cells? [End Semester 2026 (دور ثاني)]

- a. Thymine dimers from UV exposure
- b. 8-oxoguanine lesions from oxidative damage
- c. Mismatched bases
- d. Double-strand breaks from ionizing radiation

24. Which of the following DNA lesions is primarily repaired by nucleotide excision repair? [End Semester 2026 (دور ثاني)]

- a. Deaminated cytosine
- b. Thymine dimers
- c. Single-strand breaks
- d. Mismatched base pairs

25. A patient with a family history of early-onset colon cancer undergoes genetic screening. A mutation is found in a gene involved in a DNA repair system that corrects errors missed by the proofreading function of DNA polymerase, such as single-base mismatches or small insertion-deletion loops. Which repair system is defective?

[Formative 2026]

- a. Mismatch repair system
- b. Nucleotide Excision Repair
- c. Homologous Recombination
- d. Base Excision Repair



26. A cell culture experiment shows that upon induction of double-strand breaks, the cells pause at the G2/M checkpoint. What is the primary rationale for this cell cycle arrest? [Formative 2026]

- a. To ensure that all centrosomes have been fully duplicated.
- b. To prevent the segregation of damaged chromosomes during mitosis.
- c. To provide an opportunity for the synthesis of DNA primers on the lagging strand.
- d. To allow time for the complete disassembly of the nuclear envelope.

27. In a laboratory experiment, a bacterial strain is engineered to lack the dam methylase enzyme. What is the most likely consequence for DNA replication in this strain? [Formative 2026]

- a. The Mismatch Repair system will be defective.
- b. The DnaA protein will be unable to bind the origin of replication (OriC).
- c. Telomeres will shorten dramatically with each cell division.
- d. DNA polymerase I will lose its 5' to 3' exonuclease activity.

Written Questions

1. Enumerate the basic steps involved in base excision repair. (2 marks) [End Semester 2024]

2. Demonstrate the steps of nucleotide excision repair with the reference to the clinical disorder results from its defect. [End Semester 2026 (دور ثاني)]

Fighter

Doctors



8. RNA Synthesis

1. The initiation site for transcription is recognized by [End Semester 2018 (دور اول)]
 - a. α -Subunit of DNA-dependent RNA polymerase
 - b. β -Subunit of DNA-dependent RNA polymerase
 - c. Sigma factor
 - d. Rho factor
2. mRNA of prokaryotes can code for..... [End Semester 2018 (دور ثانی)]
 - a. Introns only
 - b. Many exons and introns
 - c. More than one polypeptide
 - d. Only one polypeptide
3. Sigma subunit on RNA polymerase is important for. [End Module 2019]
 - a. Binding with Ori C
 - b. Binding with gene promoter
 - c. Melting of gene promoter
 - d. Termination of transcription
4. A mRNA produced from a fragment of DNA has the sequence AAUUGGCU. The sequence of the non-template strand in the DNA that gave rise to this sequence is which one of the following? [End Module 2022]
 - a. AGCCAATT
 - b. AAUUGGCU
 - c. AATTCCGA
 - d. TTAACCGA
5. Which of the following is not a component of eukaryotic mRNA? [End Module 2022]
 - a. Poly A tail
 - b. Introns
 - c. Exons
 - d. 5' methyl guanosine capping



6. Although each specific tRNA differs from the others in its sequence of nucleotides, all tRNA molecules contain a base-paired stem that terminates in the sequence CCA at [End Module 2022]

- a. 3' terminus
- b. 5' terminus
- c. Anticodon arm
- d. Alanine tRNA

7. Which of the following statements is NOT true about tRNA? [End Semester 2022]

- a. It is the most heterogeneous type of RNA
- b. It is the smallest type of RNA
- c. It can read more than one codon
- d. The initiator is formylated in prokaryotes

8. Which of the following statements represents a defining character for RNA polymerase compared to DNA polymerase? [End Semester 2022]

- a. It has 3'-5' polymerase activity
- b. It has 5'-3' polymerase activity
- c. It needs a primer
- d. It lacks proofreading activity

9. Which of the following RNA polymerase subunits is required for its polymerase activity? [End Semester 2022]

- a. α subunit
- b. β subunit
- c. β' subunit
- d. σ subunit

10. What is the most abundant form of RNA in the cell? [Formative Exam 2023]

- a. mRNA
- b. rRNA
- c. tRNA
- d. snRNA



11. Which of the following statements is correct regarding RNA synthesis? [End Semester 2024]

- a. Transcription requires a primer
- b. DNA replication is more accurate than RNA transcription.
- c. Proofreading is required in mRNA synthesis.
- d. Eukaryotic mRNA is polycistronic

12. Which of the following statements best describes introns? [End Semester 2024]

- a. Non-coding regions removed during RNA splicing
- b. Coding regions of the gene
- c. Regulatory sequences upstream of a gene
- d. Sequences that signal termination of transcription

13. Which of the following statements is correct regarding RNA synthesis? [End Semester 2025]

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- a. Non-coding regions removed during RNA splicing
- b. Coding regions of the gene
- c. Regulatory sequences upstream of a gene
- d. Sequences that signal termination of transcription

15. In splicing in eukaryotes, what are the sequences that are removed from the primary transcript? [End Module 2026 (دور اول)]

- a. Exons
- b. Introns
- c. Codons
- d. Promoters

16. Which of the following RNAs is coding? [End Module 2026 (دور اول)]

- a. mRNA
- b. tRNA
- c. rRNA
- d. snRNA



17. Which of the following RNAs is rich in intrachain base pairing? [End Module 2026 (دور أول)]
- mRNA
 - tRNA
 - rRNA
 - snRNA
18. Which of the following is a function of the poly-A tail in eukaryotic mRNA? [End Module 2026 (دور أول)]
- It codes for a series of lysine amino acids.
 - It facilitates the initiation of transcription.
 - It protects the mRNA from degradation.
 - It is the site for ribosome binding.
19. What is a key difference between prokaryotic and eukaryotic mRNA? [End Module 2026 (دور أول)]
- Prokaryotic mRNA is monocistronic.
 - Eukaryotic mRNA has a 5' cap and a 3' poly-A tail.
 - Prokaryotic mRNA contains introns.
 - Eukaryotic mRNA is translated while being transcribed.
20. What is the main role of sigma (σ) factor in prokaryotic transcription? [End Module 2026 (دور أول)]
- Catalyzing the formation of phosphodiester bonds.
 - Terminating transcription.
 - Recognizing and binding to the promoter region.
 - Adding the 5' cap to the mRNA.
21. In splicing in eukaryotes, what are the sequences that are removed from the primary transcript? [End Module 2026]
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 - Introns
 - Codons
 - Promoters
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 - tRNA
 - rRNA
 - SnRNA



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- a. Catalyzing the formation of phosphodiester bonds.
- b. Terminating transcription.
- c. Recognizing and binding to the promoter region.
- d. Adding the 5' cap to the mRNA.

27. Which of the following best explains how a single eukaryotic mRNA can produce different protein isoforms in different tissues? [End Semester 2026 (حكومي)]

- a. Alternative splicing of pre-mRNA
- b. RNA editing of mature mRNA
- c. Tissue-specific post-translational modifications
- d. Different translation initiation sites

28. The prokaryotic promoter includes the -10 box, also known as: [End Semester 2026 (أهلية)]

- a. TATA box
- b. Pribnow box
- c. CAAT box
- d. GC box



29. Which of the following best explains how a single eukaryotic mRNA can produce different protein isoforms in different tissues? [End Semester 2026 (أهلية)]

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31. In splicing in eukaryotes, what are the sequences that are removed from the primary transcript? [End Semester 2026 (دور ثاني)]

- a. Exons
- b. Introns
- c. Codons
- d. Promoters

32. Which of the following RNAs is coding? [End Semester 2026 (دور ثاني)]

- a. mRNA
- b. tRNA
- c. rRNA
- d. snRNA

33. Which of the following best explains how a single eukaryotic mRNA can produce different protein isoforms in different tissues? [End Semester 2026 (دور ثاني)]

- a. Alternative splicing of pre-mRNA
- b. RNA editing of mature mRNA
- c. Tissue-specific post-translational modifications
- d. Different translation initiation sites

Written Questions

1. Which of the RNAs show the following criteria: (2 marks) [End Semester 2024]
 - a. accounts for the largest percentage of cellular RNA?
 - b. carries amino acids to ribosomes?
 - c. is extensively modified only in eukaryotes?
 - d. is a ribozyme in translation?
2. Compare between RNA and DNA polymerase enzymes. (2 marks) [End Semester 2024]
3. How does prokaryotic RNA pol recognize and bind the appropriate region of DNA to initiate transcription, as shown? Is a primer required in transcription? [End Semester 2026 (أهلية)]





9. Protein Biosynthesis

1. The codon which serves as translation start signal is [End Semester 2018 (دور اول)]
- a. AUG
 - b. UAG
 - c. UGA
 - d. UAA
2. Which of the following statements is NOT true about genetic code? [End Module 2019]
- a. It always codes for the same amino acid
 - b. It can be read by more than one tRNA
 - c. It is well conserved among species
 - d. The 3rd base of the codon always binds classically with the 1st base of anticodon
3. Protein biosynthesis is initiated in eukaryotes by which of the following? [End Module 2019]
- a. Shine Dalgarno sequence
 - b. Formylated initiator tRNA
 - c. AUG initiation codon
 - d. Assembly of 30s and 50s ribosomal subunit
4. The first amino acid tRNA which initiates translation in prokaryotes is [End Module 2022]
- a. Methionyl tRNA
 - b. Formyl methionyl tRNA
 - c. Tyrosine tRNA
 - d. Alanine tRNA
5. What is meant by polysomes? [End Semester 2026 (حکومی)]
- a. Multiple ribosomes on a single mRNA
 - b. Complexes of rRNA and protein
 - c. Multiple mRNAs on a single ribosome
 - d. Splicing complexes



6. A researcher is studying the initiation of protein synthesis. Which of the following initiation factors is unique to eukaryotes and is responsible for binding to the 5' cap of mRNA? [End Semester 2026 (حكومي)]

- a. IF-2
- b. eIF-4
- c. EF-Tu
- d. sigma factor

7. One among the usually asymptomatic genetic variations occurs when a base of the triplet codon in the protein coding gene is replaced by another, without any alteration in the amino acid incorporated in the polypeptide. Which among the following features of the genetic code is responsible for such silent variants? [End Semester 2026 (حكومي)]

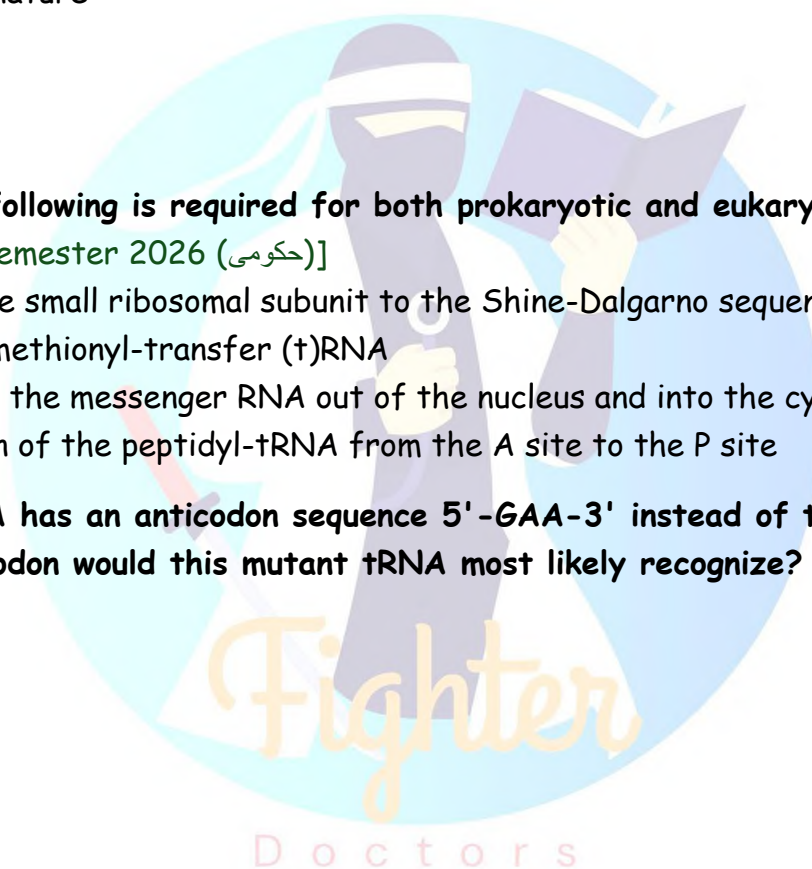
- a. Overlapping nature
- b. Unambiguity
- c. Degeneracy
- d. Universality

8. Which of the following is required for both prokaryotic and eukaryotic protein synthesis? [End Semester 2026 (حكومي)]

- a. Binding of the small ribosomal subunit to the Shine-Dalgarno sequence
- b. Formylated methionyl-transfer (t)RNA
- c. Movement of the messenger RNA out of the nucleus and into the cytoplasm
- d. Translocation of the peptidyl-tRNA from the A site to the P site

9. A mutant tRNA has an anticodon sequence 5'-GAA-3' instead of the normal 5'-UAC-3'. Which codon would this mutant tRNA most likely recognize? [End Semester 2026 (حكومي)]

- a. 5'-UUC-3'
- b. 5'-CUU-3'
- c. 5'-UUA-3'
- d. 5'-CUG-3'





10. During translation initiation in prokaryotes, the Shine-Dalgarno sequence base-pairs with: [End Semester 2026 (أهلية)]

- a. The 3' end of 16S rRNA
- b. The 5' end of 18S rRNA
- c. The anticodon of initiator tRNA
- d. The coding region of Mrna

11. A mutant tRNA has an anticodon sequence 5'-GAA-3' instead of the normal 5'-UAC-3'. Which codon would this mutant tRNA most likely recognize? [End Semester 2026 (أهلية)]

- a. 5'-UUC-3'
- b. 5'-CUU-3'
- c. 5'-UUA-3'
- d. 5'-CUG-3'

12. The P site of the ribosome holds which of the following components? [End Semester 2026 (أهلية)]

- a. The incoming aminoacyl-tRNA
- b. The tRNA carrying the growing polypeptide
- c. The empty tRNA before exit
- d. The mRNA template

13. Which of the following statements represents the term polysomes? [End Semester 2026 (دور ثاني)]

- a. Multiple ribosomes on a single mRNA
- b. Complexes of rRNA and protein
- c. Multiple mRNAs on a single ribosome
- d. Splicing complexes

14. Which of the following is required for both prokaryotic and eukaryotic protein synthesis? [End Semester 2026 (دور ثاني)]

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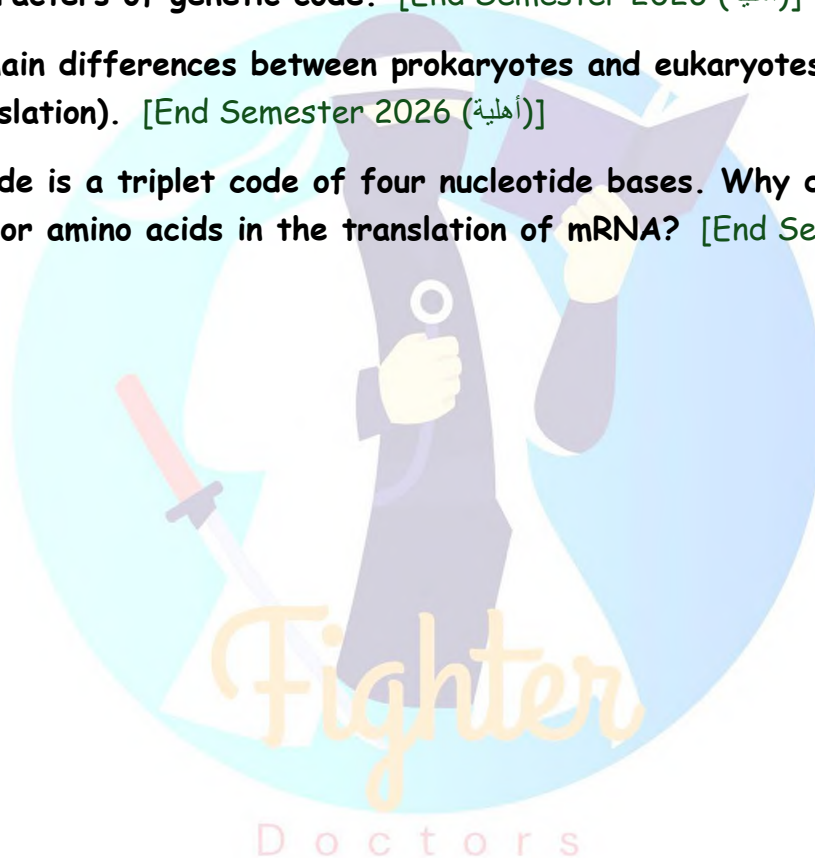


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- a. 5'-UUC-3'
- b. 5'-CUU-3'
- c. 5'-UUA-3'
- d. 5'-CUG-3'

Written Questions

1. Define the ribozyme needed during elongation in protein biosynthesis and demonstrate its role. [End Semester 2026 (حكومي)]
2. Enumerate characters of genetic code. [End Semester 2026 (أهلية)]
3. Enumerate 2 main differences between prokaryotes and eukaryotes regarding protein biosynthesis (translation). [End Semester 2026 (أهلية)]
4. The genetic code is a triplet code of four nucleotide bases. Why do only 61 of the 64 triplets code for amino acids in the translation of mRNA? [End Semester 2026 (أهلية)]





10. Mutation

1. Hemoglobin S is an example of a/an [End Semester 2018 (دور اول)]
- Silent mutation
 - Acceptable mis-sense mutation
 - Unacceptable mis-sense mutation
 - Partially acceptable mis-sense mutation
2. Substitution of an adenine base by guanine in DNA is known as [End Semester 2018 (دور ثانی)]
- Frameshift mutation
 - Transition
 - Transposition
 - Transversion
3. If the codon UAC on mRNA changes into UAG as a result of a base substitution in DNA, it will result in which type of the following mutations? [End Semester 2018 (دور ثانی)]
- Silent mutation
 - Acceptable mis-sense mutation
 - Nonsense mutation
 - Frameshift mutation
4. Substitution of the first base of a codon probably results into what type of mutation? [End Module 2019]
- Silent
 - Missense
 - Nonsense
 - Frameshift mutation
5. Which of the following is NOT a point mutation? [End Module 2019]
- Frameshift mutation
 - Nonsense
 - Missense
 - Silent



6. Which one of the following changes in the coding region of an mRNA (caused by a point mutation) would result in translation of a protein identical to the normal protein?

[End Module 2022]

- a. UCA→UAA
- b. UCA→CCA
- c. UCA→UCU
- d. UCA→ACA

7. Which of the following events occurs in response to point mutation in the 1st base of codon? [End Semester 2024]

- a. It will lead to change in the reading frame of amino acid sequences
- b. There will be no change in the amino acid composition
- c. It will lead to sudden stop in mRNA translation
- d. It will lead to a change in the amino acid composition

8. What type of mutation does not change the amino acid sequence of a protein? [End Semester 2024]

- a. Missense mutation
- b. Frameshift mutation
- c. Nonsense mutation
- d. Silent mutation

9. A 25-year-old male presents with fatigue, pallor, and shortness of breath. His medical history includes episodes of hemolytic anemia, particularly during times of stress or infection. Laboratory studies reveal a hemoglobin electrophoresis pattern consistent with sickle cell disease. Molecular testing identifies a mutation in the B-globin gene. Which type of genetic mutation is most likely responsible for the patient's condition? [End Semester 2024]

- a. Frameshift mutation
- b. Missense mutation
- c. Nonsense mutation
- d. Silent mutation

10. Which of the following events occurs in response to a point mutation in the 1st base of a codon? [End Semester 2025]

- a. It will lead to a change in the reading frame of amino acid sequences
- b. There will be no change in the amino acid composition
- c. It will lead to a sudden stop in mRNA translation
- d. It will lead to a change in the amino acid composition



11. What type of mutation does not change the amino acid sequence of a protein? [End Semester 2025]

- a. Missense mutation
- b. Frameshift mutation
- c. Nonsense mutation
- d. Silent mutation

12. A 25-year-old male presents with fatigue, pallor, and shortness of breath. His medical history includes episodes of hemolytic anemia, particularly during times of stress or infection. Laboratory studies reveal a hemoglobin electrophoresis pattern consistent with sickle cell disease. Molecular testing identifies a mutation in the β -globin gene. Which type of genetic mutation is most likely responsible for the patient's condition? [End Semester 2025]

- a. Frameshift mutation
- b. Missense mutation
- c. Nonsense mutation
- d. Silent mutation

13. A 20-year-old man with a microcytic anemia is found to have an abnormal form of β -globin (Hemoglobin Constant Spring) that is 172 amino acids long, rather than the 141 found in the normal protein. Which of the following point mutations is consistent with this abnormality? [End Semester 2026 (حکومی)]

- a. CGA $\rightarrow$ UGA
- b. UAA $\rightarrow$ UAG
- c. GCA $\rightarrow$ GAA
- d. UAA $\rightarrow$ CAA

14. In sickle cell anemia, a single base mutation (GAG $\rightarrow$ GTG) leads to glutamic acid being replaced by valine. This mutation is best classified as: [End Semester 2026 (حکومی)]

- a. Silent mutation
- b. Nonsense mutation
- c. Missense mutation
- d. Frameshift mutation



15. In sickle cell anemia, a single base mutation (GAG → GTG) leads to glutamic acid being replaced by valine. This mutation is best classified as: [End Semester 2026 (أهلية)]

- a. Silent mutation
- b. Nonsense mutation
- c. Missense mutation
- d. Frameshift mutation

16. Transition mutations involve the substitution of which of the following? [End Semester 2026 (أهلية)]

- a. G to T
- b. T to G
- c. C to T
- d. A to C

17. What is a point mutation that changes a codon to a stop codon? [End Semester 2026 (أهلية)]

- a. Silent mutation
- b. Missense mutation
- c. Nonsense mutation
- d. Frameshift mutation

18. A 20-year-old man with a microcytic anemia is found to have an abnormal form of β-globin (Hemoglobin Constant Spring) that is 172 amino acids long, rather than the 141 found in the normal protein. Which of the following point mutations is consistent with this abnormality? [End Semester 2026 (دور ثاني)]

- a. CGA → UGA
- b. UAA → UAG
- c. GCA → GAA
- d. UAA → CAA

Written Questions

1. What happens if: (2 marks) [End Semester 2024]

- a. Deletion of a base in the gene sequence.
- b. There is a mutation in mismatch repair system
- c. The exchange of a base in the coding area of a gene
- d. A point mutation occurs in UGA changed into UAA

2. Define missense mutation and describe its types with examples. [End Semester 2026 (حكومي)]

3. Define what is meant by silent mutation in a gene with reference to its effect on protein structure and explain your answer. [End Semester 2026 (دور تاني)]





11. Regulation of Gene Expression

1. **Lac operon is a cluster of** [End Semester 2018 (دور اول)]
 - a. Three structural genes
 - b. Three structural genes and their promoter
 - c. A regulatory gene, an operator and a promoter
 - d. A regulatory gene, an operator, a promoter and three structural genes
2. **Transcription of z, y and a genes of lac operon is prevented by** [End Semester 2018 (دور ثانی)]
 - a. Lactose
 - b. Allo-lactose
 - c. Repressor
 - d. cAMP
3. **In Lac operon; which of the following events does NOT occur in the presence of lactose and absence of glucose?** [End Module 2019]
 - a. Occupied operator site
 - b. Bound CAP binding site
 - c. Allolactose inhibits the repressor
 - d. Increased cAMP levels
4. **Which of the following chromatin states will be accompanied by activated transcription?** [End Module 2022]
 - a. Increased methylation and increased acetylation
 - b. Increased methylation and decreased acetylation
 - c. Decreased methylation and decreased acetylation
 - d. Decreased methylation and increased acetylation
5. **CAAT box is present in which of the following?** [End Module 2022]
 - a. Prokaryotic promoters 10 bp upstream of transcription start site
 - b. Prokaryotic promoters 35 bp upstream of transcription start site
 - c. Eukaryotic promoters 25 bp upstream of transcription start site
 - d. Eukaryotic promoters 70-80 bp upstream of transcription start site



6. In lac operon, which of the following events is active in the absence of glucose and presence of lactose? [End Semester 2022]
- Allolactose binds to repressor protein
 - Inhibition of gene i expression
 - The level of cAMP is decreased
 - The operator site is occupied
7. Histone deacetylase enzyme activates transcription in eukaryotes. [End Semester 2022]
- True
 - False
8. Histone acetylation makes which of the following statements? [End Semester 2024]
- Enhances promoter activity
 - Makes DNA more compact
 - Makes histones carry +ve charge
 - Denaturation of histones
9. In lac operon, which of the following events occurs in the presence of lactose and absence of glucose? [End Semester 2024]
- The CAP binding site is occupied with cAMP-CAP complex.
 - The expression of structural genes is inhibited.
 - The repressor binds to the operator site.
 - The expression of LacI gene is activated by the presence of allolactose.
10. Which of the following cis-acting elements is the initial binding site for RNA polymerase enzyme? [End Semester 2024]
- Terminator
 - Introns
 - Promoter
 - Exons
11. Histone acetylation is associated with which of the following statements? [End Semester 2025]
- Enhances promoter activity
 - Makes DNA more compact
 - Makes histones carry +ve charge
 - Denaturation of histones



12. In lac operon, which of the following events occurs in the presence of lactose and absence of glucose? [End Semester 2025]
- The CAP binding site is occupied with cAMP-CAP complex
 - The expression of structural genes is inhibited
 - The repressor binds to the operator site
 - The expression of LacI gene is activated by the presence of allolactose
13. Which of the following cis-acting elements is the initial binding site for RNA polymerase enzyme? [End Semester 2025]
- Terminator
 - Introns
 - Promoter
 - Exons
14. What is the primary effect of histone acetylation on nucleosome structure and function? [End Semester 2026 (حكومي)]
- Increases positive charge, strengthening DNA-histone interaction
 - Adds methyl groups to DNA, silencing genes
 - Neutralizes positive charge, loosening DNA-histone binding
 - Promotes degradation of histones
15. Which of the following describes the function of cAMP in lac operon regulation? [End Semester 2026 (حكومي)]
- It acts as an inducer by binding to the repressor
 - It binds to CAP and enhance transcription
 - It directly inhibits RNA polymerase binding
 - It degrades the repressor protein
16. Histone acetylation generally leads to which of the following events? [End Semester 2026 (أهلية)]
- Gene silencing
 - Gene activation
 - DNA methylation
 - Heterochromatin formation
17. Which enzyme is produced by the lac I gene? [End Semester 2026 (أهلية)]
- Lactose permease
 - β -galactosidase
 - Thiogalactoside transacetylase
 - Repressor protein



18. Which of the following describes the function of cAMP in lac operon regulation?

[End Semester 2026 (أهلية)]

- a. It acts as an inducer by binding to the repressor
- b. It binds to CAP, enabling it to bind DNA and enhance transcription
- c. It directly inhibits RNA polymerase binding
- d. It degrades the repressor protein

19. Which molecule acts as the inducer of the lac operon by binding to the repressor protein? [End Semester 2026 (أهلية)]

- a. Glucose
- b. cAMP
- c. Allolactose
- d. Galactose

20. In the lac operon, the repressor binds to which of the following? [End Semester 2026 (أهلية)]

- a. The promoter
- b. The operator
- c. The structural genes
- d. RNA polymerase

21. What is the primary effect of histone acetylation on nucleosome structure and function? [End Semester 2026 (دور ثاني)]

- a. Increases positive charge, strengthening DNA-histone interaction
- b. Adds methyl groups to DNA, silencing genes
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[End Semester 2026 (دور ثاني)]

- a. It acts as an inducer by binding to the repressor
- b. It binds to CAP and enhance transcription
- c. It directly inhibits RNA polymerase binding
- d. It degrades the repressor protein

Written Questions

1. Categorize the following events in the following table (Euchromatin / Heterochromatin) (2 marks): [End Semester 2024]

Methylation of CG island

Activation of histone deacetylase.

Activation of histone acetyl transferase.

Activation of DNA demethylase

2. What is meant by cis-acting and trans-acting elements? With examples (2 marks) [End Semester 2024]

3. Describe the regulatory mechanisms occur in lactose operon in the presence of both glucose and lactose and define their effect on transcription of structural genes. [End Semester 2026 (حكومي)]

4. What role do the reactions shown play in eukaryotic transcription? [End Semester 2026 (حكومي)]

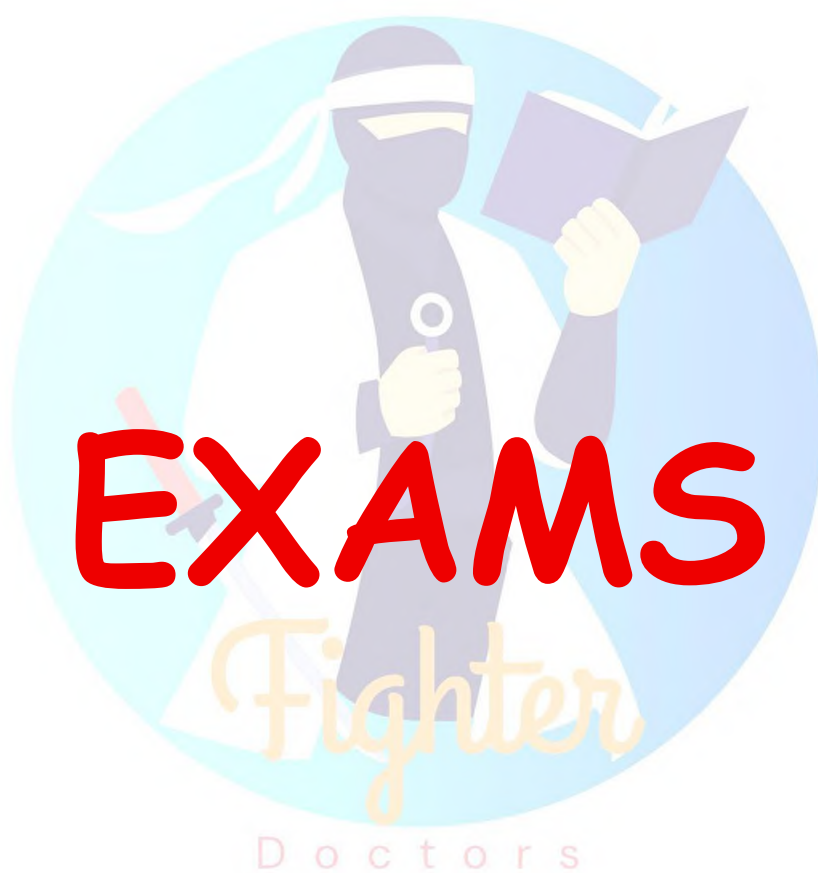
5. Describe the basic structural unit of chromatin and explain how chromatin modification plays a role in regulation of gene expression. [End Semester 2026 (أهلية)]

6. Demonstrate the structure of nucleosome and its role in regulation of gene expression. [End Semester 2026 (دور ثاني)]

12. Cancer & Tumor Markers

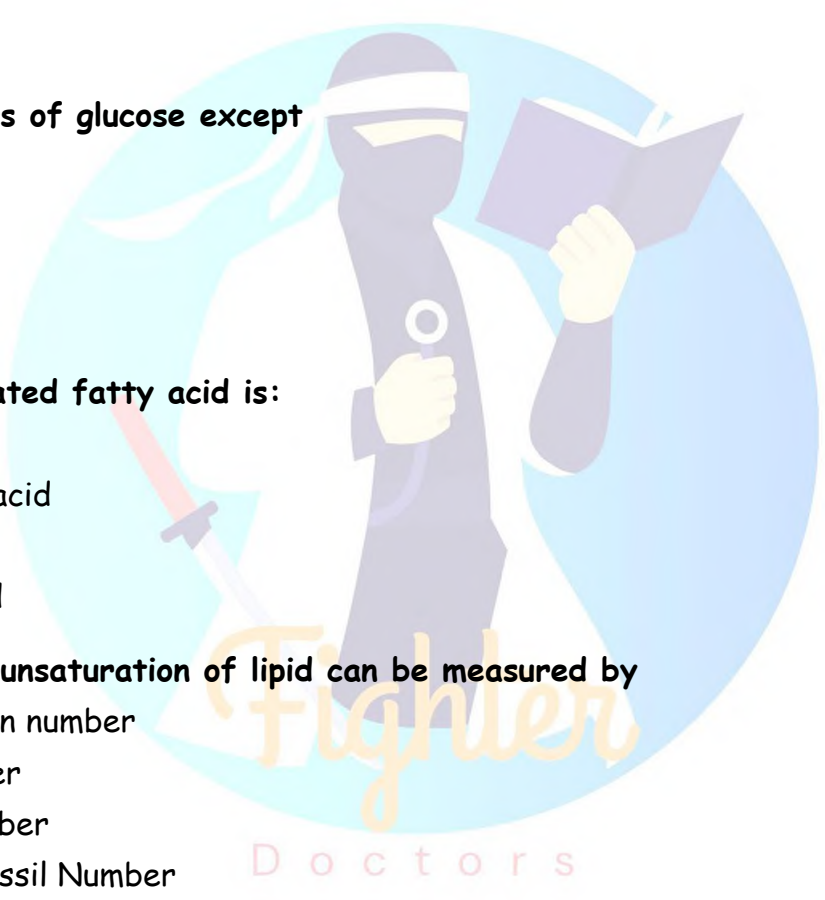
1. In cancer cells, telomerase activity is often upregulated. This allows cancer cells to: [End Semester 2026 (أهلية)]

- Replicate DNA faster
- Avoid telomere shortening and cellular senescence
- Repair DNA damage more efficiently
- Evade the immune system





1. End Semester 2018 (دور اول)

1. In glucose the orientation of the -H and -OH groups around the carbon atom 5 adjacent to the terminal primary alcohol carbon determines
 - a. D or L series
 - b. Dextrose or levorotatory
 - c. α and β anomers
 - d. Epimers
 2. Which of the following is a reducing disaccharide?
 - a. Sucrose
 - b. Trehalose
 - c. Isomaltose
 - d. Agar
 3. All are polymers of glucose except
 - a. Glycogen
 - b. insulin
 - c. Starch
 - d. Cellulose
 4. A monounsaturated fatty acid is:
 - a. Oleic acid
 - b. Arachidonic acid
 - c. Palmitic acid
 - d. Linolenic acid
 5. The degree of unsaturation of lipid can be measured by
 - a. Saponification number
 - b. Iodine number
 - c. Polenske number
 - d. Reichert Meissil Number
 6. Free fatty acids are transported in the blood
 - a. Combined with albumin
 - b. Combined with fatty acid binding protein
 - c. Combined with β -lipoprotein
 - d. In unbound free salts
 7. Cysteine, cystine and methionine are:
 - a. Essential amino acids
 - b. Non-protein amino acids
 - c. Acidic amino acids
 - d. Sulfur containing amino acids
- 



8. Rotation of plain polarized light is caused by solution of all the following except:
- Alanine
 - Glycine
 - Leucine
 - Serine
9. Isoelectric pH is the pH at which:
- Protein molecules is denatured
 - Protein molecules carries no net charges
 - Protein molecules is most soluble
 - The disulfide bonds in protein molecules are disrupted
10. In proteins the α -helix and β -pleated sheet are examples of
- Primary structure
 - Secondary structure
 - Tertiary structure
 - Quaternary structure
11. Denaturation of proteins results in
- Disruption of primary structure
 - Breakdown of peptide bonds
 - Destruction of hydrogen bonds
 - Reversible changes in the molecule
12. Facilitated diffusion:
- Requires a transmembrane protein
 - Requires ATP
 - Can move molecules against a concentration gradient
 - Is typically used to transport small nonpolar molecules
13. Enzymes, catalyzing the synthetic reactions with ATP utilization, are included into class of:
- Oxidoreductases
 - Transferases
 - Lyases
 - Ligases
14. When $[S]$ is equal to K_m , which of the following conditions exist?
- Half the enzyme molecules are bound to substrate
 - The velocity of the reaction is equal to V_{max}
 - The velocity of the reaction is independent of substrate concentration
 - Enzyme is completely saturated with substrate



15. A competitive inhibitor of an enzyme has which of the following properties?
- Is frequently a feedback inhibitor
 - Becomes covalently attached to an enzyme
 - Decreases the V_{max}
 - Interferes with substrate binding to the enzyme
16. What is the chemical name of Adenine?
- 6-Amino purine
 - 2-Amino-6-oxypurine
 - 2-Oxy-4-aminopyrimidine
 - 2,4-Dioxypyrimidine
17. The analysis of the base composition of unlabeled nucleic acid samples showed 10% adenine, 40% cytosine, 30% thymine and 20% guanine. According to such results, the source of this nucleic acid is most likely to be:
- Bacterial chromosome
 - Bacterial plasmid
 - Nuclear chromosome
 - Viral genome
18. DNA fragments are sealed by
- DNA polymerase II
 - DNA ligase
 - DNA gyrase
 - DNA topoisomerase II
19. Exposure of DNA to ultraviolet radiation can lead to the formation of
- Adenine dimers
 - Guanine dimers
 - Thymine dimers
 - Uracil dimers
20. The initiation site for transcription is recognized by
- α -Subunit of DNA-dependent RNA polymerase
 - β -Subunit of DNA-dependent RNA polymerase
 - Sigma factor
 - Rho factor
21. Hemoglobin S is an example of a/an
- Silent mutation
 - Acceptable mis-sense mutation
 - Unacceptable mis-sense mutation
 - Partially acceptable mis-sense mutation



22. The codon which serves as translation start signal is

- a. AUG
- b. UAG
- c. UGA
- d. UAA

23. Lac operon is a cluster of

- a. Three structural genes
- b. Three structural genes and their promoter
- c. A regulatory gene, an operator and a promoter
- d. A regulatory gene, an operator, a promoter and three structural genes

Answer Key

1/A	2/C	3/B	4/A	5/B	6/A	7/D	8/B	9/B	10/B
11/C	12/A	13/D	14/A	15/D	16/A	17/D	18/B	19/C	20/C
21/D	22/A	23/D							





2. End Semester 2018 (دور ثاني)

1. Which of the following is an example of disaccharide?
 - a. Fructose
 - b. Galactose
 - c. Glucose
 - d. Maltose
2. Which of the following is true about monosaccharides?
 - a. They are insoluble in water
 - b. They can be hydrolyzed
 - c. They cannot be hydrolyzed
 - d. They have sour taste
3. In which form is Glucose stored in animals?
 - a. Cellulose
 - b. Dextrins
 - c. Glycogen
 - d. Starch
4. The polysaccharide used in assessing glomerular filtration rate
 - a. Agar
 - b. Glycogen
 - c. Hyaluronic acid
 - d. Insulin
5. What is the condensed structural formula of Palmitic acid?
 - a. $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-COOH}$
 - b. $\text{CH}_3\text{-(CH}_2\text{)}_4\text{-COOH}$
 - c. $\text{CH}_3\text{-(CH}_2\text{)}_{14}\text{-COOH}$
 - d. $\text{CH}_3\text{-(CH}_2\text{)}_{16}\text{-COOH}$
6. $\Delta 9$ indicates a double bond between carbon atoms of the fatty acid
 - a. 8 and 9
 - b. 9 and 10
 - c. 9 and 11
 - d. 9 and 12
7. Unpleasant odors and taste in a fat (rancidity) can be delayed or prevented by the addition of which of the following?
 - a. Lead
 - b. Copper
 - c. Tocopherol
 - d. Ergosterol



8. Dietary fats after absorption appear in the circulation as
- HDL
 - VLDL
 - LDL
 - Chylomicrons
9. Isoelectric pH is the pH at which
- Protein molecules carry no net charges
 - Protein molecules are denatured
 - Protein molecules are most soluble
 - The disulfide bonds in protein molecules are disrupted
10. Lysine is.....
- Neutral amino acid
 - Acidic amino acid
 - Basic amino acid
 - Aromatic amino acid
11. Which of the following is a tripeptide
- Aspartame
 - Aspartate
 - Glutamate
 - Glutathione
12. During denaturation of proteins, all of the following are disrupted except..
- Primary structure
 - Secondary structure
 - Tertiary structure
 - Quaternary structure
13. Enzymes, catalyzing the synthetic reactions with ATP utilization, are included into the class of
- Ligases
 - Lyases
 - Oxidoreductases
 - Transferases
14. An allosteric enzyme influences the enzyme activity by which of the following mechanisms?
- Changing the conformation of the enzyme by binding to a site other than catalytic site
 - Changing the nature of the products formed
 - Changing the specificity of the enzyme for the substrate
 - Competing for the catalytic site with the substrate



15. A competitive inhibitor of an enzyme has which of the following properties?
- It becomes covalently attached to an enzyme
 - It decreases the V_{max}
 - It interferes with substrate binding to the enzyme
 - It is frequently a feedback inhibitor
16. In DNA, three hydrogen bonds are formed between.....
- Adenine and guanine
 - Adenine and thymine
 - Guanine and cytosine
 - Thymine and cytosine
17. The unwound strands of DNA are held apart by which of the following proteins?
- Single strand binding protein
 - Double strand binding protein
 - Rep protein
 - DNA-a protein
18. Deoxyribonucleotides are added to RNA primer by which of the following?
- DNA polymerase I
 - DNA polymerase II
 - DNA polymerase III holoenzyme
19. In mammalian cell cycle, synthesis of DNA occurs during.....
- S phase
 - G₁ phase
 - M Phase
 - G₂ phase
20. Damaged DNA caused by ultraviolet radiation can be repaired by.....
- AP endonuclease
 - DNA glycosylase
 - Mut proteins
 - uvr ABC excinuclease
21. mRNA of prokaryotes can code for.....
- Introns only
 - Many exons and introns
 - More than one polypeptide
 - Only one polypeptide
22. Substitution of an adenine base by guanine in DNA is known as
- Frameshift mutation
 - Transition



- c. Transposition
- d. Transversion

23. Transcription of *z*, *y* and *a* genes of lac operon is prevented by

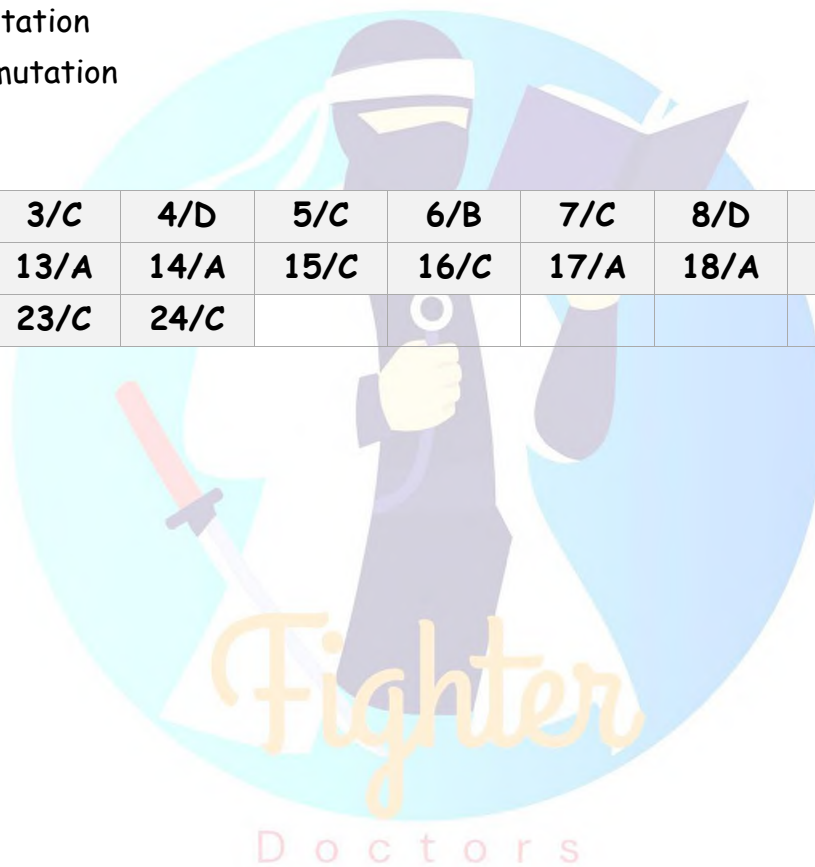
- a. Lactose
- b. Allo-lactose
- c. Repressor
- d. cAMP

24. If the codon UAC on mRNA changes into UAG as a result of a base substitution in DNA, it will result in which type of the following mutations?

- a. Silent mutation
- b. Acceptable mis-sense mutation
- c. Nonsense mutation
- d. Frameshift mutation

Answer Key

1/D	2/C	3/C	4/D	5/C	6/B	7/C	8/D	9/A	10/C
11/D	12/A	13/A	14/A	15/C	16/C	17/A	18/A	19/A	20/D
21/C	22/B	23/C	24/C						





3. End Module 2019

1. Which of the following is the Aldose sugar?
 - a. Sucrose
 - b. Ribose
 - c. Fructose
 - d. Ribulose
2. Apoenzyme is...
 - a. The inorganic part of the holoenzyme
 - b. The metal part of holoenzyme
 - c. The organic (vitamin derivative) part of holoenzyme
 - d. The protein part of holoenzyme
3. One of the following has reducing properties
 - a. Glucuronic acid
 - b. Gluconic acid
 - c. Mucic acid
 - d. Glucaric acid
4. Which of the following statements is TRUE about noncompetitive inhibitors?
 - a. They can bind to the enzyme alone or the enzyme-substrate complex
 - b. They mainly bind to the active site of enzyme
 - c. They have structural similarity between the inhibitor and the substrate
 - d. Noncompetitive inhibitors are always reversible
5. At V_{max} , of the enzyme molecules are occupied by the substrate.
 - a. 25%
 - b. 50%
 - c. 75%
 - d. 100%
6. Which of the following statements is true about allosteric enzyme?
 - a. They are presented by a sigmoidal curve when V_0 is plotted against substrate concentration
 - b. They are inhibited by substrate
 - c. They are enzymes that contain active site only
 - d. They are enzymes with a higher K_m than other enzymes
7. In a DNA molecule the thymine concentration is 30%, the guanosine concentration will be
 - a. 10%
 - b. 20%
 - c. 30%



- d. 40%
8. The complete name of the opposite structure is:
- α -D-gulopyranose
 - β -D-gulopyranose
 - α -D-gulofuranose
 - β -D-gulofuranose
9. ALL of these are correct combinations of monosaccharides to form disaccharides EXCEPT:
- Glucose + fructose = Sucrose
 - Glucose + galactose = Lactose
 - Glucose + Glucose = Maltose
 - Glucose + fructose = Lactose
10. The two sugar units of disaccharides are connected by which of the following bonds?
- O-glycosidic bond
 - N-glycosidic bond
 - Peptide bond
 - Phosphodiester bond
11. In Uridine monophosphate (UMP), the glycosidic linkage is formed between C1 of the pentose sugar and
- N1
 - N5
 - N7
 - N9
12. Which of the following statements is TRUE about melting temperature (T_m)?
- At T_m , all of the double helical structure is lost
 - DNA rich in G and C has low melting temperature
 - Unfolded DNA has a higher absorption at 260 nm UV light
 - DNA rich in A and T has high melting temperature
13. Which of the following is NOT glucosan?
- Inulin
 - Glycogen
 - Starch
 - Cellulose
14. Uracil is
- 6-Amino purine
 - 2-Amino-6-oxypurine



- c. 2-Oxy-4-aminopyrimidine
d. 2,4-Dioxypyrimidine
15. What are the two structures making up starch?
- Amylose and cellulose
 - Amylose and amylopectin
 - Amylopectin and cellobiose
 - Cellobiose and cellulose
16. The polysaccharide used in assessing the glomerular filtration rate (GFR) is
- Glycogen
 - Agar
 - Inulin
 - Hyaluronic acid
17. Condensed structural formula of Palmitic acid is
- $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-COOH}$
 - $\text{CH}_3\text{-(CH}_2\text{)}_4\text{-COOH}$
 - $\text{CH}_3\text{-(CH}_2\text{)}_{14}\text{-COOH}$
 - $\text{CH}_3\text{-(CH}_2\text{)}_{16}\text{-COOH}$
18. Which of the following enzymes is NOT required for repair of depurinated bases?
- DNA glycosylase
 - AP endonuclease
 - DNA polymerase I
 - DNA ligase
19. In mammals, the major fat in adipose tissues is
- Spingolipids
 - Cholesterol
 - Triacylglycerol
 - Phospholipids
20. Substitution of the first base of a codon probably results into what type of mutation?
- Silent
 - Missense
 - Nonsense
 - Frameshift mutation
21. Which of the following statements is NOT true about synthesis of lagging strand?
- It needs single primer
 - It is synthesized continuously
 - Its template runs from 3'-5'



- d. DNA polymerase synthesizes it in 5'-3' direction
22. Which of the following structures is a 20:2 ($\Delta 4,9$) fatty acid?
- $\text{CH}_3(\text{CH}_2)_9\text{CH}=\text{CH}(\text{CH}_2)_3\text{CH}=\text{CH}(\text{CH}_2)_2\text{COOH}$
 - $\text{CH}_3(\text{CH}_2)_2\text{CH}=\text{CH}(\text{CH}_2)_3\text{CH}=\text{CH}(\text{CH}_2)_9\text{COOH}$
 - $\text{CH}_3(\text{CH}_2)_{10}\text{CH}=\text{CH}(\text{CH}_2)_3\text{CH}=\text{CHCH}_2\text{COOH}$
 - $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}(\text{CH}_2)_3\text{CH}=\text{CH}(\text{CH}_2)_{10}\text{COOH}$
23. Removal of RNA primer in replication requires:
- 5'-3' exonuclease activity of DNA polymerase I
 - 5'-3' polymerase activity of DNA polymerase I
 - 3'-5' polymerase activity of DNA polymerase I
 - 3'-5' exonuclease activity of DNA polymerase I
24. Which of the following is $\omega 9$ fatty acid?
- Oleic acid
 - Linoleic acid
 - Linolenic acid
 - Stearic acid
25. Number of milligrams of KOH required to neutralize fatty acid present in 1 g of fat is called
- Potassium number
 - Acid number
 - Saponification number
 - Iodine number
26. Sigma subunit on RNA polymerase is important for.
- Binding with Ori C
 - Binding with gene promoter
 - Melting of gene promoter
 - Termination of transcription
27. Which of the following statements is NOT true about genetic code?
- It always codes for the same amino acid
 - It can be read by more than one tRNA
 - It is well conserved among species
 - The 3rd base of the codon always binds classically with the 1st base of anticodon
28. Hydrogenation of oleic acid produces....
- Linoleic acid
 - Linolenic acid
 - Palmitic acid
 - Stearic acid



29. Which of the following is NOT a point mutation?
- Frameshift mutation
 - Nonsense
 - Missense
 - Silent
30. What is the repair system defected in hereditary nonpolyposis colon cancer?
- Mismatch repair system
 - Double strand break system
 - Nucleotide excision repair
 - Base excision repair
31. Protein biosynthesis is initiated in eukaryotes by which of the following?
- Shine Dalgarno sequence
 - Formylated initiator tRNA
 - AUG initiation codon
 - Assembly of 30s and 50s ribosomal subunit
32. Gangliosides are complex glycosphingolipids found in
- Liver
 - Brain
 - Kidney
 - Muscle
33. The optically inactive amino acid is
- Glycine
 - Serine
 - Threonine
 - Valine
34. Histones are
- Basic proteins formed of 3 classes
 - Strongly negatively charged proteins
 - Strong positively charged protein
 - Needed to stabilize single stranded DNA structure
35. All the following are sulphur containing amino acids found in proteins except
- Cysteine
 - Cystine
 - Methionine
 - Threonine
36. During denaturation of proteins, all of the following are disrupted except
- Primary structure



- b. Secondary structure
- c. Tertiary structure
- d. Quaternary structure

37. Which of the following statements is true about Helicase?

- a. It causes unwinding of the DUE region of OriC
- b. It runs in the direction of 5'-3'
- c. It stabilizes single stranded DNA during formation of replication fork
- d. It usually binds to single stranded DNA

38. In Lac operon; which of the following events does NOT occur in the presence of lactose and absence of glucose?

- a. Occupied operator site
- b. Bound CAP binding site
- c. Allolactose inhibits the repressor
- d. Increased cAMP levels

39. Which of the following is a tripeptide?

- a. Anserine
- b. Oxytocin
- c. Glutathione
- d. Kallidin

40. Tertiary structure of a protein describes..

- a. The order of amino acids
- b. Location of disulphide bonds
- c. Loop regions of proteins
- d. The ways of protein folding

Answer Key

1/B	2/D	3/A	4/A	5/D	6/A	7/B	8/B	9/D	10/A
11/A	12/C	13/A	14/D	15/B	16/C	17/C	18/B	19/C	20/B
21/A	22/A	23/D	24/A	25/C	26/B	27/D	28/D	29/A	30/A
31/B	32/B	33/A	34/C	35/D	36/A	37/A	38/A	39/C	40/D

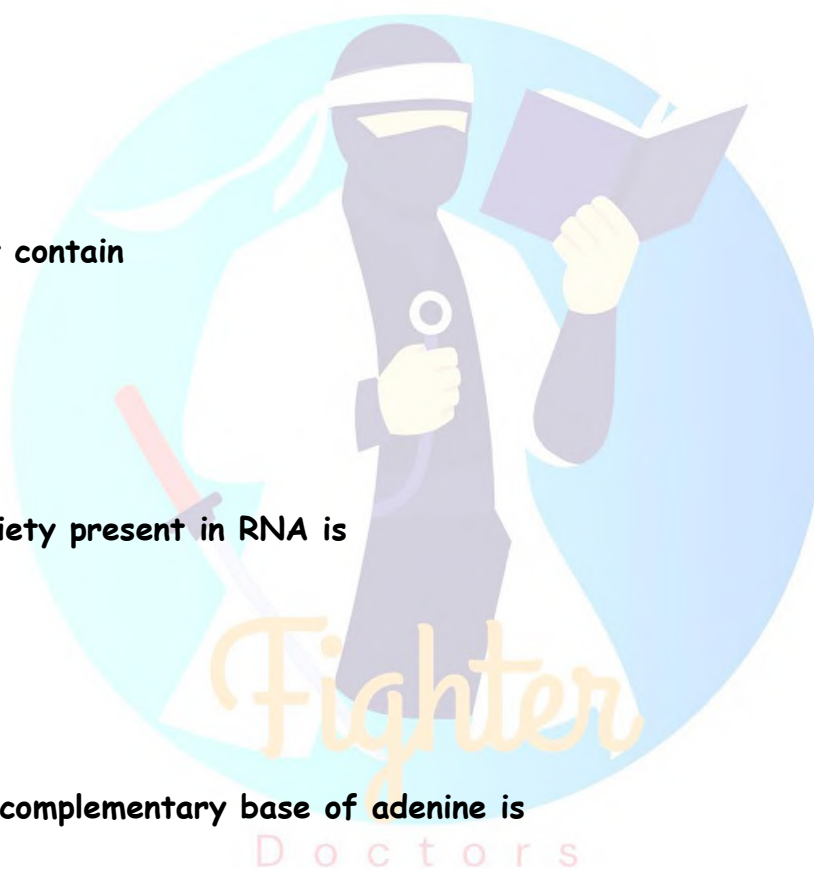


4. Formative Exam 2021

1. A sugar alcohol is:
 - a. Mannitol
 - b. Trehalose
 - c. Xylulose
 - d. Arabinose
 2. The sugar found in DNA is
 - a. Xylose
 - b. Ribose
 - c. Deoxyribose
 - d. Ribulose
 3. Ribulose is a
 - a. Ketotetrose
 - b. Aldosterone
 - c. Ketopentose
 - d. Aldopentose
 4. Spermatozoa in seminal fluid utilizes the following sugar for its metabolism
 - a. Glucose
 - b. Fructose
 - c. Sucrose
 - d. Mannose
 5. Which of the following protects our heart and kidneys from injury?
 - a. Skin
 - b. Muscles
 - c. Lubricants
 - d. Fat
 6. Fat storing cells of vertebrates are called
 - a. Hepatocytes
 - b. Astrocytes
 - c. Adipocytes
 - d. Melanocytes
 7. Main function of preen glands in birds is
 - a. Hormone secretion
 - b. Fatty acid degradation
 - c. Wax secretion
 - d. Pigment synthesis
- 



8. Cysteine, cystine and methionine are:
- Essential amino acids
 - Non-protein amino acids
 - Acidic amino acids
 - Sulfur containing amino acids
9. The basic amino acid is:
- Lysine
 - Bile acids
 - Glycine
 - Alanine
10. An -OH group is present in the side chain of
- Serine
 - Arginine
 - Lysine
 - Proline
11. RNA does not contain
- Uracil
 - Adenine
 - Thymine
 - Ribose
12. The sugar moiety present in RNA is
- Ribulose
 - Arabinose
 - Ribose
 - Deoxyribose
13. In DNA, the complementary base of adenine is
- Guanine
 - Cytosine
 - Uracil
 - Thymine
14. DNA rich in A-T pairs have
- 1 Hydrogen bond
 - 2 Hydrogen bonds
 - 3 Hydrogen bonds
 - 4 Hydrogen bonds
15. In DNA replication the enzyme required in the first step is
- DNA directed polymerase





- b. Unwinding proteins
 - c. DNA polymerase
 - d. DNA ligase
- 16. DNA fragments are sealed by**
- a. DNA polymerase II
 - b. DNA ligase
 - c. DNA gyrase
 - d. DNA topoisomerase II
- 17. For unwinding of double helical DNA,**
- a. Energy is provided by ATP
 - b. Energy is provided by GTP
 - c. Energy can be provided by either ATP or GTP
 - d. No energy is required
- 18. A nucleoside consists of**
- a. Nitrogenous base
 - b. Purine or pyrimidine base + sugar
 - c. Purine or pyrimidine base + phosphorous
 - d. Purine + pyrimidine base + sugar + phosphorous
- 19. A nucleotide consists of**
- a. A nitrogenous base like choline
 - b. Purine or pyrimidine base + sugar + phosphorous
 - c. Purine or pyrimidine base + sugar
 - d. Purine or pyrimidine base + phosphorous
- 20. A purine nucleotide is**
- a. AMP
 - b. UMP
 - c. CMP
 - d. TMP

Answer Key

1/A	2/C	3/C	4/B	5/D	6/C	7/C	8/D	9/A	10/A
11/C	12/C	13/D	14/B	15/B	16/B	17/A	18/B	19/B	20/A



5. End Module 2022

1. α -D-glucose $\rightleftharpoons$ β -D-glucose (with a change in optical rotation over time) represents which of the following phenomena?
 - a. Optical isomerism
 - b. Mutarotation
 - c. Epimerization
 - d. D and L isomerism
2. Which of the following is not a polymer of glucose?
 - a. Amylose
 - b. Inulin
 - c. Cellulose
 - d. Dextrin
3. Which of the following is a non-reducing sugar?
 - a. Isomaltose
 - b. Maltose
 - c. Lactose
 - d. Trehalose
4. What is the chemical name of guanine?
 - a. 2,4-Dioxy-5-methylpyrimidine
 - b. 2-Amino-6-oxypurine
 - c. 2-Oxy-4-aminopyrimidine
 - d. 2,4-Dioxypyrimidine
5. In mRNA, if the guanine percentage is 20%, what is the percentage of cytosine?
 - a. 20%
 - b. 30%
 - c. 40%
 - d. Cannot be estimated
6. Which one of the following changes in the coding region of an mRNA (caused by a point mutation) would result in translation of a protein identical to the normal protein?
 - a. UCA $\rightarrow$ UAA
 - b. UCA $\rightarrow$ CCA
 - c. UCA $\rightarrow$ UCU
 - d. UCA $\rightarrow$ ACA
7. What is the glycosaminoglycan which does not contain uronic acid?
 - a. Dermatan sulphate
 - b. Chondroitin sulphate
 - c. Keratan sulphate



- d. Heparan sulphate
8. Which of the following is an epimeric pair?
- Glucose and fructose
 - Glucose and galactose
 - Galactose and mannose
 - Lactose and maltose
9. In glucose the orientation of the -H and -OH groups around the carbon atom 5 adjacent to the terminal primary alcohol carbon determines which of the following
- D or L series
 - Dextro or levorotatory
 - α and β anomers
 - Epimers
10. What is the polysaccharide used in assessing the glomerular filtration rate (GFR)?
- Glycogen
 - Agar
 - Inulin
 - Hyaluronic acid
11. In mammals, what is the major fat in adipose tissue?
- Phospholipid
 - Cholesterol
 - Sphingolipids
 - Triacylglycerol
12. In humans, what is a dietary essential fatty acid?
- Palmitic acid
 - Stearic acid
 - Butyric acid
 - Linoleic acid
13. The importance of phospholipids as a constituent of cell membrane is because they possess:
- Fatty acids
 - Both polar and nonpolar groups
 - Glycerol
 - Phosphoric acid
14. Lactase enzyme belongs to which of the following enzyme classes?
- Hydrolase
 - Lyase
 - Transferase



d. Epimerase

15. In hereditary non-polyposis colorectal cancer (HNPCC), what is the defect in the patient?

- a. Inability to remove thymine dimers from DNA
- b. Inability to remove the base U from DNA
- c. Low DNA ligase activity
- d. Inability to correct mismatched bases in newly synthesized DNA

16. A mRNA produced from a fragment of DNA has the sequence AAUUGGCU. The sequence of the non-template strand in the DNA that gave rise to this sequence is which one of the following?

- a. AGCCAATT
- b. AAUUGGCU
- c. AATTCCGA
- d. TTAACCGA

17. Which of the following chromatin states will be accompanied by activated transcription?

- a. Increased methylation and increased acetylation
- b. Increased methylation and decreased acetylation
- c. Decreased methylation and decreased acetylation
- d. Decreased methylation and increased acetylation

18. Telomerase is normally active in which of the following cells?

- a. Germ cells
- b. Liver cells
- c. Pancreatic cells
- d. Cancer cells

19. Cholesterol is the precursor for the biosynthesis of which of the following?

- a. Fatty acid
- b. Prostaglandins
- c. Bile acids
- d. Sphingomyelin

20. Which phospholipid acts as surfactant?

- a. Cephalin
- b. Phosphatidylinositol
- c. Lecithin
- d. Phosphatidylserine

21. When choline of lecithin is replaced by ethanolamine, what is the product of such process?



- a. Sphingomyelin
 - b. Cephalin
 - c. Plasmalogens
 - d. Lysolecithin
22. What is meant by reversible non-competitive enzyme inhibition activity?
- a. V_{max} is increased
 - b. K_m is increased
 - c. K_m is decreased
 - d. Concentration of active enzyme is reduced
23. LDH1 and LDH2 are elevated in which of the following conditions?
- a. Myocardial infarction
 - b. Liver disease
 - c. Kidney disease
 - d. Brain disease
24. Which of the following is not a component of eukaryotic mRNA?
- a. Poly A tail
 - b. Introns
 - c. Exons
 - d. 5' methyl guanosine capping
25. What is the function of the 5'-3' exonuclease activity of DNA polymerase I?
- a. Proof-reading activity and replacement of mismatched nucleotides during replication
 - b. Removal and replacement of mismatched nucleotides after replication
 - c. Removal of RNA primers during replication
 - d. Replacement of RNA primers during replication
26. The supercoils generated during DNA unwinding are relieved by which of the following enzymes?
- a. Polymerase
 - b. Topoisomerase
 - c. Telomerase
 - d. Helicase
27. CAAT box is present in which of the following?
- a. Prokaryotic promoters 10 bp upstream of transcription start site
 - b. Prokaryotic promoters 35 bp upstream of transcription start site
 - c. Eukaryotic promoters 25 bp upstream of transcription start site
 - d. Eukaryotic promoters 70-80 bp upstream of transcription start site
28. What is the CK isoform present in cardiac muscle?
- a. BB and MB



- b. MM and MB
- c. BB only
- d. MB only

29. The first amino acid tRNA which initiates translation in prokaryotes is

- a. Methionyl tRNA
- b. Formyl methionyl tRNA
- c. Tyrosine tRNA
- d. Alanyl tRNA

30. Although each specific tRNA differs from the others in its sequence of nucleotides, all tRNA molecules contain a base-paired stem that terminates in the sequence CCA at

- a. 3' terminus
- b. 5' terminus
- c. Anticodon arm
- d. Alanyl tRNA

31. Which of the following statements is true about histones?

- a. They are basic proteins so they can interact with positively charged DNA
- b. They are basic proteins that carry negative charge
- c. They are basic proteins so they can interact with negatively charged DNA
- d. They are basic proteins that have no role in regulation of gene expression

32. Which of the following is a Sulphur containing amino acid?

- a. Methionine
- b. Leucine
- c. Valine
- d. Asparagine

Answer Key

1/B	2/B	3/D	4/B	5/D	6/C	7/C	8/B	9/A	10/C
11/D	12/D	13/B	14/A	15/D	16/C	17/D	18/A	19/C	20/C
21/C	22/B	23/A	24/B	25/A	26/B	27/D	28/D	29/B	30/A
31/C	32/A								

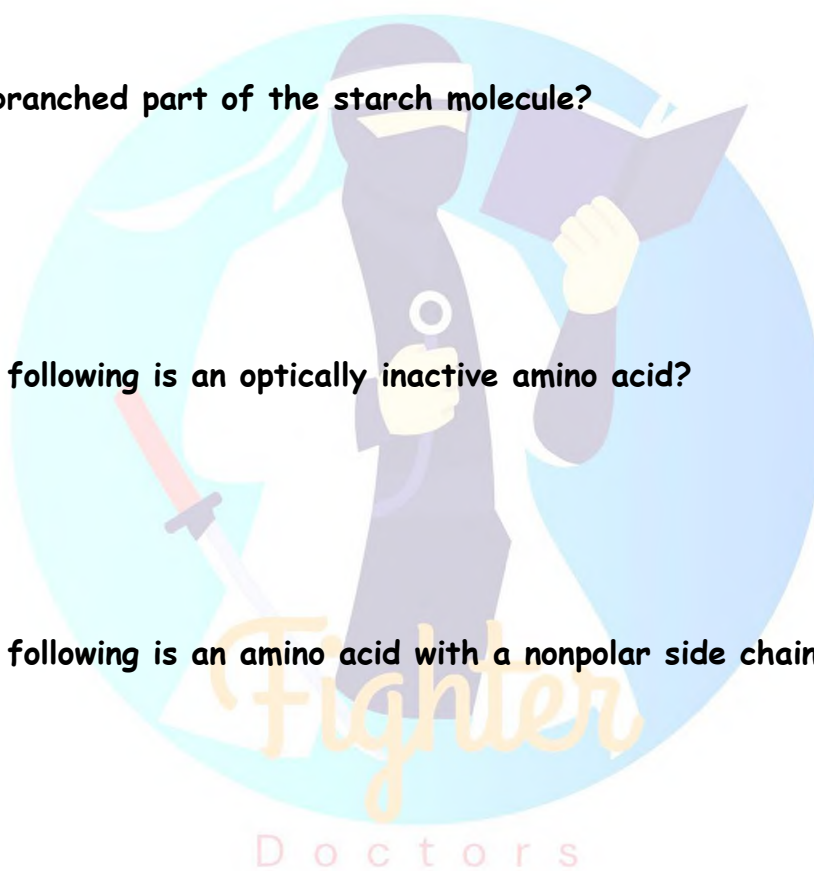


6. End Semester 2022

1. Isomers differing as a result of variations in configuration of the OH and H on carbon atoms 2, 3 and 4 of glucose are known as
 - a. Epimers
 - b. Anomers
 - c. Optical isomers
 - d. Stereoisomers
2. A disaccharide formed by a 1,1-glycosidic linkage between their monosaccharide units is
 - a. Lactose
 - b. Maltose
 - c. Trehalose
 - d. Sucrose
3. Keratan sulphate I is found in abundance in which of the following tissues?
 - a. Heart muscle
 - b. Liver
 - c. Adrenal cortex
 - d. Cornea
4. What is the carbon atom that becomes asymmetric when the straight chain form of a monosaccharide changes into ring form?
 - a. Anomeric carbon atom
 - b. Epimeric carbon atom
 - c. Isomeric carbon atom
5. The highest concentrations of fructose are found in which of the following?
 - a. Aqueous humor
 - b. Vitreous humor
 - c. Synovial fluid
 - d. Seminal fluid
6. Which of the following does not require the presence of a palindromic sequence?
 - a. Methylation sites during replication
 - b. Denatured site in gene promoter
 - c. Termination of gene transcription
 - d. Sites for restriction endonucleases
7. Which of the following statements is NOT true about tRNA?
 - a. It is the most heterogeneous type of RNA
 - b. It is the smallest type of RNA
 - c. It can read more than one codon



- d. The initiator is formylated in prokaryotes
8. Which of the following is NOT presented as an unwinding element?
- a. DUE region
 - b. Pribnow box
 - c. Hogness box
 - d. CAAT box
9. Oxidation of the aldehyde group of glucose by dilute nitric acid produces which of the following compounds?
- a. Glucuronic acid
 - b. Gluconic acid
 - c. Glucaric acid
 - d. Galactic acid
10. What is the branched part of the starch molecule?
- a. Glucose
 - b. Amylopectin
 - c. Isomaltose
 - d. Amylose
11. Which of the following is an optically inactive amino acid?
- a. Glycine
 - b. Serine
 - c. Threonine
 - d. Valine
12. Which of the following is an amino acid with a nonpolar side chain?
- a. Serine
 - b. Valine
 - c. Asparagine
 - d. Threonine
13. What is the lipoprotein associated with a high incidence of coronary atherosclerosis?
- a. LDL
 - b. VLDL
 - c. IDL
 - d. HDL
14. Defects caused by UV radiation are corrected by which of the following repair systems?
- a. Base excision repair
 - b. Nucleotide excision repair





- c. Mismatch repair
- d. Homologous recombination

15. Which of the following statements represents a defining character for RNA polymerase compared to DNA polymerase?

- a. It has 3'-5' polymerase activity
- b. It has 5'-3' polymerase activity
- c. It needs a primer
- d. It lacks proofreading activity

16. How many hydrogen bonds are there between guanine and cytosine?

- a. One
- b. Two
- c. Three
- d. Four

17. DNA unfolding is monitored by measuring the absorption of UV light at which of the following wavelengths?

- a. 250 nm
- b. 260 nm
- c. 270 nm
- d. 280 nm

18. Which of the following RNA polymerase subunits is required for its polymerase activity?

- a. α subunit
- b. β subunit
- c. β' subunit
- d. σ subunit

19. At isoelectric pH, an amino acid exists in which of the following states?

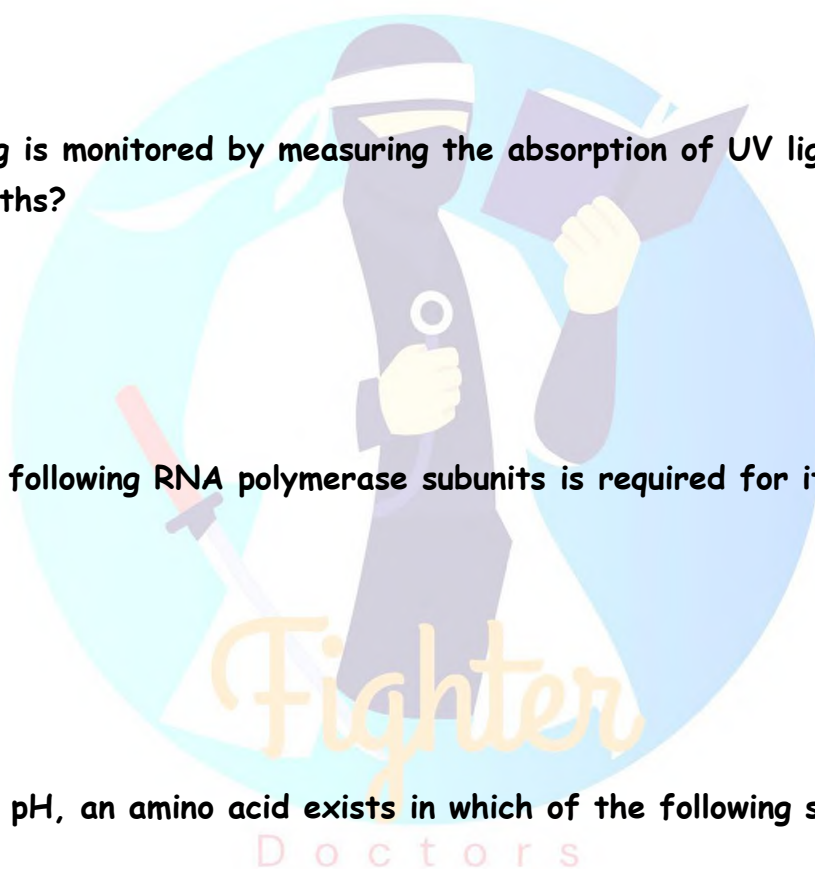
- a. Anion
- b. Cation
- c. Zwitterion
- d. Undissociated molecule

20. Primary structure of a protein is formed by which of the following bonds?

- a. Hydrogen bonds
- b. Peptide bonds
- c. Disulphide bonds
- d. Hydrophobic bonds

21. Dietary fibers are rich in which of the following?

- a. Cellulose





- b. Glycogen
 - c. Starch
 - d. Proteoglycans
22. Depot fat is composed mainly of cholesterol.
- a. True
 - b. False
23. DNA polymerase belongs to which of the following classes of enzymes?
- a. Transferases
 - b. Isomerases
 - c. Lyases
 - d. Ligases
24. Depurinated sites on DNA are corrected by which of the following factors?
- a. ABC excinuclease
 - b. DNA glycosylase
 - c. AP endonuclease
 - d. Mut proteins
25. In lac operon, which of the following events is active in the absence of glucose and presence of lactose?
- a. Allolactose binds to repressor protein
 - b. Inhibition of gene i expression
 - c. The level of cAMP is decreased
 - d. The operator site is occupied
26. Histone deacetylase enzyme activates transcription in eukaryotes.
- a. True
 - b. False
27. Which of the following DNA forms is the broadest type?
- a. A form
 - b. B form
 - c. D form
 - d. Z form
28. At T_m , of the total DNA is unwound.
- a. 100%
 - b. 75%
 - c. 50%
 - d. 25%
29. Which of the following statements is correct about synthesis of the lagging strand?



- a. It is synthesized continuously in 5'-3' direction
- b. It is synthesized specifically by DNA polymerase I only
- c. It requires multiple primers in their synthesis
- d. Its DNA template runs in the direction of 3'-5'

30. Enzymes change the equilibrium of the reaction.

- a. True
- b. False

31. Non-competitive inhibitor binds to the active site of the enzyme.

- a. True
- b. False

32. Glycerol is the type of alcohol present in wax.

- a. True
- b. False

Answer Key

1/A	2/C	3/D	4/A	5/D	6/B	7/A	8/D	9/B	10/B
11/A	12/B	13/A	14/B	15/D	16/C	17/B	18/B	19/C	20/B
21/A	22/B	23/D	24/C	25/A	26/B	27/A	28/C	29/C	30/B
31/B	32/B								

Fighter

Doctors



7. Formative Exam 2023

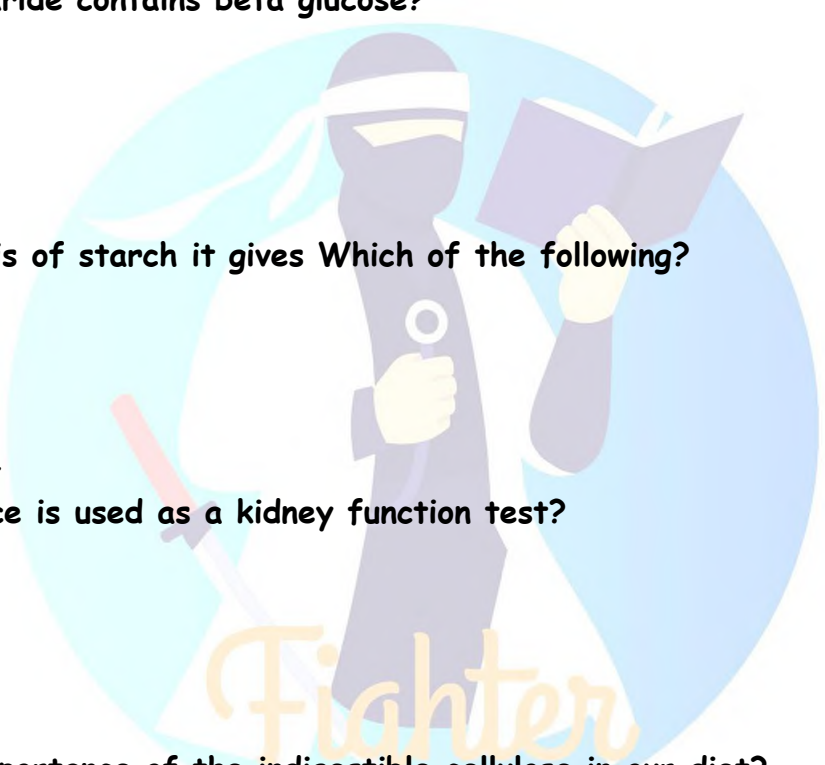
1. What is the most abundant form of RNA in the cell?
 - a. mRNA
 - b. rRNA
 - c. tRNA
 - d. snRNA
2. Which of the following is a non-reducing disaccharide?
 - a. Starch
 - b. Maltose
 - c. Sucrose
 - d. Glycogen
3. Hardening of oils nutritional value
 - a. Increases
 - b. Decreases
4. How many asymmetric carbon atoms are there in D-glucose?
 - a. Two
 - b. Three
 - c. Four
 - d. Five
5. Basic amino acids are positively charged at isoelectric point.
 - a. True
 - b. False
6. Which of the following statements is TRUE about synthesis of the lagging strand?
 - a. It is synthesized continuously
 - b. It requires one primer
 - c. It grows towards the replication fork
 - d. It is synthesized in 5'-3' direction

Answer Key

1/B	2/C	3/A	4/C	5/B	6/D				
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8. END SEMESTER 2024

1. How many carbon atoms are present in Dihydroxyacetone?
 - a. 0
 - b. 1
 - c. 2
 - d. 3
 2. Which of the following describes sucrose as a non-reducing disaccharide?
 - a. It presents in both alpha form and beta form
 - b. It's formed of alpha fructose and beta glucose linked by $\alpha 1 \rightarrow \beta 2$ linkage
 - c. It can form characteristic osazone crystals
 - d. after hydrolysis the angle of rotation becomes inverted
 3. Which disaccharide contains beta glucose?
 - a. Maltose
 - b. Isomaltose
 - c. Cellobiose
 - d. Trehalose
 4. After hydrolysis of starch it gives Which of the following?
 - a. Alpha maltose
 - b. Beta maltose
 - c. Beta glucose
 - d. Alpha isomaltose
 5. Which substance is used as a kidney function test?
 - a. Inulin
 - b. Chitin
 - c. agar
 - d. dextrose
 6. What is the importance of the indigestible cellulose in our diet?
 - a. It's hydrolysed into glucose which is absorbed in the blood
 - b. It's hydrolysed into cellobiose which is absorbed in the blood
 - c. It's hydrolysed into cellobiose which protects against constipation
 - d. As it is making stool soft and preventing constipation
- 

1.d | 2.b | 3.c | 4.a | 5.a | 6.d



12. Lung surfactant is formed mainly from the phospholipid dipalmitoyl lecithin. Which of following describes the structure of dipalmitoyl lecithin?

- a. Glycerol + 2 palmitic acids+ phosphoryl choline
- b. Glycerol + 2 palmitic acids+phosphoryl ethanolamine
- c. Sphingosine + 2 palmitic acids+ phosphoryl choline
- d. Sphingosine + 2 palmitic acids + phosphoryl ethanolamine 2 palmitic acid

13. Which of the following is a derived lipid?

- a. Carotenes
- b. Waxes
- c. Vitamen D3
- d. Ceramide

14. Which of the following amino acids would more possibly be on the surface of a membrane protein?

- a. Aspartate
- b. Lysine
- c. Valine
- d. Serine

15. At the isoelectric point, the amino acid carry?

- a. Cation
- b. Anion
- c. Un charged
- d. Dipolar

16. Which of the following describes the forces that stabilize the tertiary level of protein structure?

- a. Hydrogen bouds between components of the peptide bond is one of these forces
- b. Interaction between two cysteines forms disulfide bridges
- c. Interaction between valine, leucine, and isoleucine forms ionic interactions
- d. Interaction between aspartate and lysine forming hydrophobic interactions

17. Which level of protein structure describes how different protein subunits bind together?

- a. 1ry structure
- b. 2ry structure
- c. 3ry structure
- d. 4ry structure

18. Which of the following is considered a derived protein?

- a. Peptone
- b. Albumin
- c. Histone
- d. casein

12.a | 13.c | 14.a | 15.d | 16.b | 17.d | 18.a



19. Which of the following bonds stabilize the primary level of protein structure?

- a. Hydrogen bonds
- b. Peptide bonds
- c. Ionic bonds
- d. Disulfide bonds

20. Phosphorylation and dephosphorylation of the enzymes are known as which of the in terms?

- a. Feedback inhibition
- b. long term regulation
- c. Allosteric regulation
- d. Covalent modification

21. A 60-year-old man is diagnosed with hypertension and is started on an angiotensin-converting enzyme (ACE) inhibitor. ACE inhibitors bind to the active site of the enzyme. What be inhibition does this describe?

- a. Competitive inhibition
- b. Uncompetitive inhibition
- c. Non-competitive inhibition
- d. Feedback inhibition

22. The initial rate of an enzyme-catalyzed reaction increases with which of the following?

- a. Decreasing enzyme concentration
- b. Increasing the product concentration
- c. Increasing the substrate concentration
- d. Decreasing the pH or the temperature

23. Which of the following events occurs in response to defective mut proteins?

- a. Inability to correct pyrimidine dimers.
- b. Insensitivity to ultraviolet rays.
- c. Development of cancer in tissues with high rate of regeneration.
- d. Inability to correct deaminated cytosine.

24. Which of the following statements is correct regarding RNA synthesis?

- a. Transcription requires a primer
- b. DNA replication is more accurate than RNA transcription.
- c. Proofreading is required in mRNA synthesis.
- d. Eukaryotic mRNA is polycistronic

25. Histone acetylation makes which of the following statements?

- a. Enhances promoter activity
- b. Makes DNA more compact
- c. Makes histones carry +ve charge
- d. Denaturation of histones

19.b | 20.d | 21.a | 22.c | 23.c | 24.b | 25.a



26. What is the expected change of the apparent K_m that occurs in the presence of a noncompetitive inhibitor?

- a. Increases
- b. Decreases
- c. Remains unchanged
- d. Becomes zero

27. What is the functional form of the conjugated enzymes?

- a. Apoenzyme
- b. Coenzyme
- c. Cofactor
- d. Holoenzyme

28. Pepsinogen belongs to which of the following enzymatic categories?

- a. Zymogens
- b. Allosteric enzymes
- c. Isoenzymes
- d. Holoenzymes

29. Which of the following events occurs in response to point mutation in the 1st base of codon?

- a. It will lead to change in the reading frame of amino acid sequences
- b. There will be no change in the amino acid composition
- c. It will lead to sudden stop in mRNA translation
- d. It will lead to a change in the amino acid composition

30. In lac operon, which of the following events occurs in the presence of lactose and absence of glucose?

- a. The CAP binding site is occupied with cAMP-CAP complex.
- b. The expression of structural genes is inhibited.
- c. The repressor binds to the operator site.
- d. The expression of LacI gene is activated by the presence of allolactose.

31. Which of the following cis-acting elements is the initial binding site for RNA polymerase enzyme?

- a. Terminator
- b. Introns
- c. Promoter
- d. Exons

32. Which of the following statements best describes introns?

- a. Non-coding regions removed during RNA splicing
- b. Coding regions of the gene
- c. Regulatory sequences upstream of a gene
- d. Sequences that signal termination of transcription

26.c | 27.d | 28.a | 29.d | 30.a | 31.c | 32.a



33. What type of mutation does not change the amino acid sequence of a protein?

- a. Missense mutation
- b. Frameshift mutation
- c. Nonsense mutation
- d. Silent mutation

34. A patient undergoing chemotherapy with a topoisomerase inhibitor, This drug likely works by which of the following mechanisms?

- a. Preventing the unwinding of DNA strands
- b. Inhibiting the addition of nucleotides to the leading strand
- c. Blocking the formation of RNA primers
- d. Preventing the joining of Okazaki fragments

35. A 25-year-old male presents with fatigue, pallor, and shortness of breath. His medical history includes episodes of hemolytic anemia, particularly during times of stress or infection. Laboratory studies reveal a hemoglobin electrophoresis pattern consistent with sickle cell disease. Molecular testing identifies a mutation in the B-globin gene. Which type of genetic mutation is most likely responsible for the patient's condition?

- a. Frameshift mutation
- b. Missense mutation
- c. Nonsense mutation
- d. Silent mutation

36. Which enzyme is responsible for unwinding the DNA helix during replication?

- a. DNA polymerase
- b. Helicase
- c. Ligase
- d. Primase

37. What statement describes the synthesis of the lagging strand during DNA replication?

- a. Continuously
- b. Discontinuously in Okazaki fragments
- c. Without the need for a primer
- d. In the 3' to 5' direction

38. The fidelity of DNA replication is maintained by the proofreading activity of DNA polymerase which of the following activities?

- a. 3' to 5' exonuclease activity
- b. 5' to 3' exonuclease activity
- b. 5' to 3' polymerase activity
- d. 3' to 5' polymerase activity



39. A researcher studying cancer cells finds that certain mutations cause an increase in G-C content of the DNA. What is the consequence of increased G-C content on the DNA structure?

- a. Increased number of hydrogen bonds, making the DNA more stable
- b. Increased flexibility of the DNA helix
- c. Decreased number of hydrogen bonds, making the DNA less stable
- d. No significant effect on DNA stability

40. During denaturation of DNA, which bonds are disrupted?

- a. Phosphodiester bonds
- b. Glycosidic bonds
- c. Hydrogen bonds
- d. Peptide bond

Answer Key:

39.a | 40.c



Written

1. Compare between the followings

	Homopolysaccharides	Heteropolysaccharides
Definition		
Examples		

	Liver glycogen	Muscle glycogen
Function		
Relation to blood glucose		

	Enantiomers	Epimers
Definition		
Examples		

2. at the opposite structure of two triglyceride molecules and answer the following questions:(2 marks)

[توجد صورة لجزيئين من الـ Triglyceride]

Is triglyceride 1 mixed or simple?

Is the fatty acid at position 3 of triglyceride 1 omega 3 or omega 6?

What is the name of the fatty acid in triglyceride 2?

Explain why triglyceride 1 has a higher iodine value than triglyceride 2?

3. Mention the importance of cholesterol for the human body? (2 marks)

.....

.....

4. Explain with examples hydrophilic uncharged amino acids? (2 marks)

.....

.....

5. Describe the process of protein denaturation (Definition & effects on proteins) (3 marks)

.....

.....

6. Describe what is meant by quaternary level of protein structure and enumerate its stabilizing force

.....

.....

7. Enumerate the basic steps involved in base excision repair. (2 marks)

.....

.....



8. Categorize the following events in the following table (2 marks):

1. Methylation of CG island
2. Activation of histone deacetylase.
3. Activation of histone acetyl transferase.
4. Activation of DNA demethylase

Euchromatin Heterochromatin

9. The following picture describes the relationship between the enzyme action and the substrate Describe this relationship and demonstrate how to increase the rate of the reaction in figure (panel)(2 marks)

[توجد صورتان توضحان الإنزيمات والركائز (Substrate and Enzyme)]

10. Compare between competitive and non-competitive inhibitors. (2 marks)

Side of comparison Competitive Non-competitive

11. What happens if: (2 marks)

- a. Deletion of a base in the gene sequence.
- b. There is a mutation in mismatch repair system
- c. The exchange of a base in the coding area of a gene
- d. A point mutation occurs in UGA changed into UAA

12. Which of the RNAs show the following criteria: (2 marks)

- a. accounts for the largest percentage of cellular RNA?
- b. carries amino acids to ribosomes?
- c. is extensively modified only in eukaryotes?
- d. is a ribozyme in translation?

13. What is meant by cis-acting and trans-acting elements? With examples (2 marks)

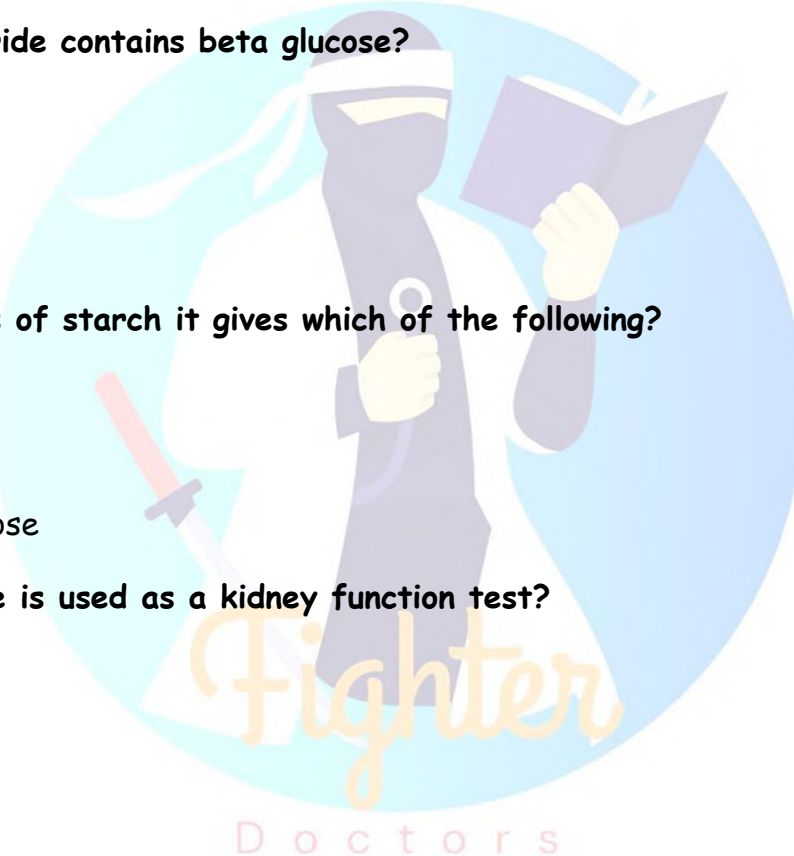
14. Describe the basic structural unit of chromatin structure. (2 marks)

15. Compare between RNA and DNA polymerase enzymes. (2marks)

DNA polymerase	RNA polymerase



9. End Semester 2025

1. How many carbon atoms are present in Dihydroxyacetone?
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 - b. 1
 - c. 2
 - d. 3
 2. Which of the following describes sucrose as a non-reducing disaccharide?
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 - a. It's hydrolysed into glucose which is absorbed in the blood
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- c. Hydrogen bonds
- d. Peptide bond

Answer Key

1/D	2/B	3/C	4/A	5/A	6/D	7/A	8/C	9/A	10/D
11/B	12/D	13/A	14/B	15/D	16/A	17/C	18/C	19/B	20/A
21/C	22/D	23/A	24/D	25/A	26/C	27/A	28/D	29/A	30/B
31/B	32/B	33/A	34/A	35/C					

Fighter

Doctors



10. End Module 2026 دور اول

1. Which of the following statements best describes glucose?

- a. It is a C-4 epimer of galactose.
- b. It is a ketose and usually exists as a furanose ring in solution.
- c. It is produced from dietary starch by the action of α -amylase.
- d. It is utilized in biological systems only in the L-isomeric form.

2. Which of the following best describes starch?

- a. It cannot be digested by humans.
- b. It is composed of unbranched amylose and branched amylopectin.
- c. It is composed of unbranched amylose and unbranched amylopectin.
- d. It is a D-glucose polymer.

3. Ionic (Electrostatic) bond can be formed between side chains of which of the following?

- a. Alanine & leucine
- b. Lysine & aspartate
- c. Leucine & valine
- d. Aspartate & glutamate

4. What is the protein bond that is stable even in denaturation?

- a. Peptide bond
- b. H-bond
- c. Disulphide bond
- d. Hydrophobic interaction

5. Which of the following statements is characteristic for cellulose compared to starch?

- a. It is a glucose polymer.
- b. It can be digested by humans.
- c. It is the main source of carbohydrates in diet.
- d. It can't be used as a source of energy.

6. Methionine & threonine are classified as which of the following?

- a. Phenol containing amino acids
- b. Acidic amino acids
- c. Basic amino acids
- d. Essential amino acids

7. Which is considered an optically inactive amino acid?

- a. Serine
- b. Valine
- c. Glycine
- d. Alanine

8. Which amino acid contains an imino group?

- a. Serine



- b. Valine
- c. Proline
- d. Alanine

9. In prion diseases, what is characteristic for the infectious form of prion protein (PrP^{Sc})?

- a. Differs from PrP^C in primary amino acid sequence
- b. Is more susceptible to proteolytic degradation
- c. Contains a higher proportion of β -pleated sheet structure
- d. Is encoded by a different gene than PrP^C

10. Which of the following protein abnormalities is most characteristic of Alzheimer disease?

- a. Conversion of α -helices into β -sheets in prion protein
- b. Accumulation of neurofibrillary tangles composed of hyperphosphorylated tau protein
- c. Formation of Lewy bodies composed of α -synuclein
- d. Accumulation of misfolded prion protein (PrP^{Sc}) in neurons

11. In uncontrolled diabetes, which sugar alcohol tends to accumulate in tissues like the lens, contributing to complications (e.g., cataracts and neuropathy)?

- a. Mannitol
- b. Xylitol
- c. Sorbitol
- d. Ribitol

12. Which polysaccharide consists of β -(1 $\rightarrow$ 4)-linked glucose units?

- a. Amylose
- b. Glycogen
- c. Cellulose
- d. Amylopectin

13. Which of the following sugars lacks reducing properties?

- a. Lactose
- b. Fructose
- c. Maltose
- d. Sucrose

14. Hydrolysis of sucrose by the enzyme sucrase (invertase) yields which monosaccharides?

- a. Glucose and galactose
- b. Two molecules of glucose
- c. Fructose and galactose
- d. Glucose and fructose

15. Which of the following is an anomeric pair?

- a. D-glucose and L-glucose
- b. D-glucose and D-fructose



c. α -D-glucose and β -D-glucose

d. α -D-glucose and α -L-glucose

16. What is the general formula of dextrose?

a. $C_5H_{10}O_5$

b. $C_6H_{10}O_5$

c. $C_6H_{10}O_6$

d. $C_6H_{12}O_6$

17. Spermatozoa in seminal fluid utilizes which of the following sugars for its metabolism?

a. Glucose

b. Fructose

c. Sucrose

d. Mannose

18. Which is a disaccharide linked by a β -1,4 glycosidic linkage?

a. Lactose

b. Sucrose

c. Cellulose

d. Maltose

19. Which level of protein structure is defined by the linear sequence of amino acids?

a. Primary

b. Secondary

c. Tertiary

d. Quaternary

20. In a protein engineering experiment, a scientist replaces all the internal leucine residues (hydrophobic) with glutamate (negatively charged). What is the most likely effect on the protein's tertiary structure?

a. Protein misfolding and loss of function.

b. Increased stability due to new salt bridges.

c. Conversion of the protein from globular to fibrous.

d. No significant change.

21. Considering the isoelectric point (I.P), what would be the net charge of a protein with an I.P of 8.5 when placed in a solution at pH 7.4?

a. Positive

b. Negative

c. Neutral (Zwitterion)

d. Cannot be determined

22. Which statement describes the use of glycogen during starvation?

a. Liver glycogen is conserved for muscle use

b. Muscle glycogen is converted to glucose for the brain

c. Liver glycogen is broken down to maintain blood glucose



d. Muscle glycogen increases due to lack of energy use

23. Which of the following is a set of hydrophobic amino acids?

- a. Aspartic - valine - isoleucine
- b. Arginine - lysine - methionine
- c. Hydroxyproline - histidine - Glutamine
- d. Valine - methionine - isoleucine

24. What is a key difference between a native protein and its denatured form, regarding its ability to:

- a. Increase the viscosity of a solution.
- b. Act as an antigen and induce an immune response.
- c. Expose its hydrophobic groups to the solvent.
- d. Be composed of a sequence of amino acids.

25. The conversion of α -D-Glucose (+112 $\rightarrow$ +52.5) and β -D-glucose (+19 $\rightarrow$ +52.5) is defined by which of the following?

- a. Stereoisomerism
- b. Optical isomerism
- c. Mutarotation
- d. Epimerization

26. What is the product(s) of reduction of glucose?

- a. Dulcitol
- b. Sorbitol
- c. Mannitol
- d. Mannitol and sorbitol

27. The main cause of lactose intolerance is directly attributed to which of the following?

- a. An autoimmune reaction to casein in milk
- b. A deficiency of the enzyme sucrase
- c. A deficiency of the lactase enzyme
- d. Impaired galactose transport into enterocytes

28. Which of the following levels of protein structure describes the relationship among several subunits?

- a. Primary
- b. Secondary
- c. Tertiary
- d. Quaternary

29. Amino acids containing one amino group and one carboxylic group are known as ...

- a. Polar
- b. Acidic
- c. Neutral
- d. Basic



30. Which of the following statements is TRUE regarding invert sugar?

- a. It is the hydrolytic product of lactose sugar.
- b. It is a non-reducing mixture.
- c. It exhibits mutarotation.
- d. It deviates the angle of polarized light to the right.

31. If a DNA sample from a species contains 20% Thymine, what is the percentage of Guanine?

- a. 20%
- b. 30%
- c. 40%
- d. 60%

32. What is the main function of histone H1?

- a. Forming the core of the nucleosome.
- b. Binding to linker DNA.
- c. Acetylating other histone proteins.
- d. Serving as the scaffold for the 30nm fiber.

33. Which of the following DNA forms is a left-handed helix?

- a. A form
- b. B form
- c. Z form
- d. A and B form

34. In a DNA sample, which result would be consistent with Chargaff's findings?

- a. $A = G$; $G = T$
- b. $A + G = C + T$
- c. $A + T = G + T$
- d. $A + C = G + C$

35. Which of the following represents the chemical structure of adenine?

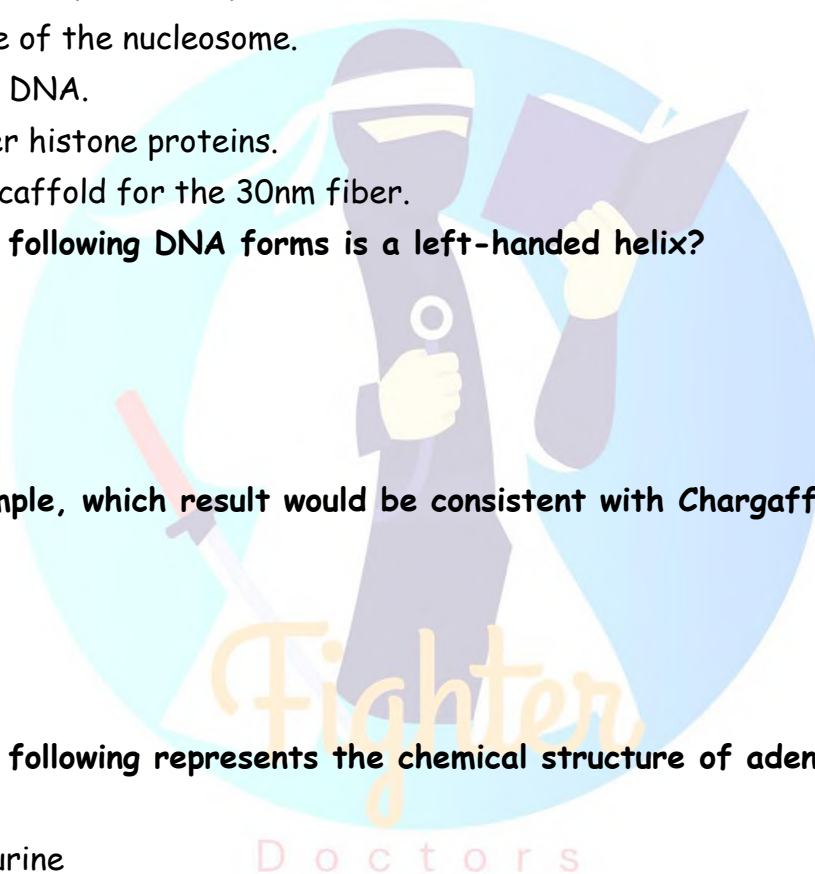
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- b. 2-amino-6-oxypurine
- c. 6-oxypurine
- d. 2,6-dioxypurine

36. A mutation in the gene for the DnaA protein in E. coli that renders it unable to bind ATP would most directly affect which stage of replication?

- a. Elongation
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37. A DNA sequence reads 5'-ATCG-3' on one strand. What is the sequence of the complementary strand?

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39. What is the enzyme that joins the small fragments of the lagging strand together into a continuous strand?

- a. DNA gyrase
- b. DNA ligase
- c. DNA polymerase
- d. Primase

40. Which of the following statements is true about the nucleotide structure?

- a. The pentose sugar is present in the a configuration
- b. The pyrimidine nitrogenous base is a bicyclic structure.
- c. The glycosidic linkage is formed between N1 of purine and C1 of sugar.
- d. The phosphate group is attached to carbon 5 on the pentose sugar.

41. Which of the following statements is true about the phosphodiester bond?

- a. It is only present in the structure of DNA not RNA.
- b. It is responsible for formation of the DNA double helical structure.
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42. Which of the following is NOT true about the DNA double helical structure?

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- b. Introns
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50. A mutation that inactivates the 3' to 5' exonuclease activity of DNA polymerase would result in?

- a. A slower rate of DNA synthesis.
- b. An increased rate of mutation.
- c. Inability to remove RNA primers.
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51. Which of the following RNAs is coding?

- a. mRNA
- b. tRNA
- c. rRNA
- d. snRNA

52. Which of the following correctly lists the order of enzyme action at the replication fork?

- a. Helicase → Primase → DNA Polymerase → Ligase
- b. Primase → Helicase → DNA Polymerase → Ligase
- c. DNA Polymerase → Primase → Helicase → Ligase
- d. Ligase → Primase → DNA Polymerase → Helicase

53. In eukaryotes, DNA replication occurs during which phase of the cell cycle?

- a. G1 phase
- b. S phase
- c. G2 phase
- d. M phase

54. Which of the following enzymes requires a palindromic sequence?

- a. Rho protein
- b. Dam methylase
- c. Reverse transcriptase
- d. Primase

55. Which of the following RNAs is rich in intrachain base pairing?

- a. mRNA
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56. Which of the following is a function of the poly-A tail in eukaryotic mRNA?

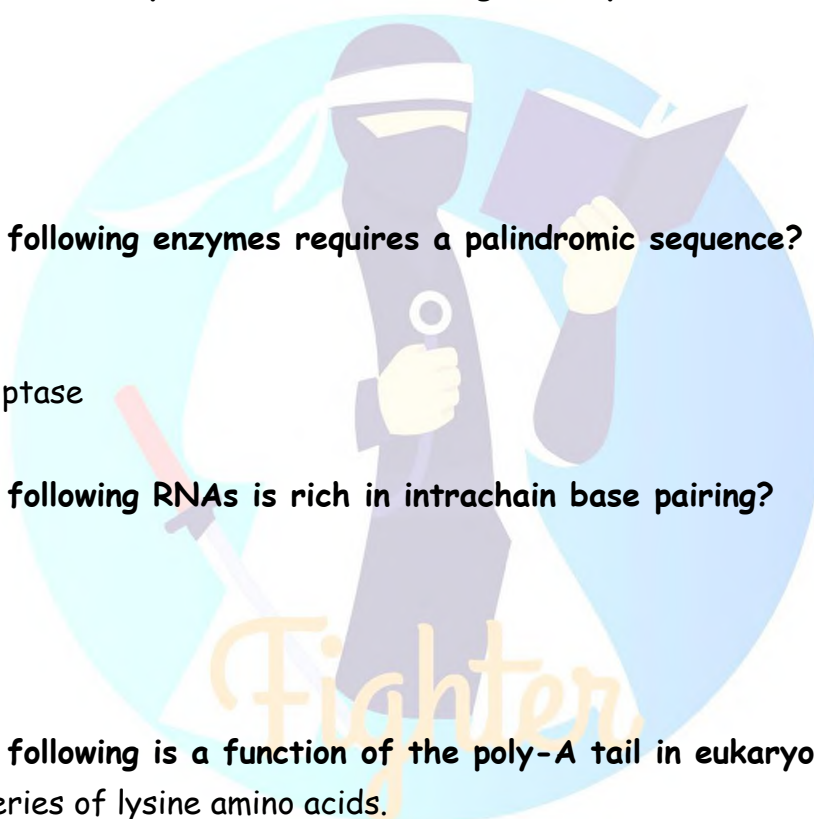
- a. It codes for a series of lysine amino acids.
- b. It facilitates the initiation of transcription.
- c. It protects the mRNA from degradation.
- d. It is the site for ribosome binding.

57. What is a key difference between prokaryotic and eukaryotic mRNA?

- a. Prokaryotic mRNA is monocistronic.
- b. Eukaryotic mRNA has a 5' cap and a 3' poly-A tail.
- c. Prokaryotic mRNA contains introns.
- d. Eukaryotic mRNA is translated while being transcribed.

58. What is the main role of sigma (σ) factor in prokaryotic transcription?

- a. Catalyzing the formation of phosphodiester bonds.
- b. Terminating transcription.
- c. Recognizing and binding to the promoter region.





d. Adding the 5' cap to the mRNA.

59. What is the amino acid commonly entering into the structure of histones?

- a. Arginine
- b. Aspartate
- c. Glutamate
- d. Alanine

60. Which deamination indicated below would lead to a mutation in a resulting protein if not repaired?

- a. T to U
- b. C to U
- c. G to A
- d. U to C

Answer Key

1/C	2/B	3/B	4/A	5/D	6/D	7/C	8/C	9/C	10/B
11/C	12/C	13/D	14/D	15/C	16/D	17/B	18/A	19/A	20/A
21/A	22/C	23/D	24/B	25/C	26/B	27/C	28/D	29/C	30/C
31/B	32/B	33/C	34/B	35/A	36/C	37/C	38/B	39/B	40/D
41/C	42/D	43/D	44/D	45/C	46/C	47/C	48/B	49/B	50/B
51/A	52/A	53/B	54/B	55/B	56/C	57/B	58/C	59/A	60/C

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11. End Module 2026

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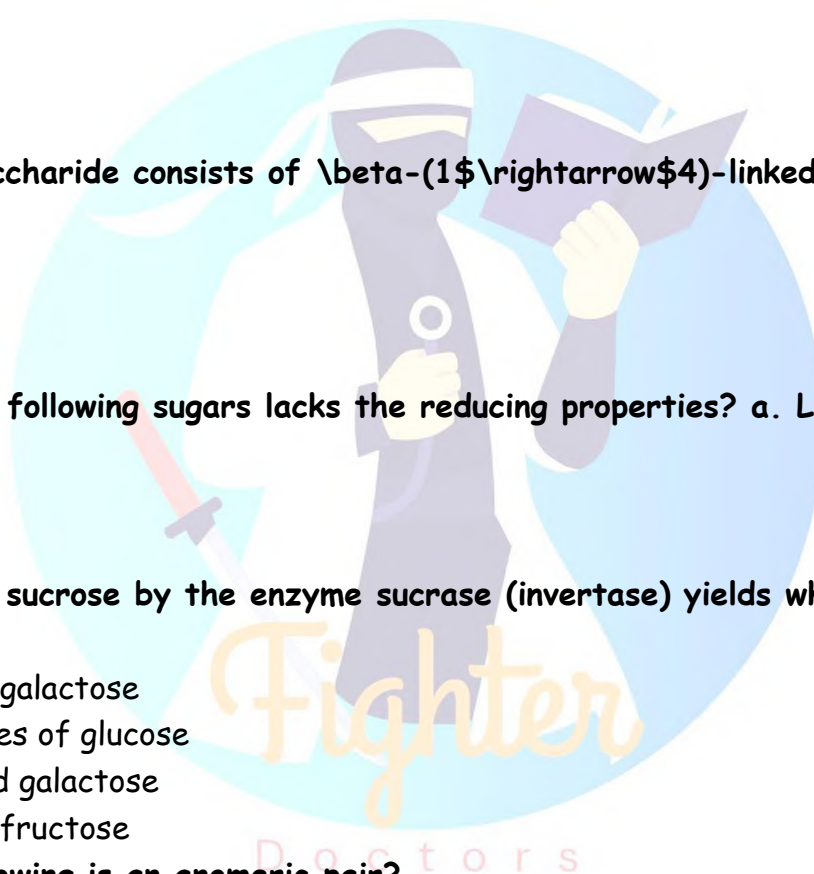
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- c. alpha-D-glucose and beta-D-glucose
- d. alpha-D-glucose and alpha-L-glucose

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- a. $C_3H_6O_3$
- b. $C_4H_8O_4$
- c. $C_5H_{10}O_5$
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- It facilitates the initiation of transcription.
 - It protects the mRNA from degradation.
 - It is the site for ribosome binding.
57. What is a key difference between prokaryotic and eukaryotic mRNA?
- Prokaryotic mRNA is monocistronic.
 - Eukaryotic mRNA has a 5' cap and a 3' poly-A tail.
 - Prokaryotic mRNA contains introns.
 - Eukaryotic mRNA is translated while being transcribed.
58. What is the main role of sigma (σ) factor in prokaryotic transcription?
- Catalyzing the formation of phosphodiester bonds.
 - Terminating transcription.



- c. Recognizing and binding to the promoter region.
- d. Adding the 5' cap to the mRNA.

59. What is the amino acid commonly entering in the structure of histones? a. Arginine

- b. Aspartate
- c. Glutamate
- d. Alanine

60. Which deamination indicated below would lead to a mutation in a resulting protein if not repaired?

- a. T to U
- b. C to U
- c. G to A
- d. U to C





12. End Semester 2026 حکومی

1(Phenylketonuria usually is due to a defect in the enzyme phenylalanine hydroxylase and this results in the reduced synthesis of an aromatic hydroxyamino acid. The levels of which of the following amino acids is decreased in the body in this condition?

- a. Tyrosine
- b. Tryptophan
- c. Histidine
- d. Serine

2(Which of the following amino acids differs from the other in the nature of their amino group?

- a. Proline
- b. Glycine
- c. Leucine
- d. Alanine

3(In Sickle Cell Anemia, the nucleotide alteration leads to substitution of a glutamate residue with valine at the 7th position in the β -globin chain. Which of the following best describes this substitution?

- a. A hydroxy-amino acid is replaced by a simple amino acid
- b. A polar amino acid is replaced by a non-polar amino acid
- c. An aliphatic amino acid is replaced by an aromatic amino acid
- d. A dietary essential amino acid is replaced by a non-essential amino acid

4(A particular point mutation results in disruption of the α -helical structure in a segment of the mutant protein. What is the most likely change in the primary structure of the mutant protein?

- a. glutamate to aspartate.
- b. lysine to arginine.
- c. methionine to proline.
- d. valine to alanine.

5(Which of the following interactions is NOT involved in the stabilization of tertiary structure of proteins?

- a. Peptide bonds
- b. Hydrophobic interactions
- c. Hydrogen bonds
- d. Ionic interactions

6(A scientist observes that a normally soluble globular protein becomes insoluble and loses its biological function after being heated. What is the explanation for this case?

- a. The primary structure was hydrolyzed
- b. The protein denatured, exposing hydrophobic groups
- c. New peptide bonds were formed
- d. The protein gained a quaternary structure

7(Which of the following best describes a functional plasma enzyme?

- a. Its concentration in plasma is normally very low compared to tissues.
- b. It increases in plasma during tissue damage or disease.
- c. Its substrate is always present in the blood.
- d. It has no known physiological function in the blood.



- 8(In a patient with hereditary nonpolyposis colorectal cancer (HNPCC), DNA sequencing reveals a mutation in the hMSH2 gene. Which of the following replication errors would accumulate most prominently in this patient's cells? a. Thymine dimers from UV exposure
b. 8-oxoguanine lesions from oxidative damage
c. Mismatched bases
d. Double-strand breaks from ionizing radiation
- 9(A patient with myocardial infarction has elevated creatine kinase MB (CKMB). Which enzyme class does creatine kinase belong to? a. Oxidoreductase
b. Transferase
c. Hydrolase
d. Ligase
- 10(Which amino acid is a site for N-linked glycosylation of proteins? a. Serine
b. Threonine
c. Asparagine
d. Tyrosine
- 11(The "lock and key" model of enzyme action proposes that:
a. The active site changes shape to fit the substrate
b. The substrate and active site are complementary in shape from the start
c. Enzymes are only specific for one type of reaction, not substrate shape
d. Cofactors are necessary for substrate binding
- 12(Which of the following correctly describes a coenzyme?
a. An inorganic metal ion that activates enzymes
b. An organic helper molecule, often a vitamin derivative
c. The holoenzyme with its non-protein part
d. The protein portion of an enzyme
- 13(The iodine number of fat or oil is a direct measure of its a. Average chain length of fatty acids
b. Degree of unsaturation
c. Saponification value
d. free fatty acid content
- 14(Which lipid is a component of the inner mitochondrial membrane? a. Phosphatidylcholine
b. Phosphatidylethanolamine
c. Cardiolipin
d. Sphingomyelin
- 15(Denaturation of proteins involves loss of which of the following? a. Primary structure
b. Ionic interactions
c. Amino acid sequence
d. Peptide bonds
- 16(Phospholipids are an essential structural component of which of the following?
a. Liver glycogen stores
b. Lung surfactant
c. Hemoglobin in red blood cells
d. Bone matrix



- 17(Which of the following is a storage form of lipid in human body? a. Phospholipid
b. Glycolipid
c. Triacylglycerol
d. Cholesterol)
- 18(Which of the following is a monounsaturated fatty acid? a. Oleic acid
b. Linoleic acid
c. Linolenic acid
d. Stearic acid)
- 19(Which of the following is essential fatty acid? a. Oleic acid
b. Linolenic
c. Palmitic acid
d. Stearic acid)
- 20(Unpleasant odour and taste in a fat (rancidity) can be prevented by addition of which of the following? a. Lead
b. Copper
c. Vitamin E
d. Ergosterol)
- 21(Which of the following is a derived lipid?
a. Fatty acid
b. Triacylglycerol
c. Phospholipid
d. Glycolipid)
- 22(Complete hydrogenation of linolenic produces which of the following? a. Arachidonic acid
b. Linoleic acid
c. Oleic acid
d. Stearic acid)
- 23(Linolenic acid belongs to which of the following classes? a. omega-12
b. omega-9
c. omega-6
d. omega-3)
- 24(A patient with constipation is advised to increase dietary fiber. Which polysaccharide increases stool bulk?
a. Starch
b. Glycogen
c. Cellulose
d. Inulin)
- 25(Phosphatidic acid on hydrolysis yields which of the following contents?
a. Glycerol, fatty acids, phosphoric acid, choline
b. Glycerol, fatty acids, phosphoric acid
c. Glycerol, fatty acids, phosphoric acid, Glucose
d. Sphingosine, fatty acids, phosphoric acid)
- 26(Deoxyribose lacks oxygen at which carbon? a. C1
b. C2)



- c. C3
- d. C4

27(In the cyclic form of glucose (pyranose), the anomeric carbon is most directly involved in which of the following?

- a. Determining whether the sugar is a pentose or hexose
- b. Formation of alpha and beta anomers via the orientation of its hydroxyl group
- c. Defining the sugar as D or L based on its configuration
- d. Phosphorylation during glycolysis

28(Glucuronic acid is physiologically important primarily because of which of the following?

- a. It serves as a major energy source for brain cells.
- b. It is a key intermediate in glycolysis.
- c. It conjugates with toxins and drugs to increase their excretion
- d. It forms the backbone of cellulose in plant cell walls.

29(In terms of glycosidic bond composition, how do starch and glycogen differ? a. Starch contains only alpha-1,4 bonds, while glycogen contains both alpha-1,4 and alpha-1,6 bonds.

- b. Both contain alpha-1,4 and alpha-1,6 bonds, but glycogen has more frequent alpha-1,6 bonds.
- c. Glycogen contains beta-1,4 bonds, while starch contains alpha-1,4 bonds.
- d. Starch contains alpha-1,6 bonds at branching points, while glycogen contains only alpha-1,4 bonds.

30(A biology student is examining three polysaccharides: starch, cellulose, and glycogen.

Which of the following statements accurately distinguishes their glycosidic linkages?

- a. Starch and cellulose contain beta-1,4 linkages, while glycogen contains alpha-1,4 linkages.
- b. Starch and glycogen contain alpha-1,4 linkages, while cellulose contains beta-1,4 linkages.
- c. All three contain alpha-1,4 linkages, but only glycogen has alpha-1,6 branches.
- d. Cellulose and glycogen contain both alpha-1,4 and alpha-1,6 linkages, while starch contains only beta-1,4 linkages.

31(Enzyme A has a K_m of 0.1 mM for its substrate, while Enzyme B has a K_m of

10 mM for the same substrate. At a substrate concentration of 0.5 mM: a. Both enzymes will operate at their maximum velocities

- b. Enzyme A will have a higher activation level than Enzyme B
- c. Enzyme B will have a higher activation level than Enzyme A
- d. Both enzymes will have equal activity

32(A palindromic DNA sequence is characterized by which of the following? a. Reading the same forward and backward on the same strand

- b. Self-complementary within each strand with twofold symmetry
- c. Containing only one type of nucleotide
- d. Being present only in viral DNA

33(A researcher heats a solution of double-stranded DNA and observes a sudden increase in UV absorbance at 260 nm. Which of the following best explains this phenomenon?

- a. The phosphodiester bonds are breaking, releasing free nucleotides



- b. The hyperchromic effect due to unstacking of bases
 - c. The DNA is being hydrolyzed into nucleosides
 - d. The DNA is forming Z-DNA conformation
- 34(What type of bond is cleaved when DNA is hydrolyzed by nucleases? a. Glycosidic bonds
- b. Hydrogen bonds
 - c. Phosphodiester bonds
 - d. Disulfide bonds
- 35(How is the phosphodiester bond formed between two nucleotides in a DNA strand?
- a. Between the 5' phosphate of one nucleotide and the 3' OH of the next
 - b. Between the 3' phosphate of one nucleotide and the 5' OH of the next
 - c. Between two nitrogenous bases via hydrogen bonds
 - d. Between the sugar of one nucleotide and the base of the next
- 36(Which of the following best describes the composition of the nucleosome core particle?
- a. DNA wrapped around histone H1 only
 - b. DNA wrapped around a tetramer of histones H3 and H4
 - c. DNA wrapped around an octamer of histones H2A, H2B, H3, and H4
 - d. DNA wrapped around non-histone scaffolding proteins
- 37(What is the primary effect of histone acetylation on nucleosome structure and function?
- a. Increases positive charge, strengthening DNA-histone interaction
 - b. Adds methyl groups to DNA, silencing genes
 - c. Neutralizes positive charge, loosening DNA-histone binding
 - d. Promotes degradation of histones
- 38(What is meant by polysomes?
- a. Multiple ribosomes on a single mRNA
 - b. Complexes of rRNA and protein
 - c. Multiple mRNAs on a single ribosome
 - d. Splicing complexes
- 39(A researcher is studying the initiation of protein synthesis. Which of the following initiation factors is unique to eukaryotes and is responsible for binding to the 5' cap of mRNA?
- a. IF-2
 - b. eIF-4
 - c. EF-Tu
 - d. sigma factor
- 40(Which of the following describes the function of cAMP in lac operon regulation?
- a. It acts as an inducer by binding to the repressor
 - b. It binds to CAP and enhance transcription
 - c. It directly inhibits RNA polymerase binding
 - d. It degrades the repressor protein
- 41(One among the usually asymptomatic genetic variations occurs when a base of the triplet codon in the protein coding gene is replaced by another, without any alteration in the amino acid incorporated in the polypeptide. Which among the following features of the genetic code is responsible for such silent variants?



- a. Overlapping nature
 - b. Unambiguity
 - c. Degeneracy
 - d. Universality
- 42(Which of the following is required for both prokaryotic and eukaryotic protein synthesis?
- a. Binding of the small ribosomal subunit to the Shine-Dalgarno sequence
 - b. Formylated methionyl-transfer (t)RNA
 - c. Movement of the messenger RNA out of the nucleus and into the cytoplasm
 - d. Translocation of the peptidyl-tRNA from the A site to the P site
- 43(A 20-year-old man with a microcytic anemia is found to have an abnormal form of \beta-globin (Hemoglobin Constant Spring) that is 172 amino acids long, rather than the 141 found in the normal protein. Which of the following point mutations is consistent with this abnormality?
- a. CGA → UGA
 - b. UAA → UAG
 - c. GCA → GAA
 - d. UAA → CAA
- 44(During DNA replication, which of the following best explains why DNA polymerases require an RNA primer, whereas RNA polymerases do not?
- a. Unlike RNA polymerase, DNA polymerases only add nucleotides to an existing 3'-OH group.
 - b. RNA primers provide higher fidelity for DNA synthesis.
 - c. DNA polymerases lack the ability to unwind DNA, unlike RNA polymerases.
 - d. RNA polymerases use ribonucleotides that do not require priming.
- 45(A mutant tRNA has an anticodon sequence 5'-GAA-3' instead of the normal 5'-UAC-3'. Which codon would this mutant tRNA most likely recognize?
- a. 5'-UUC-3'
 - b. 5'-CUU-3'
 - c. 5'-UUA-3'
 - d. 5'-CUG-3'
- 46(In sickle cell anemia, a single base mutation (GAG → GTG) leads to glutamic acid being replaced by valine. This mutation is best classified as:
- a. Silent mutation
 - b. Nonsense mutation
 - c. Missense mutation
 - d. Frameshift mutation
- 47(Which of the following best explains how a single eukaryotic mRNA can produce different protein isoforms in different tissues?
- a. Alternative splicing of pre-mRNA
 - b. RNA editing of mature mRNA
 - c. Tissue-specific post-translational modifications
 - d. Different translation initiation sites
- 48(A new antibiotic is designed to selectively inhibit bacterial DNA replication by targeting the enzyme that synthesizes RNA primers. Which enzyme would be the target?
- a. DNA polymerase I
 - b. DNA polymerase III
 - c. Primase



d. DNA ligase

49(How does the function of DNA polymerase I differ from DNA polymerase III during lagging strand synthesis?

- a. Pol III synthesizes the Okazaki fragments, while Pol I synthesizes the RNA primers.
- b. Pol III removes the RNA primers, while Pol I fills the resulting gaps with DNA.
- c. Pol III synthesizes the Okazaki fragments, while Pol I removes and replaces RNA primers.
- d. Pol III joins Okazaki fragments, while Pol I proofreads the newly synthesized DNA.

50(Which of the following DNA lesions is primarily repaired by nucleotide excision repair?

- a. Deaminated cytosine
- b. Thymine dimers
- c. Single-strand breaks
- d. Mismatched base pairs



**QUESTION 2 : Answer the following questions(5x2=10 marks)**

1. Compare between sucrose and invert sugar regarding structure and reducing properties.
2. Demonstrate an example of sugar alcohol of clinical importance.
3. Enumerate two biochemical defects in Alzheimer's disease.
4. List two differences between alpha helix and beta sheet.
5. List two differences between Amylose and amylopectin.

Question 3: Answer the following questions: (5 × 2 = 10 marks)

1. Enumerate four mechanisms of enzyme catalysis.
2. Compare between Omega 3 and omega 6 fatty acids.
3. You are studying the enzyme dihydrofolate reductase, a target for cancer chemotherapy. You test two potential drugs, Drug X and Drug Y, for their inhibitory effects. The following kinetic data are obtained:
Without inhibitor: $K_m = 1.5 \text{ {M}}$, $V_{max} = 60 \text{ {pmol/min}}$
With Drug X: $K_m = 6.0 \text{ {M}}$, $V_{max} = 60 \text{ {pmol/min}}$
With Drug Y: $K_m = 1.5 \text{ {M}}$, $V_{max} = 20 \text{ {pmol/min}}$
• Classify the type of inhibition exhibited by Drug X and Drug Y. Justify your answer based on the changes in K_m and V_{max} .
4. List two main differences between triacylglycerol and wax.
5. Describe the regulatory mechanisms occur in lactose operon in the presence of both glucose and lactose and define their effect on transcription of structural genes.

Question 4: Answer the following questions: (5 × 2 = 10 marks)

1. Compare between synthesis of leading and lagging strand and tabulate your answer.
2. Define missense mutation and describe its types with examples.
3. What is the functional significance of A- and T-rich DNA compared to G- and C-rich DNA?
4. Define the ribozyme needed during elongation in protein biosynthesis and demonstrate its role.
5. What role do the reactions shown play in eukaryotic transcription?



13. End Semester 2026 أهلية

- 1-Which of the following best describes a functional plasma enzyme?
- Its concentration in plasma is normally very low compared to tissues.
 - It increases in plasma during tissue damage or disease.
 - Its substrate is always present in the blood.
 - It has no known physiological function in the blood.
- 2-Which of the following best describes the kinetics of an enzyme-catalyzed reaction in the presence of a non-competitive inhibitor?
- The inhibitor can bind to both free enzyme and the ES complex; K_m unchanged; V_{max} decreased
 - The inhibitor can bind to both free enzyme and the ES complex; K_m decreased; V_{max} unchanged
 - The inhibitor can bind only to free enzyme; K_m unchanged; V_{max} decreased
 - The inhibitor can bind only to free enzyme; K_m decreased; V_{max} unchanged
- 3-The difference between holoenzyme and apoenzyme is that holoenzyme:
- Contains its cofactor
 - Is fully active
 - Is phosphorylated
 - Is glycosylated
- 4-Which of the following best describes the "induced fit" model of enzyme-substrate interaction?
- The substrate exactly fits into the rigid active site
 - The enzyme changes conformation upon substrate binding
 - The substrate changes shape to match the enzyme
 - Only the correct substrate can enter the active site
- 5-Enzyme A has a K_m of 0.1 mM for its substrate, while Enzyme B has a K_m of 10 mM for the same substrate. At a substrate concentration of 0.5 mM:
- Both enzymes will operate at their maximum velocities
 - Enzyme A will have a higher activation level than Enzyme B
 - Enzyme B will have a higher activation level than Enzyme A
 - Both enzymes will have equal activity
- 6-What is the organic non-protein part of conjugated enzymes?
- Coenzyme
 - Zymogen
 - Co-factor
 - Holoenzyme
- 7-In a patient with hereditary nonpolyposis colorectal cancer (HNPCC), DNA sequencing reveals a mutation in the MSH2 gene. Which replication error would accumulate most prominently in this patient's cells?
- Thymine dimers from UV exposure
 - 8-oxoguanine lesions from oxidative damage
 - Mismatched bases
 - Double-strand breaks from ionizing radiation
- 8-Histone acetylation generally leads to which of the following events?
- Gene silencing



- b. Gene activation
 - c. DNA methylation
 - d. Heterochromatin formation
- 9-Which enzyme is produced by the lac I gene?
- a. Lactose permease
 - b. β -galactosidase
 - c. Thiogalactoside transacetylase
 - d. Repressor protein
- 10-What is the main glycosidic bond in starch?
- a. β -1,4 linkage
 - b. α -1,6 linkage
 - c. α -1,4 linkage
 - d. β -1,6 linkage
- 11-Which of the following describes the function of cAMP in lac operon regulation?
- a. It acts as an inducer by binding to the repressor
 - b. It binds to CAP, enabling it to bind DNA and enhance transcription
 - c. It directly inhibits RNA polymerase binding
 - d. It degrades the repressor protein
- 12-The prokaryotic promoter includes the -10 box, also known as:
- a. TATA box
 - b. Pribnow box
 - c. CAAT box
 - d. GC box
- 13-During translation initiation in prokaryotes, the Shine-Dalgarno sequence base-pairs with:
- a. The 3' end of 16S rRNA
 - b. The 5' end of 18S rRNA
 - c. The anticodon of initiator tRNA
 - d. The coding region of Mrna
- 14-A mutant tRNA has an anticodon sequence 5'-GAA-3' instead of the normal 5'-UAC-3'. Which codon would this mutant tRNA most likely recognize?
- a. 5'-UUC-3'
 - b. 5'-CUU-3'
 - c. 5'-UUA-3'
 - d. 5'-CUG-3'
- 15-In sickle cell anemia, a single base mutation (GAG $\rightarrow$ GTG) leads to glutamic acid being replaced by valine. This mutation is best classified as:
- a. Silent mutation
 - b. Nonsense mutation
 - c. Missense mutation
 - d. Frameshift mutation
- 16-Which of the following best explains how a single eukaryotic mRNA can produce different protein isoforms in different tissues?
- a. Alternative splicing of pre-mRNA
 - b. RNA editing of mature mRNA
 - c. Different translation initiation sites
 - d. Tissue-specific post-translational modifications
- 17-Which molecule acts as the inducer of the lac operon by binding to the repressor protein?
- a. Glucose



- b. cAMP
 - c. Allolactose
 - d. Galactose
- 18-In the lac operon, the repressor binds to which of the following? a. The promoter
- b. The operator
 - c. The structural genes
 - d. RNA polymerase
- 19-Spontaneous deamination of cytosine results in which of the following? a. Thymine
- b. Uracil
 - c. Guanine
 - d. Adenine
- 20-Transition mutations involve the substitution of which of the following? a. G to T
- b. T to G
 - c. C to T
 - d. A to C
- 21-What is a point mutation that changes a codon to a stop codon? a. Silent mutation
- b. Missense mutation
 - c. Nonsense mutation
 - d. Frameshift mutation
- 22-The P site of the ribosome holds which of the following components? a. The incoming aminoacyl-tRNA
- b. The tRNA carrying the growing polypeptide
 - c. The empty tRNA before exit
 - d. The mRNA template
- 23-In humans, which of the following is a dietary essential fatty acid? a. Palmitic acid
- b. Stearic acid
 - c. Oleic acid
 - d. Linoleic acid
- 24-What is the nitrogenous base in lecithin? a. Ethanolamine
- b. Choline
 - c. Serine
 - d. Betaine
- 25-The prokaryotic promoter includes the -10 box, also known as: a. TATA box
- b. Pribnow box
 - c. CAAT box
 - d. GC box
- 26-In mismatch repair (MMR), how does the cell distinguish the newly synthesized strand from the template strand in *E. coli*?
- a. By the presence of RNA primers
 - b. By methylation of the parental strand
 - c. By the presence of single-strand binding proteins
 - d. By the direction of DNA synthesis
- 27-In cephalin, choline is replaced by which of the following? a. Serine



- b. Ethanolamine
- c. Betaine
- d. Sphingosine

28-Which of the following enzymes has 5'-3' exonuclease activity that removes RNA primers?

- a. DNA polymerase I
- b. DNA polymerase III
- c. Primase
- d. Ligase

29-In prokaryotes, the main enzyme responsible for DNA elongation is: a. DNA polymerase I

- b. DNA polymerase II
- c. DNA polymerase III
- d. DNA polymerase A

30-A medical researcher is performing electrophoresis on a mixture of amino acids at pH 6.0. Glycine (pI ~6.0) remains stationary, while aspartic acid (pI

~3.0) migrates toward the anode. This is because:

- a. Both amino acids are zwitterions at pH 6.0
- b. Glycine has a net negative charge and aspartic acid has a net positive charge at pH 6.0
- c. Glycine has no net charge, while aspartic acid has a net negative charge at pH 6.0
- d. Glycine has no net charge, while aspartic acid has a net positive charge at pH 6.0

31-A mutation in DNA polymerase III that eliminates its 3'-5' exonuclease activity would result in:

- a. Increased replication speed
- b. Decreased mutation rate
- c. Increased mutation rate
- d. Inability to synthesize DNA

32-Which checkpoint ensures DNA replication is complete before mitosis? a. G1 checkpoint

- b. G2 checkpoint
- c. M checkpoint
- d. S checkpoint

33-The difference between leading and lagging strand synthesis is primarily due to:

- a. Different DNA polymerases on each strand
- b. The antiparallel nature of DNA and the 5'→3' synthesis direction
- c. Different primer requirements
- d. Different proofreading mechanisms

34-During DNA replication, proofreading occurs by: a. 3'→5' exonuclease activity of DNA polymerase

- b. Nucleotide excision repair
- c. 5'→3' exonuclease activity of DNA polymerase
- d. Mismatch repair enzyme

35-In cancer cells, telomerase activity is often upregulated. This allows cancer cells to:

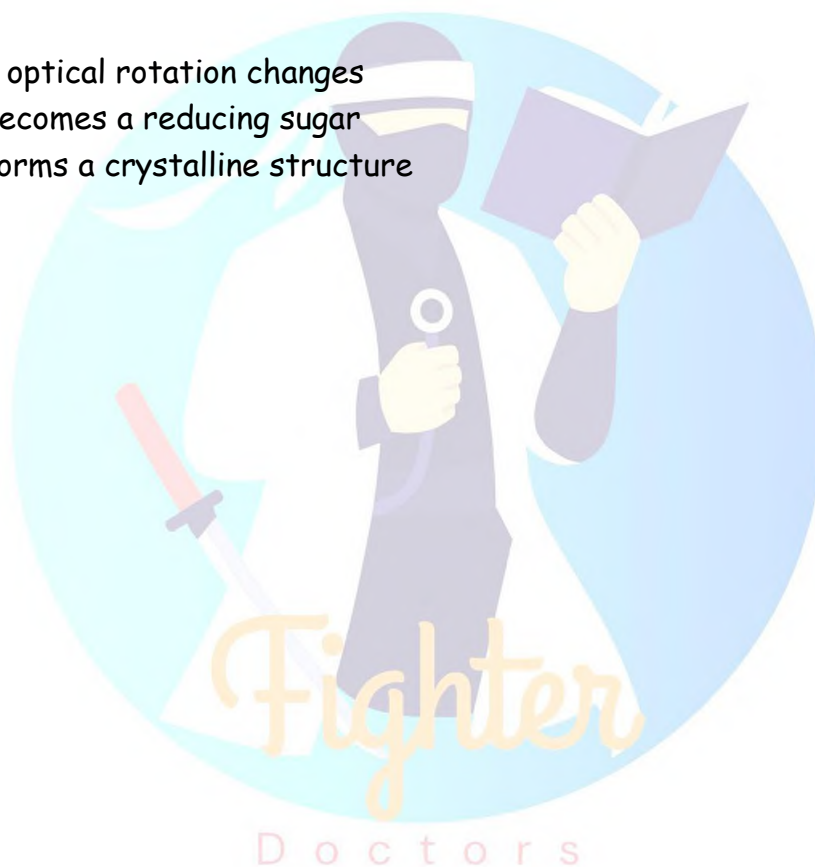
- a. Replicate DNA faster
- b. Avoid telomere shortening and cellular senescence



- c. Repair DNA damage more efficiently
d. Evade the immune system
- 36-Histones are rich in which amino acids?
- a. Acidic amino acids (Asp, Glu)
b. Basic amino acids (Lys, Arg)
c. Sulfur-containing amino acids (Cys, Met)
d. Aromatic amino acids (Phe, Tyr)
- 37-A palindromic DNA sequence is characterized by:
- a. Reading the same forward and backward on the same strand
b. Self-complementarity within each strand with twofold symmetry
c. Containing only one type of nucleotide
d. Being present only in viral DNA
- 38-The imino group is present in which of the following amino acids?
- a. Serine
b. Valine
c. Proline
d. Alanine
- 39-All of the following are basic essential amino acids except:
- a. Histidine
b. Leucine
c. Lysine
d. Arginine
- 40-Which of the following is a health risk associated with hydrogenated fats?
- a. It produces toxic gases
b. It increases the risk of atherosclerosis
c. It makes the oil rancid
d. It destroys vitamins
- 41-Which type of fatty acid is most susceptible to rancidity?
- a. Oleic acid
b. Stearic acid
c. Palmitic acid
d. Linolenic acid
- 42-D-Glucose and D-Galactose are considered which of the following isomers?
- a. Anomers
b. Epimers
c. Enantiomers
d. Ketoses isomers
- 43-Which of the following is a primary component of structural fat?
- a. Triglycerides
b. Cholesterol
c. Free fatty acids
d. Glycerol
- 44-Which of the following is a ketose sugar?
- a. Glucose
b. Galactose
c. Fructose
d. Ribose
- 45-Cellulose is indigestible by humans because it contains:



- a. α -1,4 linkages
 - b. β -1,4 linkages
 - c. α -1,6 linkages
 - d. β -1,6 linkages
- 46-Single-stranded binding proteins (SSBs) function to:
- a. Synthesize RNA primers
 - b. Prevent reannealing of separated strands
 - c. Join Okazaki fragments
 - d. Proofread newly synthesized DNA
- 47-What is the shape of the curve for allosteric enzyme kinetics?
- a. Hyperbolic
 - b. Straightline
 - c. Sigmoidal (S-shaped)
 - d. U-shaped
- 48-Why is the product of sucrose hydrolysis called "invert sugar"?
- a. Because it changes color
 - b. Because the optical rotation changes
 - c. Because it becomes a reducing sugar
 - d. Because it forms a crystalline structure



**Question 2: Answer the following questions: (5 × 2 = 10 marks)**

1. Compare between liver and muscle Glycogen as regard Fate and function in a table.
2. Demonstrate an example of sugar alcohol of clinical importance.
3. Enumerate two biochemical defects in Alzheimer's disease.
4. List two differences between alpha helix and beta sheet.
5. List Two differences between Amylose and amylopectin.

Question 3: Answer the following questions: (5 × 2 = 10 marks)

1. Rancidity: Define and enumerate factors protecting against it.
2. Enumerate two important functions of phospholipid.
3. A 55-year-old male presents to the emergency department with chest pain. Serum enzyme analysis shows elevated total lactate dehydrogenase (LDH) and creatine kinase (CK). Further isoenzyme analysis reveals:
 - ➔ LDH1: 45% (normal: 15-30%)
 - ➔ LDH2: 30% (normal: 30-40%)
 - ➔ LDH3: 15% (normal: 15-25%)
 - ➔ LDH4: 5% (normal: 5-10%)
 - ➔ LDH5: 5% (normal: 5-10%)
 - ➔ CK-MB: 12% of total CK (normal: <5%)
 - Which isoenzyme pattern is indicative of myocardial infarction? Explain and define the structure of the dominant isoenzyme.
4. Enumerate characters of genetic code.
5. Methotrexate is a commonly used drug that inhibits the dihydrofolate reductase enzyme. From the diagram below, describe the type of inhibition in this case and justify your answer.

Question 4: Answer the following questions: (5 × 2 = 10 marks)

1. Enumerate 2 main differences between prokaryotes and eukaryotes regarding protein biosynthesis (translation).
2. Describe the basic structural unit of chromatin and explain how chromatin modification plays a role in regulation of gene expression.
3. Mention the role of each of the following enzymes:
 - a- Helicase
 - b- Primase
4. The genetic code is a triplet code of four nucleotide bases. Why do only 61 of the 64 triplets code for amino acids in the translation of mRNA? ⑤ How does prokaryotic RNA pol recognize and bind the appropriate region of DNA to initiate transcription, as shown? Is a primer required in transcription?



14. Final 2026 دور ثاني

Total Marks: 55 Marks

Time: 3 Hours

Question 1: Choose the correct answer and shadow your answer in the answer table:
(50 × 0.5 marks = 25 marks)

1. Which of the following amino acids is known as aliphatic?

- a. Tyrosine
- b. Tryptophan
- c. Histidine
- d. Serine

2. In Sickle Cell Anemia, the nucleotide alteration leads to substitution of a glutamate residue with valine at the 6th position in the β -globin chain. Which of the following best describes this substitution?

- a. A hydroxy-amino acid is replaced by a simple amino acid
- b. A polar amino acid is replaced by a non-polar amino acid
- c. An aliphatic amino acid is replaced by a non-essential amino acid
- d. A dietary essential amino acid is replaced by an aromatic amino acid

3. A particular point mutation results in disruption of the α -helical structure in a segment of the mutant protein. What is the most likely change in the primary structure of the mutant protein?

- a. glutamate to aspartate
- b. lysine to arginine
- c. methionine to proline
- d. valine to alanine

4. Which of the following best describes about the kinetics of an enzyme catalyzed reaction in the presence of a non-competitive inhibitor?

- a. The inhibitor can bind to both free Enzyme and the ES complex; K_m unchanged; V_{max} decreased
- b. The inhibitor can bind to both free Enzyme and the ES complex; K_m decreased; V_{max} unchanged
- c. The inhibitor can bind only to free Enzyme; K_m unchanged; V_{max} decreased
- d. The inhibitor can bind only to free Enzyme; K_m decreased; V_{max} unchanged

5. What is the difference between holoenzyme and apoenzyme is that holoenzyme:

- a. Contains its cofactor
- b. Is fully active
- c. Is phosphorylated
- d. Is glycosylated

6. Which of the following best describes a functional plasma enzyme?

- a. Its concentration in plasma is normally very low compared to tissues.
- b. It increases in plasma during tissue damage or disease.
- c. Its substrate is always present in the blood.
- d. It has no known physiological function in the blood.



7. In a patient with hereditary nonpolyposis colorectal cancer (HNPCC), DNA sequencing reveals a mutation in the hMSH2 gene. Which of the following replication errors would accumulate most prominently in this patient's cells?

- a. Thymine dimers from UV exposure
- b. 8-oxoguanine lesions from oxidative damage
- c. Mismatched bases
- d. Double-strand breaks from ionizing radiation

8. To which of the following classes lactase enzyme belongs to?

- a. Oxidoreductases
- b. Transferases
- c. Hydrolases
- d. Ligases

9. Enzyme A has a K_m of 0.1 mM for its substrate, while Enzyme B has a K_m of 10 mM for the same substrate. At a substrate concentration of 0.5 mM:

- a. Both enzymes will operate at their maximum velocities
- b. Enzyme A will have a higher activation level than Enzyme B
- c. Enzyme B will have a higher activation level than Enzyme A
- d. Both enzymes will have equal activity

10. What is the organic non-protein part of the conjugated enzymes?

- a. Coenzyme
- b. Zymogen
- c. Co-factor
- d. Holoenzyme

11. Which of the following amino acids differs relatively to its structure?

- a. Proline
- b. Glycine
- c. Leucine
- d. Alanine

12. What is the form of an amino acid at its isoelectric point?

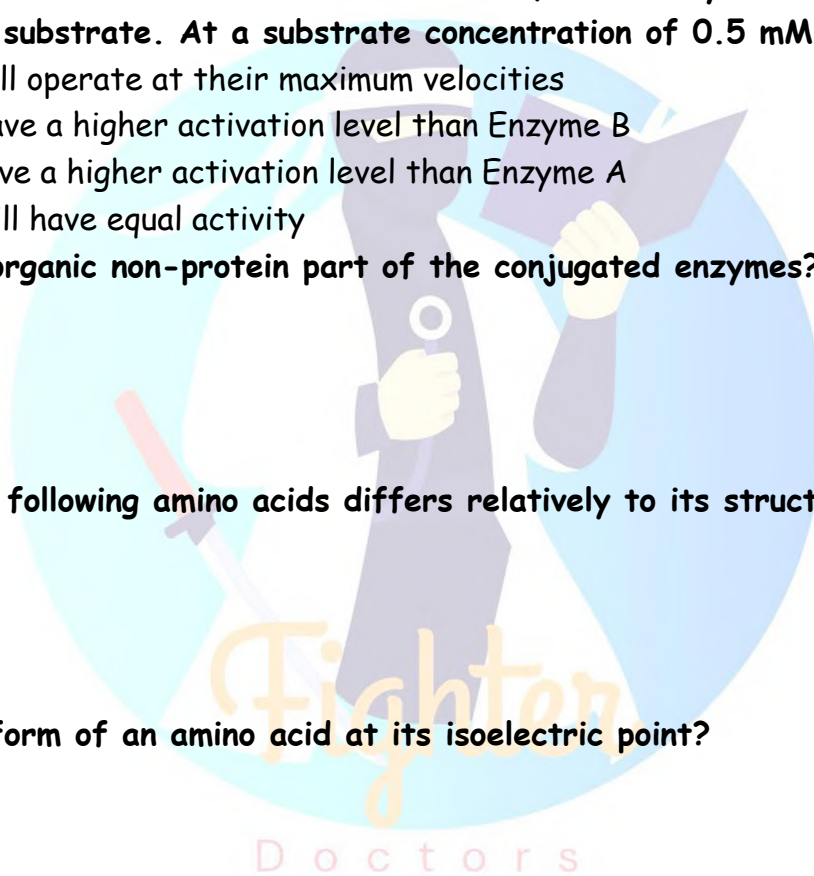
- a. Cation
- b. Anion
- c. Zwitterion
- d. Not charged form

13. The helical structure of any polypeptide chain is mainly derived from which of the following?

- a. Hydrogen bonds
- b. Disulfide linkage
- c. Hydrophobic interaction
- d. Peptide linkage

14. Which lipid is a component of the inner mitochondrial membrane?

- a. Phosphatidylcholine
- b. Phosphatidylethanolamine





- c. Cardiolipin
- d. Sphingomyelin

15. Denaturation of proteins involves loss of which of the following?

- a. Primary structure
- b. Ionic interactions
- c. Amino acid sequence
- d. Peptide bonds

16. Which of the following is a storage form of lipid in human body?

- a. Phospholipid
- b. Glycolipid
- c. Triacylglycerol
- d. Sulfolipid

17. Which of the following is an essential fatty acid?

- a. Oleic acid
- b. Linolenic acid
- c. Palmitic acid
- d. Stearic acid

18. Which of the following is a derived lipid?

- a. Fatty acid
- b. Triacylglycerol
- c. Phospholipid
- d. Glycolipid

19. Complete hydrogenation of linolenic produces which of the following?

- a. Arachidonic acid
- b. Linoleic acid
- c. Oleic acid
- d. Stearic acid

20. Linolenic acid belongs to which of the following classes?

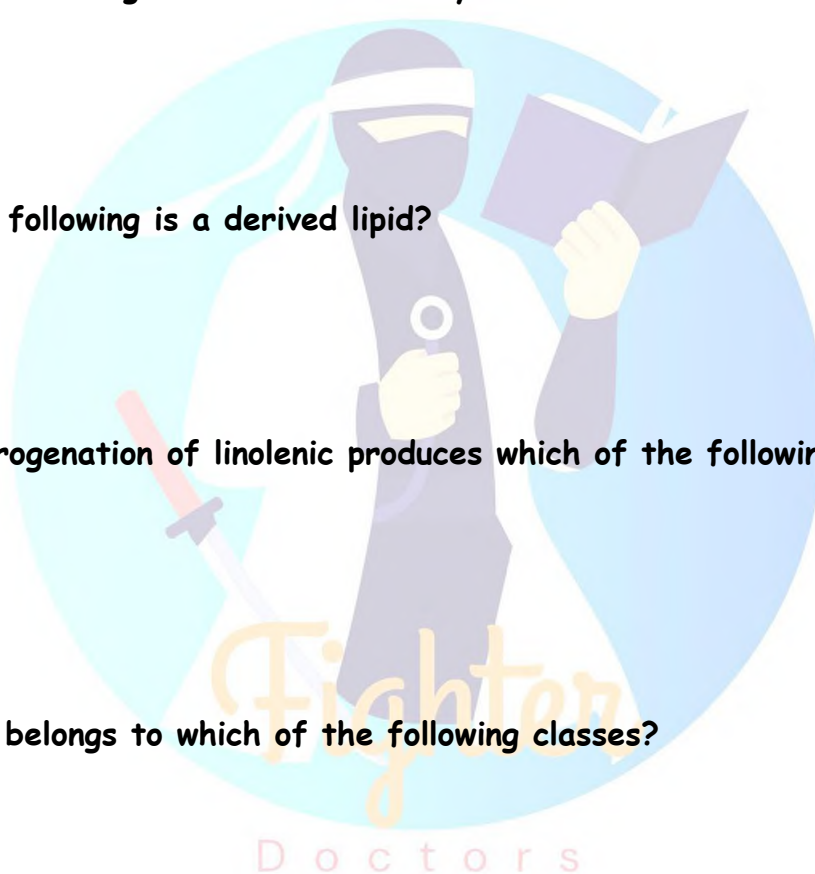
- a. ω -12
- b. ω -9
- c. ω -6
- d. ω -3

21. A patient with constipation is advised to increase dietary fiber. Which of the following carbohydrates is recommended in this case?

- a. Starch
- b. Glycogen
- c. Cellulose
- d. Inulin

22. Hydrolysis of phosphatidic acid by phospholipases yields which of the following contents?

- a. Glycerol, fatty acids, phosphoric acid, choline
- b. Glycerol, fatty acids, phosphoric acid





c. Glycerol, fatty acids, phosphoric acid, Glucose

d. Sphingosine, fatty acids, phosphoric acid

23. Deoxyribose lacks oxygen at which carbon?

a. C1

b. C2

c. C3

d. C4

24. In the cyclic form of glucose (pyranose), the anomeric carbon is most directly involved in which of the following?

a. Determining whether the sugar is a pentose or hexose

b. Formation of α and β anomers via the orientation of its hydroxyl group

c. Defining the sugar as D or L based on its configuration

d. Phosphorylation during glycolysis

25. Enzyme inhibition caused by a substance resembling substrate molecule is known as which of the following?

a. Competitive inhibition

b. Non-competitive inhibition

c. Feedback inhibition

d. Allosteric inhibition

26. A biology student is examining three polysaccharides: starch, cellulose, and glycogen. Which of the following statements accurately distinguishes their glycosidic linkages?

a. Starch and cellulose contain β -1,4 linkages, while glycogen contains α -1,4 linkages.

b. Starch and glycogen contain α -1,4 linkages, while cellulose contains β -1,4 linkages.

c. All three contain α -1,4 linkages, but only glycogen has α -1,6 branches.

d. Cellulose and glycogen contain both α -1,4 and α -1,6 linkages, while starch contains only β -1,4 linkages.

27. In uncontrolled diabetes, which sugar alcohol tends to accumulate in tissues like the lens, contributing to complications (e.g., cataracts and neuropathy)?

a. Mannitol

b. Xylitol

c. Sorbitol

d. Ribitol

28. Which of the following sugars lacks the reducing properties?

a. Lactose

b. Fructose

c. Maltose

d. Sucrose

29. Which of the following statements is TRUE regarding invert sugar?

a. It is the hydrolytic product of lactose sugar.

b. It is a non-reducing mixture.

c. It exhibits mutarotation.

d. It deviates the angle of polarized light to the right.



30. A researcher heats a solution of double-stranded DNA and observes a sudden increase in UV absorbance at 260 nm. Which of the following best explains this phenomenon?

- a. The phosphodiester bonds are breaking, releasing free nucleotides
- b. The hyperchromic effect due to unstacking of bases
- c. The DNA is being hydrolyzed into nucleosides
- d. The DNA is forming Z-DNA conformation

31. What type of bond is cleaved by restriction endonucleases?

- a. Glycosidic bonds
- b. Hydrogen bonds
- c. Phosphodiester bonds
- d. Disulfide bonds

32. How is the phosphodiester bond formed between two nucleotides in a DNA strand?

- a. Between the 5' phosphate of one nucleotide and the 3' OH of the next
- b. Between the 3' phosphate of one nucleotide and the 5' OH of the next
- c. Between two nitrogenous bases via hydrogen bonds
- d. Between the sugar of one nucleotide and the base of the next

33. What is the main function of histone H1?

- a. Forming the core of the nucleosome.
- b. Binding to linker DNA.
- c. Acetylating other histone proteins.
- d. Serving as the scaffold for the 30nm fiber.

34. What is the primary effect of histone acetylation on nucleosome structure and function?

- a. Increases positive charge, strengthening DNA-histone interaction
- b. Adds methyl groups to DNA, silencing genes
- c. Neutralizes positive charge, loosening DNA-histone binding
- d. Promotes degradation of histones

35. Which of the following statements represents the term polysomes?

- a. Multiple ribosomes on a single mRNA
- b. Complexes of rRNA and protein
- c. Multiple mRNAs on a single ribosome
- d. Splicing complexes

36. Which of the following represents the chemical structure of adenine?

- a. 6-aminopurine
- b. 2-amino 6-oxypurine
- c. 6-oxypurine
- d. 2, 6 dioxypurine

37. A mutation in the gene for the DnaA protein in E. coli that renders it unable to bind ATP would most directly affect which stage of replication?

- a. Elongation
- b. Termination



- c. Initiation
- d. Proofreading

38. A 20-year-old man with a microcytic anemia is found to have an abnormal form of β -globin (Hemoglobin Constant Spring) that is 172 amino acids long, rather than the 141 found in the normal protein. Which of the following point mutations is consistent with this abnormality?

- a. CGA $\rightarrow$ UGA
- b. UAA $\rightarrow$ UAG
- c. GCA $\rightarrow$ GAA
- d. UAA $\rightarrow$ CAA

39. Which of the following describes the function of cAMP in lac operon regulation?

- a. It acts as an inducer by binding to the repressor
- b. It binds to CAP and enhance transcription
- c. It directly inhibits RNA polymerase binding
- d. It degrades the repressor protein

40. In splicing in eukaryotes, what are the sequences that are removed from the primary transcript?

- a. Exons
- b. Introns
- c. Codons
- d. Promoters

41. A mutation that inactivates the 3' to 5' exonuclease activity of DNA polymerase would result in?

- a. A slower rate of DNA synthesis.
- b. An increased rate of mutation.
- c. Inability to remove RNA primers.
- d. Failure to join Okazaki fragments.

42. Which of the following RNAs is coding?

- a. mRNA
- b. tRNA
- c. rRNA
- d. snRNA

43. In dTMP, what is the atom in the nitrogenous base forming glycosidic linkage with ribose sugar?

- a. Carbon no 1
- b. Carbon no 5
- c. Nitrogen no 1
- d. Nitrogen no 9

44. Which of the following is required for both prokaryotic and eukaryotic protein synthesis?



- a. Binding of the small ribosomal subunit to the Shine-Dalgarno sequence
- b. Formylated methionyl-transfer (t)RNA
- c. Movement of the messenger RNA out of the nucleus and into the cytoplasm
- d. Translocation of the peptidyl-tRNA from the A site to the P site

45. Which of the following best explains how a single eukaryotic mRNA can produce different protein isoforms in different tissues?

- a. Alternative splicing of pre-mRNA
- b. RNA editing of mature mRNA
- c. Tissue-specific post-translational modifications
- d. Different translation initiation sites

46. A new antibiotic is designed to selectively inhibit bacterial DNA replication by targeting the enzyme that synthesizes RNA primers. Which enzyme would be the target?

- a. DNA polymerase I
- b. DNA polymerase III
- c. Primase
- d. DNA ligase

47. How does the function of DNA polymerase I differ from DNA polymerase III during lagging strand synthesis?

- a. Pol III synthesizes the Okazaki fragments, while Pol I synthesizes the RNA primers.
- b. Pol III removes the RNA primers, while Pol I fills the resulting gaps with DNA.
- c. Pol III synthesizes the Okazaki fragments, while Pol I removes and replaces RNA primers.
- d. Pol III joins Okazaki fragments, while Pol I proofreads the newly synthesized DNA.

48. Which of the following DNA lesions is primarily repaired by nucleotide excision repair?

- a. Deaminated cytosine
- b. Thymine dimers
- c. Single-strand breaks
- d. Mismatched base pairs

49. During DNA replication, which of the following best explains why DNA polymerases require an RNA primer, whereas RNA polymerases do not?

- a. Unlike RNA polymerase, DNA polymerases only add nucleotides to an existing 3'-OH group.
- b. RNA primers provide higher fidelity for DNA synthesis.
- c. DNA polymerases lack the ability to unwind DNA, unlike RNA polymerases.
- d. RNA polymerases use ribonucleotides that do not require priming.

50. A mutant tRNA has an anticodon sequence 5'-GAA-3' instead of the normal 5'-UAC-3'. Which codon would this mutant tRNA most likely recognize?

- a. 5'-UUC-3'
- b. 5'-CUU-3'
- c. 5'-UUA-3'
- d. 5'-CUG-3'



Question 2: Answer the following questions: (5 × 2 = 10 marks)

A. Compare between starch and cellulose regarding their structure, function and tabulate your answer.

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B. Describe the differences between liver and muscle glycogen regarding function, importance and fate of hydrolysis.

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C. Define polysaccharides with reference to its classification with examples.

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D. Enumerate two biochemical defects or changes that occur in protein denaturation.

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E. List two differences between primary and secondary structure of polypeptide chain.

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Question 3: Answer the following questions: (5 × 2 = 10 marks)

A. Enumerate four main differences between functional and non-functional plasma enzymes.

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B. Define rancidity and describe causes and effects.

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C. Methotrexate is a competitive inhibitor for the enzyme dihydrofolate reductase, a target for cancer chemotherapy. Illustrate the main characters of this inhibitor with the reference to its effect on enzyme kinetics.

D. List two main differences between triacylglycerol and wax in a table.

E. Define what is meant by silent mutation in a gene with reference to its effect on protein structure and explain your answer.

Question 4: Answer the following questions: (5 × 2 = 10 marks)

A. Compare between synthesis of leading and lagging strand and tabulate your answer.

B. Demonstrate the steps of nucleotide excision repair with the reference to the clinical disorder results from its defect.

C. Demonstrate the structure of nucleosome and its role in regulation of gene expression.

D. What is the functional significance of palindromic sequence?



15. Formative 2026

1. A patient with a family history of early-onset colon cancer undergoes genetic screening. A mutation is found in a gene involved in a DNA repair system that corrects errors missed by the proofreading function of DNA polymerase, such as single-base mismatches or small insertion-deletion loops. Which repair system is defective?
- Mismatch repair system
 - Nucleotide Excision Repair
 - Homologous Recombination
 - Base Excision Repair
2. A patient with a high fever is brought to the emergency department. The elevated body temperature is likely denaturing many plasma and cellular proteins. Which level of protein structure would remain intact even after the hydrogen bonds, disulfide bridges, and hydrophobic interactions stabilizing the protein's native conformation are disrupted by the heat?
- Secondary structure
 - Quaternary structure
 - Tertiary structure
 - Primary structure
3. A cell culture experiment shows that upon induction of double-strand breaks, the cells pause at the G2/M checkpoint. What is the primary rationale for this cell cycle arrest?
- To ensure that all centrosomes have been fully duplicated.
 - To prevent the segregation of damaged chromosomes during mitosis.
 - To provide an opportunity for the synthesis of DNA primers on the lagging strand.
 - To allow time for the complete disassembly of the nuclear envelope.
4. A researcher is studying a mutant hemoglobin protein where a single point mutation replaces a neutral amino acid with a charged one. This creates a novel interaction between two residues on the protein's surface that disrupts its quaternary structure. This new, pathological bond would most likely form between the side chains of which pair of amino acids?
- Alanine & leucine
 - Leucine & valine
 - Aspartate & glutamate
 - Leucine and aspartate
5. A 7-year-old boy presents with short stature, a bird-like facial appearance, and premature graying. Genetic testing confirms a mutation in a protein involved in the removal of RNA primers during DNA replication and in DNA repair. Which enzyme is most likely deficient?
- DNA ligase
 - Primase
 - DNA polymerase III
 - DNA polymerase I
6. In a laboratory experiment, a bacterial strain is engineered to lack the dam methylase enzyme. What is the most likely consequence for DNA replication in this strain?
- The Mismatch Repair system will be defective.
 - The DnaA protein will be unable to bind the origin of replication (OriC).
 - Telomeres will shorten dramatically with each cell division.
 - DNA polymerase I will lose its 5' to 3' exonuclease activity.



7. A new antiviral drug is designed to be a nucleoside analog that lacks a 3' hydroxyl group. When incorporated into a growing DNA strand by a viral polymerase, it terminates further elongation. Which property of DNA polymerase explains why a similar drug would be highly toxic to human cells if it were incorporated?

- a. DNA polymerase has 5' to 3' polymerase activity.
- b. DNA polymerase can only synthesize DNA in the 5' to 3' direction on both strands.
- c. DNA polymerase requires a primer with a free 3' hydroxyl group to add new nucleotides.
- d. DNA polymerase has 3' to 5' proofreading exonuclease activity.

8. A structural biologist is examining the sharp turn in a protein's backbone, which is facilitated by the unusual cyclic structure of one of its constituent amino acids. This amino acid's side chain is bonded to its own amino group, making it a secondary amine. Which amino acid contains this imino group?

- a. Proline
- b. Valine
- c. Alanine
- d. Serine

9. A 55-year-old man presents with rapidly progressive dementia and myoclonus. Creutzfeldt-Jakob disease (CJD) is suspected. A brain biopsy is analyzed, and the pathogenic form of the prion protein (PrP^{Sc}) is found to be highly resistant to degradation by proteases. This resistance is best explained by which of the following structural properties of PrP^{Sc}?

- a. It has a different amino acid sequence than the normal cellular prion protein (PrP^C).
- b. It contains a higher proportion of β -pleated sheet structure.
- c. It is more susceptible to proteolytic degradation.
- d. It is encoded by a different gene than PrP^C.

.1a | 2.d | 3.b | 4.d | 5.d | 6.a | 7.c | 8.a | 9.b